

Ali BOULANOUAR TAZDAYT D TNEZDAYT XI TMAZIIT-TABELDIT N AT IGLI

Tabeldit ha-ay d awal n weyrem n Igli, yella wi yeštīnin tacellhit. At Weyrem la ttfen deg tezdayt xi ifassen-nsen d tiymas-nsen, lādya tazdayt d igget xi isekla iqdimen aṭṭas. Ttekker-d deg tenezruft, geddi i yelmed bnaḍem tifellaht d wekraz, yeẓzi-tt ɛla xater tettarew aṭṭas, d yiniwen-nnes yer-sen azal d amsadur d ameqqran. Bla ma nettu tizmert-nnes i ušṭemmel n tezyelt d fad. Al idennat Tabeldit tella ul tettwari-c. Ha-ay d ma i yuggjen aṭṭas n iwawalen ad zwan. Ass-u aḍlis-u i yeryey xi tezdayt d leqdem (asurif) d ameqqran yel zdat. Ad yenfee tabeldit deg Igli deg umezwar d yenfae Tamaziyt qae.

Tazdayt yer-s azal deg tudert n midden n At iyermaḍen xef-s tuttut-nnsen, d yiṭṭaḍ-nsen, d waska nsen d tidrin d yixxamen-nsen la d uṛaṛ-nsen. Tazdayt ha-ay d azamul n umsuref d rreḥma, yella iggen n wenzi yettini : « tazdayt nḍer-tt xi tyunt, ad x-k-d-tenḍer iyniwen ». Anamek-nnes : Tazdayt d taṣebbart. Ssawra deg-s ujar n 900.000 n tezdayin ttatfent deg 343 ddguga an : Buḥeqqus, Tingezza, tadmamt... netta-t ha-ay d isebdad n leflaḥet deg tneẓdayt. Tazdayt yer-s azal d ameqqran yel umekraz ; yettig-tt an yiggen x lwacun-nnes. Yettucc-as isem an yigget xi yessi-s. Llan ddguga n tezdayin tteqqaren-asent x isemawen : Yelli-s-n-Lhaḡ, Tin-n-Ḡuher, Tin Gezza. Iwḍen al geddi i dewwlen tbaraken i wi yezyen tazdayt d al ttezzan i wi s-temmut tezdayt.

رأت عيناي النور شتاء 1977، نشأت وترعرعت في مدينة يتواصل جل ساكنيها بالأمازيغية (تبلديت) تقع هذه المدينة إلى 75 كم شمال ولاية بني عباس، زاولت دراستي في المدرسة الابتدائية للذكور. سنة 1989 انتقلت لاستكمال دراستي في متوسطة اقلي لفترة ثلاث سنوات ثلث خلالها شهادة التعليم المتوسط بتفوق، انتقلت بعدها إلى داخلية ثانوية العلامة ابن البيطار ببني عباس، ثلاث سنوات توجت بالحصول على شهادة البكالوريا سنة 1995. دراستي الجامعية كانت بمدينة سيدي بلعباس أين قضيت خمس سنوات للحصول على شهادة مهندس دولة في البيولوجيا من جامعة "الجيلالي اليابس" سنة 2000. اشتغلت بتدريس مادة علوم الطبيعة في ثانوية العقيد عثمان بشار.

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Ali BOULANOUAR

TAZDAYT D TNEZDAYT

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ⵜⴰⴳⴷⴰⵢⵜ ⴰⵢⵍⵉ ⵜⴰⵎⴰⴷⵓⵔⵜ
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Tigemmi
ⵜⴰⵎⴰⴷⵓⵔⵜ



2023

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ASQAMU UNNIG N TIMMUZZA

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Tazdayt d tnezdayt xi
tmaziyt-tabeldit n At Igli

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Arraz n Uselway n Tegduda
i Tsekla d Tutlayt Tamaziɣt
- 2023 -



Ali BOULANOUAR

Tazdayt d tnezdayt xi
tmaziyt-tabeldit n At Igli

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Tazwart tamatut

Tazdayt d igget xi isekla iqdimen aṭṭas. Ttekker-d deg tnezruft, gedd i yelmed bnaɛdem tifellaḥt d wekraz, yezzi-tt ɛla xater ttaɛrew aṭṭas. iyniwen-nnes yer-sen azal¹ amsadur² d ameqqran, bla ma nettu tizmert-nnes i usteḥmel n tezyelt d fad. Tazdayt yer-s azal deg tudert n midden n At iɣermawen xef-s tuttut-nnsen, d yiṛṛaḍ-nsen, d waska nsen d tidrin d yixxamen-nsen la d uṛaṛ-nsen. Tazdayt ha-ay d azamul n umsuref 3 d rreḥma. , yella iggen n wenzi yettini : « tazdayt nḍer-tt xi tyunt, ad x-k-d-tenḍer iyniwen ». Anamek-nnes : Tazdayt d taṣebbart.

Ssawra deg-s ujar n 900.000 n tezdayin ttatfent deg 343 dɗguga an : Buɣeqqus, Tingezza, tadmamt... netta-t ha-ay d isebɗad n leflaḥet deg tnezdayt. Tazdayt yer-s azal d ameqqran yel umekraz ; yettig-tt an yiggen x lwacun-nnes. Yettucc-as isem an yigget xi yessi-s. Llan dɗguga n tezdayin tteqqaren-asent x isemawen : Yelli-s-n-Lhaḡ, Tin-n-Ḡuher, Tin Gezza. Iwḍen al geddi i dewwlen tbaraken i wi yezyen tazdayt d al ttezzan i wi s-temmut tezdayt.

Zzman yella umekraz kull wass-nnes yettzegga³ deg tanzdayt, yettqama netta d tezdayin-nnes ujar n ma

¹ Azal : *valeur*.

² Amsadur : *alimentaire*.

³ Yetzegga : *passer*.

yettqama netta d twacunt-nnes, yettzzar-ten xi tallast⁴ yel tallast.

Tazdayt tuṣṣiḥt ul tehlek-c tettaṛew ujar n 10 n iziwayen, kull aziwa deg-s jar 15 d 20 Kg n yiniwen. Amala tazdayt igget tettaṛew jar 150 d 200 Kg, azal n yiniwen yettawed xef dollar al 11,32 \$ i yeggen n kilu. Tsemma iggen n tezdayt terra-d 2264 \$ an malek neznez 24 bettiyet n upitrul.

Kull tiyawṣiwin tiqburiñ yebbreñt (ṣwant)⁵ tzeggant smiyat-nsen n tmaziṣt ; tazdayt teffreg⁶ aṭṭas n iwawalen x uyebbu d tsellek-d awal n Tbeldit xi wenfal d weybar.

Iyermawen n Ssawra zzman qaε ttuy-ten tessawalen tamaziṣt. Ass-u teqyem-d yir deg Igli, Mazzer d ukeb⁷ deg Tayit (Berbi). Qaε iyermawen yenniḍen teyber deg-sen Tmaziṣt. Walayenni⁸ tazdayt d tnezdayin, ddguga n yiniwen qeyment tṭfent ismawen-nsent xi tmaziṣt an : Aziwa, Taqernift, Tadmamt, Tingezza⁹ Tamtert, Idiy, Ammas¹⁰...

Tira d nettat yettaḡḡen ilel n tmura ad iqyem ul yettyebbar-c. At Igli ttsawalen awal n Tbeldit sswen-t deg uyi n yemmat-nsen.

Neččnin nessen tanfust n Tayunt¹¹ n Racid i d-isellken Tahiruglifit seg tmettant. Tayunt n Racid ten Dyurit¹² tella deg Memphis deg Miser tettwari deg aseggas n 196, zdat n tarwa n sidna eisa, s tlata n yilsawen: Tahiruglifit n

⁴ Tallast : *obscurité*.

⁵ Iyber / aybar/ yber : *disparition*.

⁶ Tfrg : *protéger*.

⁷ Akeb = ḡrus.

⁸ Walaynni : *mais*.

⁹ Tadmamt, Tingezza : deguga n yeniwen.

¹⁰ Tamtert, Idiy, Ammas : iyermawen deg Sawra.

¹¹ Tayunt : *pierre*.

¹² Dyurit : *type des roches*.

Iferɛunen, Tahiraṭiqit d Tayunanit yettwalga-d deg useggas n 1799. Tahiruglifit n iferɛunen tella tnegger, tza azeggu-nnes lukan mačči d taɣunt n Racid i tt-d-isellken.

Ger-neɣ neččin, Tazdayt ha-ay d taɣunt n Racid i wawal n Tbeldit. Tamaziɣt-Tabeldit ɣreɣren iɣewran-nnes deg tmurt an iɣewran n tezdayt taɣebbart ; tinsa-nnes deg waman, tameгна-nnes deg timsi.

Awttas-inu xi lxedmet-u :

1. Yellan cra n iwawalen n Tabeldit qeymen beṣaḥ la d iggen ul yessin lmeɣna-nsen an smiyyat n tnezdayin ixex ad nilga ma yettinin.
2. Ad wteɣɛ sen n ijuḍaḍ xi iggen n tɣunt :
3. Amezwar ad aryey Tabeldit maki ad tawed lwacun-inu ;
4. d west sen ad sslemdeɣ i medden maki ad ɣɣyen tazdayt d fergen-tt xi waṭṭannen d yixxa ;
5. Ad nesres afus deg ufus maki ad nalli xi tmaziɣt deg Dzayer d Tefriqt ;
6. Ad ssikyey midden ɣel amihi n tesmaḍalt ;
7. Ad nessimɣer awal n tmaziɣt : Ad nsittef iwawalen d ijdiden xi Tabeldit ɣel Tmaziɣt d Tmaziɣt ɣel Tbeldit.

Adlis-u yebḍa ɣel 06 n yixfawen :

- Amezwar yettsawal xi tezdayt : amezruy-nnes, amyezwer-nnes deg taxonomie, tiwtilin n twennaḍt n wukkur-nnes,
- Ixef west sen yettsawal xi iyniwen : timura i yetteznuzun iniwen, awwaḍ n yiniwen, ankaḍ, tasnelɣa n yiyniwen...

- Ixef west krađ yettsawal xi waṭṭanen n tezdayt ;
- Ixef west kuṛ yettsawal xi ixxa ;
- Ixef west semmus yettsawal xi Ifarisen n tezdayt ;
- Ixef west sḍis yettsawal xi tinezdayen n Ssawra.

- I -

Tazdayt n yiniwen

1. Tazdayt deg umezruy (*histoire*)

1.1. Fel Ibabiliyen

Tamurt n Sen-Iyezran¹³ nettat d amkan amezwar n ukur n tezdayt ucin-as ccan d ameqqran əla xater tella tessutuc-ten d ad teşarađ-ten. Tella tezdayt yer-sen an ddat-sen¹⁴, yir yemmat-sen tucca-sen ayi, d tazdayt tucca-sen iyniwen (Tunuyt 1).



Tunuyt 1 : (a) Ajellid Ameqqran yettsutuc iyniwen i yisemyan-nnes¹⁵ (b) Tamettuť tessuđuđ mmi-s adday n tezdayt tiřew sen n iziwayen (tamđelt n Karatépé.) (Awines wis VII ngum n sidna eisa)

1.2. Fel Ifereunen

Imiřriyen xi zzman ucin-as ccan i yiniwen d tezdayt mani tužuđt-nnes¹⁶ tetťef xi ujenna saddu n Lqahira, al wadday deg Nnuba d tnežruft n Siwa msedřarent netnin d yiřzer n Nnil.

¹³ Misopotamie.

¹⁴ Ddat-sen : yemmat-sen.

¹⁵ Ismey / isemyan = akli / aklan, *esclave(s)*.

¹⁶ Žži / tužuđt / ; Kab. : zzu / uzu ; *cultiver / culture*.

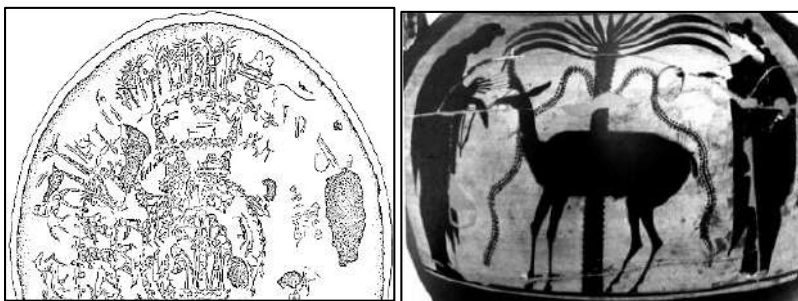
Tella tettqabal sen-n-tulut (2/3) n isekla n tuẓuḍt, mani ila gent d cceğret n lğennet Ha-ay d axwin¹⁷ i tligid qae tugniwin n hyuḍa-nsen eemren xi tezdayen. (Tunuyt 2).



Tunuyt 2 : (a) Taşwired n hyeḍ deg i temḍelt n Sennedjemgi tamurt n Théba (awines wis XIII^e-XII^e ngum n sidna eisa) tesskun-d amjar n yirden d wenkaḍ¹⁸ n yiniwen. (b) tazdayt deg temḍelt n Memphite.

1.3. Fel Infniqiyen

Netnin xi imezwar i yeẓyen tizdayen d sreḥlen-ten ɣel tmura-iḍen Ha-ay d xi mani i tusu smeyyet-nnes deg lbutanik *Phoenix dactylifera* (Tunuyt 3).



Tunuyt 3 : n Ifniqiyin teskunu-d tazdayt mlih.

¹⁷ Kab. : ɣef waya ; c'est pourquoi.

¹⁸ Nkeḍ / ankaḍ : récolter les dattes / récolte.

2. Amyezwer n *taxonomie*

Amezwar i yessersen isem n tezdayt deg lbutanik Hay d Carl Von Linné aseggas n 1734, tutef deg igget n twacult¹⁹ n isekla i iɛlmen newwar, d leɛlufet-nnes deg-s iggen n cceqfet. Twacult n tezdayen deg-s ujar n 2500 n tlemsin tegrew deg 215 n tewsitin²⁰. (Tafelwit 1).

Tazdayt ttinin-as deg lbutanik *Phoenix dactylifera*

– *Phoenix* : tusa-d xi ujdiɖ n Phoenix yettkemmed²¹
deffer-s yetteffey-d xi iɣed an tezdayt ad terɣu
deffer-s ad tedder, uc-as ɣir aman.

– *Dactylifera* : tusa-d xi dactylus. Xi tlatinit ttinin-tt
i yiɖuɖan, an iyniwen deg uziwa.

Tawacult n Arecaceae tgerɣew aɭas n tezdayen-iden yella ɣer-sen aɭas n faydet i bnadem, an cocotier (*Cocos nucifera*L) i xef-s ssufuɣen tlussi, udi, ccrab d uɣi n coco.

Tizdayen n yiniwen ɣer-sent aɭas n lemnafes-iden ladɣa²²i tmura n Tefriqt ugafa an iyniwen, tiffawin²³, tuttut n welli d ileɣman (uqqicen), aska²⁴ n tidrin²⁵, azɖa n ijertilen, tiseksa, tɔbuqa...atg²⁶.

¹⁹ Tawacult : *famille*.

²⁰ *Genres*.

²¹ *Kab.* : yettruɣu.

²² *Kab.* : ladɣa - *surtout*.

²³ Tuffa / tiffawin : *palmes*.

²⁴ Aska : lebni : *construction*.

²⁵ Tiddart / tidrin ; *Kab.* : axxam / ixxamen ; *maison(s)*.

²⁶ atg : ar taggara.

Tafelwit 1 : Amyezwer botanique n tazdayt.

Asettef klasik	
Tagelda	Plantae
Tafesna	Arecales
Twacult	Arecaceae
Twasit	<i>Phoenix</i>
Talmest	<i>Phoenix dactylifera</i> L., 1734
Asettef phylogénétique	
Tafesna	Arecales
Twacult	Arecaceae

3. Aẓwer²⁷ d usrebbu n tezdayen deg tmurt

3.1. Azurez n tezdayen

Tekker-d tezdayt tamezwart n xla g timura i yehman d imukan n uyurar deg tnezruft tameqqrant Taerabt ladiya (surtout) deg Tmurt-n-sen-Iyezran (Leiraq), mani i tella tissi i tezdayt n gudu²⁸ beṣṣeḥ llan iyniwen-nnes d ikeɛbac xi ta n tuẓuḍt.

Tazdayt n gudu tusa-d xi aṭṭas n uxelled²⁹ maki ad sseknun³⁰ taɣara³¹ n yiniwen d ad ssegmun-ten³².

3.2. Asrebbu n tezdayt n yiniwen deg Dzayer

Tuẓuḍt n tezdayt tettef qa3 timura i d-yusen adday n Waṭlas Aṣeḥrawi, xi rjem³³ n Merruk, (uṭṭu n tfuyt³⁴) al rjem

²⁷ Aẓwer ; Kab. : aẓar : origine.

²⁸ Gudu ; Kab. : tura ; imir-a ; imar-u.

²⁹ Axelled ; croisement.

³⁰ Sseknun ; md : sseknun tira-nnek ; Kab. : seggem.

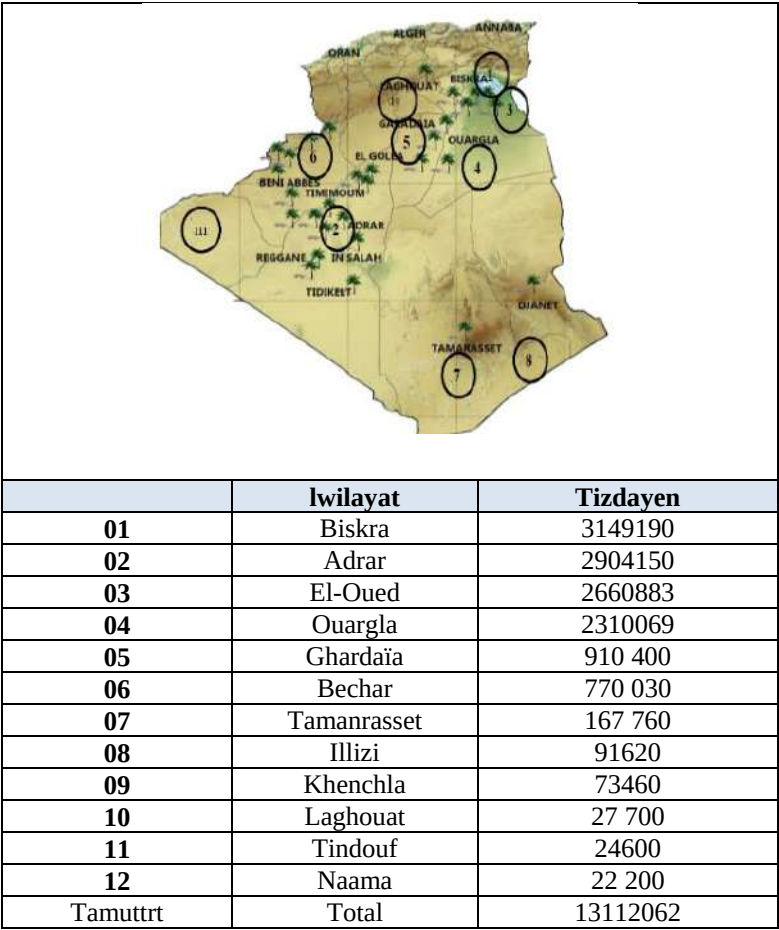
³¹ Taɣara : qualité.

³² Gmu / ssegmun.

³³ Rjem ; Kab. : tilist / tilisa.

³⁴ Uṭṭu n tfiyt : yuḍa, tuḍa / uṭṭu ; Kab. : aṭelluy n yiṭij, n tafukt.

n Tunes d Libya (anqar n tfuyt³⁵). D al tetṭaf xi wadday n Waṭlas aṣḥrawi al Reggan d Tamenṣast, tqama lḡiḥt n usammer Ha-ay d mani i yellan aṭṭas n tezdayin ladya surtout Wergla, Tibeskert d Suf (BOUGUEDOURA, 1991). (Tunuyt 4).



Tunuyt 4 : Imukan n wukkur n tezdayt deg Dzayer (Taaṣliyet).

³⁵ Anqar n tfuyt : Kab. : anqar n tfukt.

4. Tiwtilin n twennaḍt n wukkur n tezdayin

4.1 Lxedmat n wanezwu (*effet climatique*)

4.1.1 Tazyelt (Tunuḡt 5)

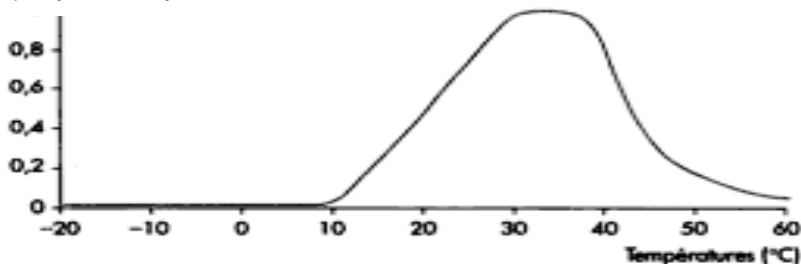
Tazdayt n yiniwen d ddeg³⁶ n isekla i yettṣebren i lḥemman (thermophile) tazmert tameqqrant n weymay teqqas tazyelt n 30°C. Tkatert n ulac(0) n tenfest n tizizewt nettat d 10°C.

Ajric i yettmutti jar -15 d -6°C yettæddem tazdayt laḡya wul-nnes mani i tettiqqar tmegna n teffawin d tezdayt tkenna d wefxas n teqlwanin yettas-d d ameggaru.

Tazdayt tttṣebbar i wejric d lḥemman ɛla xater wul-nes yuden³⁷ xi teffawin n walaw d uzequr-nnes yuden x usan yer-s lqimet d tameqrant deg usemḍi³⁸ n tezdayt.

Lḥemman n 18°C nettat d tkatert ulac (0) n weymay n teqlwanin, tinni³⁹ n yiniwen tissi yer(x) 120 al 200 n wassen (yer)x leḡsab n wemkan d lḥumman d ddeg n tezdayt. (BOUGUEDOURA, 1991).

Aḡmay n tezdayt



Tunuḡt 5 : Tkatert n weymay n tezdayt (PEYRON, 2000).

³⁶ Ddeg : ṣṣenf : *variété*.

³⁷ Aden ; yuden / tmadunt : *Kab.* : del, idel / tadimt, taduli.

³⁸ Asemḍi / usemḍi : *Kab.* : asismeḍ / usismeḍ.

³⁹ Yenwu / tinni : *Kab.* : tiwwin / yewwa.

4.1.2. Tifawt

Tazdayt n yiniwen tsebber aṭṭas i tfawt dax-win⁴⁰i s-tt-qqaren xi trumit "héliophile". Deg i tuẓudt n tezdayen nettaggej⁴¹ jar tezdayt d wetmas 10 lmitrat makin ad tifaw mlih lhalt jar-atsent d ammi i ttawaḍen yiniwen bekri. (BOUNAGA, 1985).

4.1.3. Tuft

Tazdayt n yiniwen tettaki⁴² aṭṭas nettat d tuft⁴³ n ujenna ladya surtout alextin⁴⁴ i llant ttderdugent tqelwanin ; d lweqt n tinni n yiniwen. Allay ameqqran n tuft yetteshegga⁴⁵ tudda (at hewwa) la transpiration n yiniwen d mi ay tteḍdalen tinni-nsen menbeḍ ttuṭṭan⁴⁶ d urcimen⁴⁷. Iyniwen uṣbiḥen ttwankaḍen deg imukan mani i tettili tuft n ujenna takeḍbuct (udun⁴⁸ n 40%) (BOUNAGA, 1985).

4.2. Acal

Tazdayt teqqas tamurt ad yili ucal-nnes yexfif an ijdi beṣṣeḥ ad tezmer ad tekker deg qae dḍguga n ucal, aluḍ, tafza, tarrist⁴⁹ la d lkerked⁵⁰. Tazdayt tsebber i tiffudreyt⁵¹ n

⁴⁰ Dax-win, daḡ-win ; Kab. : daḡenni. *C'est pourquoi*. Tamawt : Di Igli asekkil « ḡ » ur d-yegri ara aṭas di taggara n wawal ; yettuḡal d « ε » md : ččie, swie ; neḡ tettḡal d asekkil n « x ». md : idxes : adyes, dax : daḡ ; yexs / iteqqas ; yeḡs / yeqqas.

⁴¹ Ggej / ttaggej : Kab. : eḡḡ / ttaḡḡa.

⁴² Aki / ttaki : Kab. : ḥulfu / ttḥulfu, *sentir*.

⁴³ Tuft : *humidité*.

⁴⁴ Alextin : *au moment ; quand*.

⁴⁵ Hwa, yehwa / yettheḡga / yettesheḡga : ader, yuder / yettader / yettsader.

⁴⁶ Uḍa, yuḍa / ttuṭṭa [ttuḍḍa] : Kab. : ṡli, yeṡli / tteṡlay.

⁴⁷ Urcimen : iyniwen bezgen xi wallay n tuft : Kab. : urkimen / d agurbez n tmter ibezgen s nnda i d-yetteṡlayen.

⁴⁸ Udun : *moins*, Kab. : ddaw, qel.

⁴⁹ Tarrist : d acal yella deg-s bezzaf n *le calcaire* / Kab. : anners.

⁵⁰ lkerked : Kab. : akerkar.

⁵¹ Tiffudreyt, tiḍrayt. Tiffudreyt ajenna n tiḍrayt. Kab. : temrey.

tisent tawed al 15%, ajenna n 30% tettiqqar⁵², amekraz ixes-as ad iyez anegraw (le système) n drainage matta i la yettsaw⁵³ x waman i yedran.
(PEYRON, 2000).

Tafelwit 2 : Tesskunu tiwtilin n wakkur n tezdayt.

Tiwtilin n u akur n tazdayt	
Tkatert n ukkur (ilem n tfesniwt n tizzizewt)	10 °C
Azmar i tafiwt d weymay uşbiḥ.	Jar 0°C d 50°C maca tuşbiḥt n wakkur tettili jar 32 - 38°C,
Azmar i wejric	Qae tuffa : - 9°C Tamegna n tuffa : - 6°C
Asteḥmel n uyurar d tiqqart	Aṭas n iseggasen beşṣaḥ tarwa (fructification) tettili dṛus.
Aman	Jar 15 000 d 20 000 m ³ /ha x leḥsab n tidrayt n waman d ddeg n ucal
Amẓar (pluie) uxcin ⁵⁴ i yetteeddmén tazdayt	Deg lweqt n umersid ⁵⁵ d umeggaru n tinni n yiniwen
Asteḥmel n tisent : ▪ Tazdayt takeebuct ▪ Tazdayt tameqqrant	▪ jar 3 d 6 g/l ▪ jar 9 d 10 g/l
Acal	Tazdayt tekker-d deg qae ddeguga n ucal beşṣaḥ teqqas acal i yexfifen.

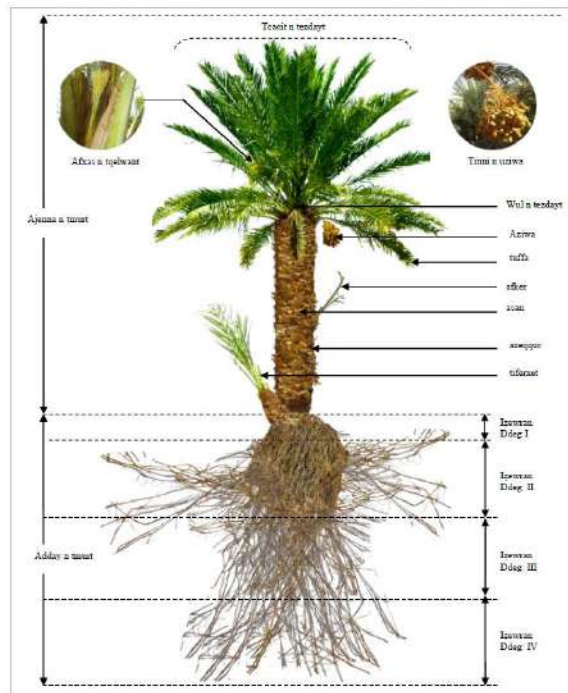
⁵² Teqqur / tettiqqar : teqqur / tettqar.

⁵³ Sew / tessaw / tissi Kab. : sew / tess / tissit.

⁵⁴ Amẓar uxcin : pluie en dehors de la période des précipitations normales : pluies nuisibles.

⁵⁵ Amersid : polinisation. Kab. : ddekkar.

5. Tasnelya⁵⁶ n Tezdayt (Tunuyt 6)



Tunuyt 6 : Tasnelya n Tezdayt (Taaşliyet).

5.1. İzewran (Tunuyt 6)

Azeqqur n tezdayt x u wadday yettili yuf an bşel, nnefs-nnes ajenna n tmurt d wenniɗen adday, x-s⁵⁷ ay-d tteffyen izewran zzrebɛen bezzaf, mfagament⁵⁸ netnint d qaɛ n ccjur tinident. Deg izewran n tezdayt nettlegga 4 n ddguga :

⁵⁶ Tasnelya : *morphologie* (Ref : *amawal berber disctionnary*).

⁵⁷ X-s = xef-s / yef-s.

⁵⁸ Mfagam / mfagament : mxalafent.

Ddeg I⁵⁹

Izewran n tneffut : ttilin deg tinset⁶⁰ n tezdayt dehren-d yel ujenna, ttilin d izdaden⁶¹ d uqsifen tteffyen-d x lbeşlet n uzeqqur, d ul ttujren-ca (ne dépassent pas) 0.20 à 0.25 lmitrat adday n ucal.

Deg II

Izewran n tuttut ttilin meţgen⁶² aţţas ladya deg d tnezdayin⁶³ i deg-s tettili tuţuđt n tezdayin waħd-s netnin ha-ay-d sam⁶⁴ ameqqran deg izewran, teggen ujar n 55% deg izewran n tezdayt qae.

Ddeg III

Izewran n tissi d uţemmem⁶⁵ n waman tizzegreţ-nsen tenċeđ nettat d weswir n waman adday n tmurt.

Ddeg VI

Ddeg-u n izewran yetteli d akeεbuc ladya al axtin (ha-ay-d melmi= alarmi/allami) i tella tissi x ujenna.
(BOUGUEDOURA, 1991).

5.2. Azeqqur (Tunuýt 6)

Izeyran⁶⁶ n tezdayt ttfaten 15 lmitrat n tzeqrert, tazdayt i itessin mliħ i tffat rayon n uzeqqur-nnes iggen n (0l mitr). X ujenna n uzeqqur tteffyen-d tiffawin, x wadday teffyen-d

⁵⁹ Ddeg : şşenf. *Variété, qualité.*

⁶⁰ Tinset / tinsa. *Pieds.*

⁶¹ Azdad / izdaden : izewran izdaden: *racines fines.* Kab. : arqaq / irqaqen.

⁶² Mţeg / meţgen. Kab. : jbed / nnejbaden.

⁶³ Tanezdayt / tinezdayin : *palmeraie(s).*

⁶⁴ Sam = řreşyun : *la part.* Amedya : sam-inu n tezdayin, sam-inu n weysum.

⁶⁵ Źemmem / aţemmem : *sucer.*

⁶⁶ Azyer / izeyran : *tronc.*

izewran. Asan yusa-d an ɛračča telwi x uzeqqur tesmuđu tazdayt deg unebdu makin ul tiqqar-ca.

Tiffawin i yemmun⁶⁷, deffer n weqrađ-nsent tqama tameгна-nsent deg uzeqqur ttinin-as *tigernaf*. Deg wammas n uzeqqur tikelt tikelt nettlegga-d deg-s ifekrawn (Tunuýt 6). Tiferxatin teffýent-d ajenna n beřlet n uzeqqur x-sen yettewaden tužudt n tezdayen. (BOUGUEDOURA, 1991).

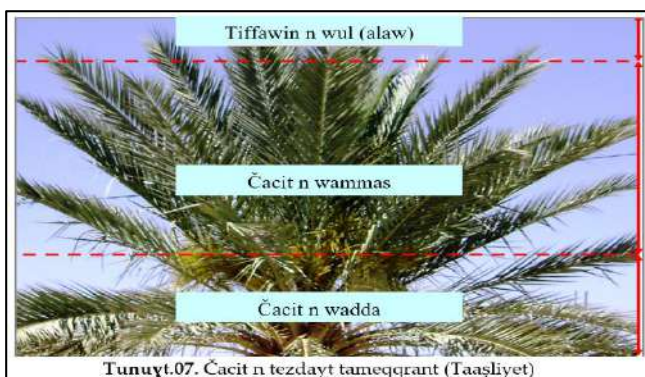
5.3. Ajemmar

Yettas-d deg wul n tezdayt, yettili d amellal netta i yettagjen tazdayt ad tezzegređ d ad tessewseε x-s a-ay-d teffýent tiffawin, ifekrawn d tferxatin.

5.4. Čacit n tezdayt (tunuýt 7)

Qae tiffawin tizeyzawin n tezdayt tgerwent an čacit, gi tezdayt tameqqrant n wujar n 10 n iseggasen tig x 50 al 200 tuffa.

Tiffawin teddrent x 03 al 07 n iseggasen x leřsab n ddeg d tissi n tezdayt. Ad nezmer ad nmeyyez :



⁶⁷ Mmu / yemmu / yemmun = mmet / yemmut / yemmuten.

-Čacit n wadday : tettili d tameqqrant gi tezdayin tiwessarın ;

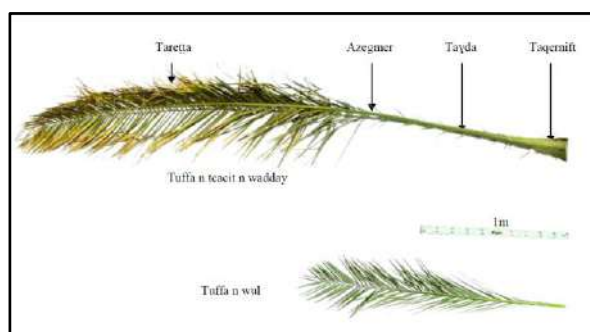
-Čcacit n wammas tettili d temeçqqrant gi tezdayin tikeεbac ;

-Tiffawin n wul ttılın d tikeεbac d tilemmađın areqqem-nsent yettılı d azezwaw yeçseđ. (MUNIER, 1973)

5.5. Tiffawin (Tunuýt 8)

Tiffawin ha-ay-d tıfıay⁶⁸ n tezdayt, kul tuffa yettılı deg-s aţas n trettıwın neđdent⁶⁹ gi teýda igget teqqel⁷⁰ gi ten-nıđen, maylent tmeddant x izegmar al tawđed ýel tqernıft. Deg wadday n tuffa tıllınt qeşent men beed tdewwlent-d d tırettıwın. Tiffawın teffıent-d x wul n tezdayt kul aseggas ttekkrent-d x 10 al 30 n tıffawın tıjıdın.

Tazdayt tameqqrant awel⁷¹ nettehlek-ca⁷² tig jar 100 al 125 n tıffawın tızezwawın.
(MUNIER, 1973)



Tunuýt 8 : ayawas n tuffa n tezdayt (Taaşliyet).

⁶⁸ Tıfırıt / tıfıray. md : tıfırıt n tıra. Tıfırıt n ccejret. Tıfıray = latay) .

⁶⁹ Nđed / yenđed / menđed. Kab. : nđed / yenđed / imenđed.

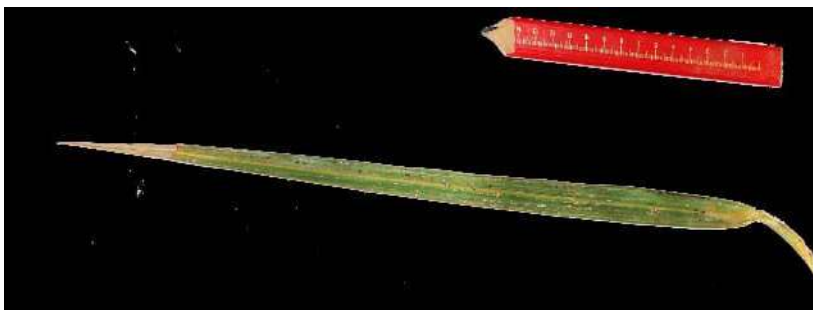
⁷⁰ Teqqel / tamuýlı. kb : tımuqel / tamuýlı.

⁷¹ Awel / ul. Kab. : awer / ur.

⁷² Awel nettehlek-ca. ul telli tehlek : indemne, saine. Kab. : ur tennehlak.

5.6. Tareṭṭa (Tawlaft 1)

Tuffa ha-ay d tífrit n tezdayt. Tuffa igget yella deg-s ujar n 200 n treṭṭwin, ttillint qeṣhent yel wadday d al tikeɛbac d al timeqqranin yel ujenna. Tareṭṭa tizzegreṭt-nnes tettfat 30 sm.



Tawlaft 1 : teskunu-d tizzegreṭ n treṭṭa.

5.7. Azegmer (Tawlaft 2)

Yettig jar 15cm d 20 cm, ttillin yel wadday n tuffa, maki ad testeḥmel tazdayt ṣṣehd d tiqqart tireṭṭwin ḍḥant-d d izegmar maki ul ḍeyɛent-ca aman. Izegmar ttsellken tazdayt x ileyman, d tẓukki⁷³.



Tawlaft 2 : Teskunu-d tizzegreṭ n uezegmer (Taaṣliyet).

⁷³ Tẓukki : d ijḍaḍ ikeɛbac i yettḍerran tazdayt (tiḥbibet, tsallaft, sibbu...). Ijḍaḍ : d imeqqranen an : atbir, tjarfet.

5.8. Uzbir (Tawlaft 3)

Ha-ay d tayda n tuffa tameqqurt, al axtin i x-s kksen tirettwin i tqama deg-s yir izegmar. Deffer n wenkađ⁷⁴ i ten-gerwen tisednan⁷⁵ d al sserkayen⁷⁶ xef-sen timsi n lkucet maki ad ssenwent⁷⁷.



Tawlaft 3 : Teskunu-d tizzegreṭ n treṭṭa (Taaşliyet).

5.9. Taqernift (Tawlaft 4)

Taqernift ha-ay d mi ay i d-yettqaman xi tuffa taqdimt i yeqquren deffer n uşeggeṭ⁷⁸, amekraz deffer n wenkađ yettesneqqa tizdayin-nnes mlih, itebbi iyemmay⁷⁹, d al yettzebbber tiffawin i yemmun.



Tawlaft 4 : Teskunu-d Taqernift tameqqurt n tzdayt (Taaşliyet).

⁷⁴ Nkeđ / yenkeđ / ankađ : *Kab.* : lqeđ / yelqeđ / alqađ.

⁷⁵ Tisednan : timeṭṭuṭin.

⁷⁶ Sserku timsi / aserki n temsi. *Kab.* : ssker / askar n tmess.

⁷⁷ Enw / ssenw. *Kab.* : eww / seww.

⁷⁸ Şeggeṭ / aşeggeṭ : aḷbar n tiffawin i yeqquren.

⁷⁹ Iyemmay : d iyniwen uḷan jar tiffawin d tqernaf wass n wenkađ.

5.10. Asan⁸⁰ (Tawlaft 5)

Igu an ƣračča, yelwi x uzeqqur. Yettqetƣa ƣawsa d tameqqrant deg usemđi n tezdayt. Yettarra-tt-d tšebber i fad. Tazdayt tawessart tnetƣter san-nnes deffer n wuƣtu n tqernaf mani i yettqama uzeqqur d aqeddid (d ašeryan). San deg i tmura n tezdayin ttqetƣan xef-s⁸¹aƣƣas n ƣawsiwin an iseƣwan, ccennet d tiħlasin... atg.



Tawlaft 5 : Teskunu-d asan n tzdayt (Taašliyet).

5.11. Tiferxatin (Tunuƣt 6)

Tizdayin tikeebac teggent x 60 cm al 200cm tekkrent-d ƣel wadday n uzeqqur, deg uyur n Magu-Ibrir. Itekkes-ten amekraz maki ad sen-ieawed tuƣuđt. Xef-sent i ttsektaren tizdayin asexuellement. Tiferxet tettili d tisya-s⁸² i dda-s⁸³ génétiquement d phynotypiquement. Ha-ay d axwin amekraz ul yetteƣzi-ca x leeluf, əla xaƣer leelufet tettili txelled. Lukan ad tekker ad tucc awtem niƣ⁸⁴ taxelđit.

⁸⁰ Asan : yettwanna ƣel At Igli : « san ».

⁸¹ Xef-s : yettwanna ƣel At Igli : x-s.

⁸² Issi / yettissi / issay : *Kab.* : ddem / iteddem / tuddma, cabeh / yettcabah / acabeh / tizzya-s.

⁸³ I dda-s : i yemma-s.

⁸⁴ Niƣ : yettwanna ƣel At Igli « niə ».

5.12. Ifekrawen⁸⁵ (Tunuyt 6)

Teqqaren-asen xi tɹumit *les gourements*, tekkren-d deg wammas n uzeqqur itekkes-ten amekraz makki ul ttɛeddmɛn-ca tazdayt. Teffyɛn-d ifekrawen x wadday n tqɛrniɛt.

5.13. Aziwa (Tawlaft 6)

Yettwezzen x 15 al 40 kg, ttgerwen deg-s izriren⁸⁶ ɛemmren x iyniwen. Tettaɹew igget n tezdayt al 14 n iziwayen. Teffyɛn-d iziwayen x wul n tezdayt deg uyur n furar d al tqaman al uyur n ctember. Deffer n tinni, yettnɛkkeɛ-ten amekraz xi wɛmjer.



Tawlaft 6 : Teskunu-d aziwa n tɛdayt (Taaɣliyet).

⁸⁵ Afker / ifekrawen : (af + kker ? *mauvaise croissance*). D tiferxet tekker-d ɣel ujenna deg uzeqqur. Tɛeddem, tqerra tazdayt : *excroissance* ; *un tissu avec une division cellulaire anormale*.

⁸⁶ Azrir / izriren. *Kab.* : azrar / izurar.

5.14. Azrir (Tawlaft 7)

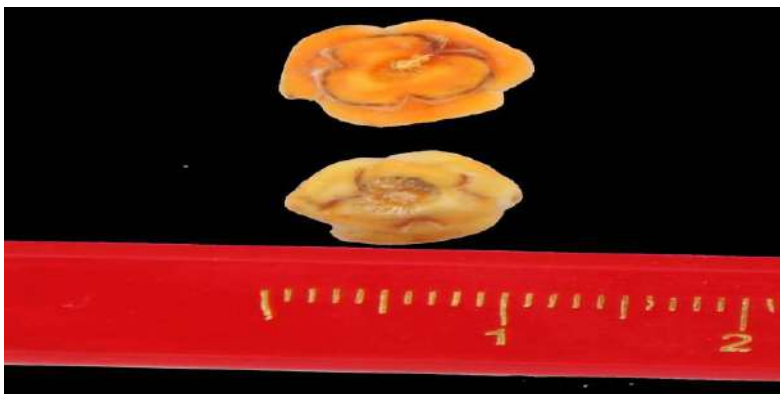
Deg yiggen n uziwa n tlegga-d bezzaf n yezriren, kul azirir yetttli deg-s jar ujar n 10 n yiniwen ṭṭfen deg-s xi ṭyuffet.



Tawlaft 7 : Azrir n uziwa (Taaşliyet).

5.15. Ṭyuffet (Tawlaft 8)

Netta-t i iteṭṭfen ayniw ɣel wezrir n uziwa. Yettli llunnes d ačini, dyametr-nnes ittig jar 08 mm d 11 mm. Gi tnewwart tetti ha-ay d lkursi yettuf-d ayniw d nettat tettqama an tmadunt.



Tawlaft 8 : Teskunu-d ṭyuffet n winiw (Taaşliyet).

5.16. Aqelwan (Tawlaft 9)

Llun-nnes yettili d ačini zdat ma yeqqur, ha-ay d tameгна n uziwa tteṭtef-t deg wul n tezdayt. Ad yezmer ad yissi⁸⁷ al 100 n yezriren, x lehsab n ddeg n tezdayt. Tiẓzegreṭ-nnes x 1m yel ujenna.



Tawlaft 9 : Teskunu-d aqelwn n uziwa (Taaşliyet).

5.17. Tazraṛayt (Tawlaft 10)

Ha-ay d aziwa i yeqquren, ul deg-s yelli-ca iyniwen. Zzman llant ferḍen x-s tiddart d tenwalt.



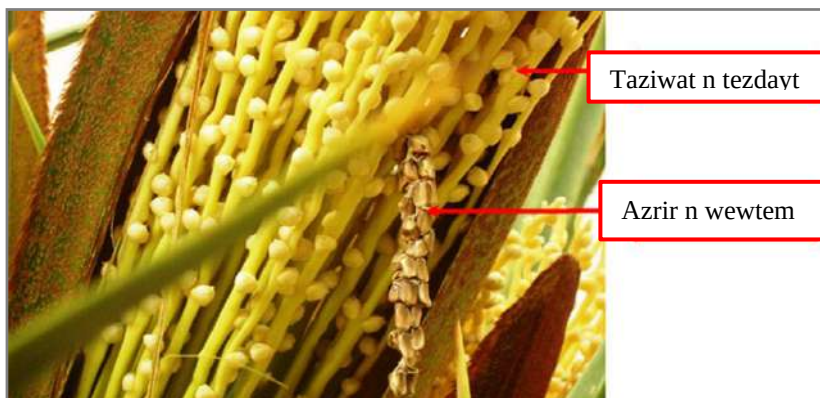
Tawlaft 10 : Tazraṛayt n uziwa (Taaşliyet).

⁸⁷ Yettissi. *Kab.* : iteddem.

6. Tiziwayin n tezdayt

Tazdayt d ddeg n ccjur i iteggen nnewwar. Mani i d-nettledda amersiɗ n wewtem d tziwat n tezdayt (tawtemt) kul iggen weɣd-s, mfagamen, taziwat tettili tuden x iggen n wadan teqqaren-as taqelwant. Tanewwart n tezdayt tettili d tamellalt, timlelt-nnes terrez. Tettig jar 3 d 4 mm n diamètre, d al tettili teɣlem 03 n tecnafin, d 03 n wadanen igget i yettyemyen maki ad tettucc lḥebbet n weyniw. Tettqama tnewwart n tezdayt x lkursi llun-nnes yettili d azeyzaw yerzem gi tecnafin. Tanewwart n tezdayt tettili temfagam nettat d tnewwart n wewtem mani tameggarut-u tettili teɣger ul tucc-ca tizdal. Amersiɗ yettiget aɗu niy ixxa (insectes). Gi tnezdayin, mani tettili leflaḥet d tameqqrant iggen n wewtem ad yaweɗ i 100 n tezdayin. (Tunuyt 9)

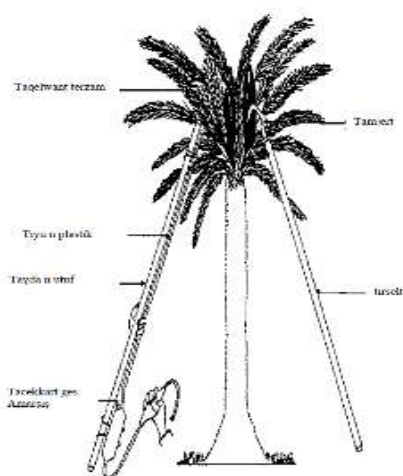




Tawlaft 11 : Amerşıd x ufus (Taaşliyet).

6.2. Amerşıd x lmacinet n uzuzer

Al axentin tefxes tqelwant n umerşıd, ad nisi aziwa d nezzuzer-t ajenna n buşıyyar makin ad negrew nnebyet⁹⁰ n umerşıd. Amerşıd i yettwagerwen ad t-nxelled netta d ukeb n farina (1V/ 2V), menbeed ad nezzuzer x lmacinet (Tunuyt 10).



Tanuyt 10 : Amerşıd x lmacinet n uzuzer (ZAID, 2002).

⁹⁰ Nnebyet. Kab. : ayebar. Poussière.

6.3. Asifeḍ⁹¹ n umerṣiḍ (Tawlaft 11)

Al axtin i yejjenjem umerṣiḍ, yetthewwa-d⁹² amekraz xi tezdayt yettẓalla x rrṣul (*xeḥ-s tẓallit d sslam*) menbeed yettessellem tazdayt i-s igu amersṣiḍ. Yessattaf azegmer deg tqernift, maki tella tesskunu-d tugna (ṭn ...).



Tawlaft 12 : Teskunu-d ajenjem n u amersṣiḍ (Taaṣliyet).

6.4. Tazdayt deg useggas

Tazdayt teẓẓar⁹³ deg useggas aṭṭas n ineffilen ⁹⁴: tiẓegreṭ deg Seɛdssɛud⁹⁵, tettig amersṣiḍ gi tefsut, tnawwen yiniwen-nnes deg unebdu d al ttnekḍen-ten deg uyur n tubé.

⁹¹ Asifeḍ = ajenjem = asmeddu. Taggara.

⁹² Yetthewwa. Tettwanna [yetthegga].

⁹³ Ẓer / teẓẓar / yeẓru / teẓru. Dder / tettidir / yedder / tedder / tudra.

⁹⁴ Nfel / ssenfel / ineffilen (*changements*).

⁹⁵ Seɛdssɛud : ameggarun n tejrect. Anzi : « Seɛdssɛud, aman tguren deg uqeccuḍ ».

Tafelwit 3 : Tazdayt deg useggas.

	Iyyaren											
	J	F	M	A	M	J	J	A	S	O	N	D
Asgenfu n tezdayt	X										X	X
Aymay n tziwat adday n tuffa		X										
Ufuy n tgelwanin		X										
Afxas n teqlwanin		X	X									
Amersid			X									
Ḥebb-lellu ⁹⁶				X								
ilebbuzen izeyzawen						X						
llebbuzen iwrayen							X	X				
Iyniwen i yiwɗen (i yenwun)								X	X			

7. Dguga n tezdayin deg Dzayer

Deg Dzayer llan ujar n 934 n ddeguga n tezdayin. Mfagament gi tiffawin, deg izegmar, deg tiẓẓegreṭ d yiniwen. Ɣir deg Ttwat d Ssawra ad nezmer ad d-neḥseb al 364 n ddeg. Bla Iyzer n Ɣir (175 n ddeg), Suf (56 n ddeg) d ay d-yeqqimen llan deg Werġlen. Ssbab n wenfal n ddeguga n tezdayen ha-ay d tuẓuḍt xi tielufin mani i d-tettucc ddeg n tanefla⁹⁷ d ajdid, teqqaren-as *taxeldit*⁹⁸.

⁹⁶ Ḥebb-lellu : ḥebb + alelluc (ḥebb + nnewwar).

⁹⁷ Tanefla : *croisement*.

⁹⁸ Taxeldit : *hybride*.

Tafelwit 4 : Ddguga n tizdayen deg Dzayer.

U	Ddeg	Ljihet	U°	Ddeg	Ljihet
1	Tazeggayt/Hmira	Anzul utrim	22	Tadmamt	Anzul utrim
2	Tin gazza	Anzul utrim	23	Admam	Anzul utrim
3	Tumliha	Anzul utrim	24	Accaw	Anzul utrim
4	Bufeqqus	Anzul utrim	25	Bba Mexluf	Anzul utrim
5	Aherdan	Anzul utrim	26	Ccix Mhemmed	Anzul utrim
6	Tin Juher	Anzul utrim	27	Taxeldit	Anzul utrim
7	Tin Nnaşer	Anzul utrim	28	Aherdan n Mazzer	Anzul utrim
8	Taqerbuct	Anzul utrim	29	Yelli-s n Lhağ	Anzul utrim
9	Uqqiren/Lyers	Anzul utrim	30	Lmeğhul	Anzul utrim
10	Deglet-Nur	Anzul agmuđan	31	Tazeggayt	Anzul agmuđan
11	Tawrayt	Anzul agmuđan	32	Əbdlecziz	Anzul agmuđan
12	Tsibbi	Anzul agmuđan	33	Tarmunt	Anzul agmuđan
13	Zagray	Anzul agmuđan	34	Tinisin	Anzul agmuđan
14	Degla tamellalt	Anzul agmuđan	35	Tertaga	Anzul agmuđan
15	Degla-beyda	Anzul agmuđan	36	Messuhi	Anzul agmuđan
16	Takermust-Hewwa	Anzul agmuđan	37	Taneşlit	Anzul agmuđan
17	Takermust-Şşefra	Anzul agmuđan	38	Tanetbuct	Anzul agmuđan
18	Tawadant	Anzul agmuđan	39	Tawrayt	Anzul agmuđan
19	Taferzayt	Anzul agmuđan	40	Tafezwin	Anzul agmuđan
20	Tagneyya	Anzul agmuđan	41	Tacerwit	Anzul agmuđan
21	Tahemrayt	Anzul agmuđan	42	Tamuxxaxt	Anzul agmuđan

8. Dguga n wagam d tissi deg tenzdayen

Temfagamen dguga n tissi n tezdayin x leḥsab n lḥemman, ddeg n waman, ddeg n ucal :

- Tiferxatin tikeεbac wass x wass bla azgal ;
- Ala xtin i yella leεmer n tezdayin jar 1 d 4 n iseggasen ad tsew tezdayt iggen n tikelt deg 03 n iwassen.
- Ad tsew tezdayt iggen n tikelt deg ssimanet (umilas) al axtin yettfat 05 n iseggasen.

Tazdayt i yettarwen yexeε-as 100 L n waman ad s-tettwacc deg yiggen n tikelt d x leḥsab n waddad (ḥalet) n tezdayt mani ixεε-as i umekraz ad izayed niḡ ad yessenqes. Lewqat i s-ixaε i umekraz ad izayed aman :

- Zdat n umerεiḡ makin ad dredqent tqelwanin fisseε ;
- Deg lweqt n tinni n yiniwen ;
- Deffer n wenkaḡ srid⁹⁹ ;
- Lewqat i s-ixaε i umekraz I d-isenqεε aman :
- Al axtin llan waman n tmurt ttalin-d ḡel ujenna, an lebbardat¹⁰⁰ ;
- Zdat n wenkaḡ ;
- Tajrect mani i yettili lḥal yesmed ;
- Uffuy n nnewwar.

⁹⁹ Srid : *direct*.

¹⁰⁰ Lebbarda / lebbardat : aman ttalin qerrben, ttalin ḡel ujenna.

8.1. Ayrur¹⁰¹ (Tawlaft 13.a)

Yessitef-t-d yel Igli Sidi Belæetman deg useggas n 1600. Yettwaxdem x iqecdan n tammayt, itettef-t iggen tteqqaren-as *anagam*. Ayrur yettili yersu x sen n şşwaleb ttfen x lqaf yel lqaf¹⁰² n tanut. Ajenna-n sen ttesrusun jar 2 al 3 n tizedmin, an tververt (plateforme), yettbedded xef-sent unagam. Deg şşwaleb teqqnen 02 n tersal tteqsaħent-tent x 04 n isebdad tettef *lqadi* yettili yejhed. Ayrur deg tettef deg sen n tmaska, ha-ay d sen n tilluħin neğrent d neqbent x sen n ġihat d al ttwaţţfent x lemlex (ilem n welxem) deg i tmaska teqqnen 08 n iseydisen x iggen n wesyun-iden ittig al 8m teqqaren-as *azerzef*. Tellmen-t x usan n wewtem n tezdayt. Tiseydisin ttilin duħen x iggen-iden n uqeccuħ ttinin-as *amerwed*. Asyun n weyrur teqqaren-as *zilem*. Teddaren-t x tiggi¹⁰³, tameгна-n nes teqqen deg sen n cennet : igget yel ujenna d tanniħen yel wadday.

Cennet tacnaft-n nes ttinin-as *tarma*. Tettili tedru x zerzef d al yettili deg-s 04 n timezyin ttinin-asent tiyaţţen. Teqqnen deg-sent zilem.

Al axtin i yewwet unagam ayrur, yettqelleb aman deg i tefrawt x ujenna ttinin-as *iney*. Yetttili yerjel¹⁰⁴ d iwsee, d x wadday nettlegga-d *ajdel*. Yettili yerzem d imayel makin ad yeyħa aman n wagam deg i tijent.

8.2. Tsaylalt¹⁰⁵ (Tawlaft 13.c)

Tisya-s i weyrur, ttemfagam id-s yir akeb elaxater yella deg-s x iggen n lġiht taħunt makin ad tessexxif agam, d yella

¹⁰¹ Ayrur : *Fr. polie*.

¹⁰² Kab. : seg yiri yer yiri.

¹⁰³ Tiggi : lħelfa n yijdi.

¹⁰⁴ Yerjel. Kab. : yergel.

¹⁰⁵ Tsaylalt : lħjet tugel.

deg wemkan n cennet tagnint teddaren-tt xi iqelwanen d treṭṭwin. Tsaylalt u deg-ca (maci) an weyrur, tsekkān-tt x uluḍ deg wemkan n iqecḍan n temmayt.

8.3. Lyeryaz (Tawlaft 13.b)

Maki ad yekkes ufellah xi man-nnes ccqa, yessenfel anagam x weyyul, aserdun niḡ alḡem. Ayrur i deg-s tettili zzaylet teqqaren-as *lyeryaz*.



Tawlaft 13 : Teskunun-d (a) ayrur (b) lyeryaz d (c) tsaylalt.

8.4. Tafeggart

X leḡsab n Hirodot teḍher-d tfeggart leḡwayeh n Furs deg uwines wes 05 zdat n tarwa n sidna Σisa. Deg anegraw-u tsuffuyen-d aman n wadday n tmurt maki ad swen tizdayin, tafeggart ha-ay d ssenslet n tunin.

Jar-atsent targa n 1m, igget tḡeṭṭa gi ten-iḍen. Aḡza n tfeggart yettbedda x weḡza n tanut tameqqrant. Teggen-as tamḡart thewwan-d xef-s aman. teqqazen-as sneslet n tunin-iḍen n werwaḡ d wefraḍ. Jar-atsent x 50m al 500m, deg umeggaru n temḡart nettlegga-d tijent mani i tgerwen aman. Menbaed teggen taqesrit (Tawlaft 14.b) kul iggen tettucc-as sam-nnes x leḡseb n lweqt.

- Tayirt (Tawlaft 14.a) ha-ay d ssaet n ḥsab n lweqt, yettili d sṣṭel n wezdaḥ inqeb al axtin i iḥammer yettqama 15 mn makin ad yexwa.
- Argaz : ha-ay d 09 n tṭirat, d yettig sen n swayee d rrbee (2:15mn)
- Nnubet : ha-ay d 45 n tṭirat niṭ ḥdaec n sswayee d rrbee (11:15mn) yella sen n nnubet igget deg yiḍ d ten-iḍen deg wass.

Zzman deg At Ṣebbas deg tiṭ n Sidi Ṣetman llant 04 n tqebbal mani kul taqbilt yettseḥḥa-as-d 10 n nnubat.



Tawlaft 14 : Teskunu-d (a) tiyiret d (b) taqasrit n tfuggart.

8.5. Abadu

Anagam yettbedda tiyta n weyrur zdat n wenqar n tfiyt. Al axtin i yettæemmer tijent x waman, itekkes-as adyaq kul ffyen-d x u wenfif d al tegguren deg lqendrat al d awden yel igemmunen, şaru¹⁰⁶ ha-ay d x d win i t-rejlen d trezmen igemmunen. Amekraz itezzi deg igemmunen-nnes izeyza (lexdar) n useggas.



Tawlaft 15 : Tescunu-d (a) lbasan (b) tanut (c) abadu.

¹⁰⁶ Aşaru : tnast. Kab. : asaru / tasarut.

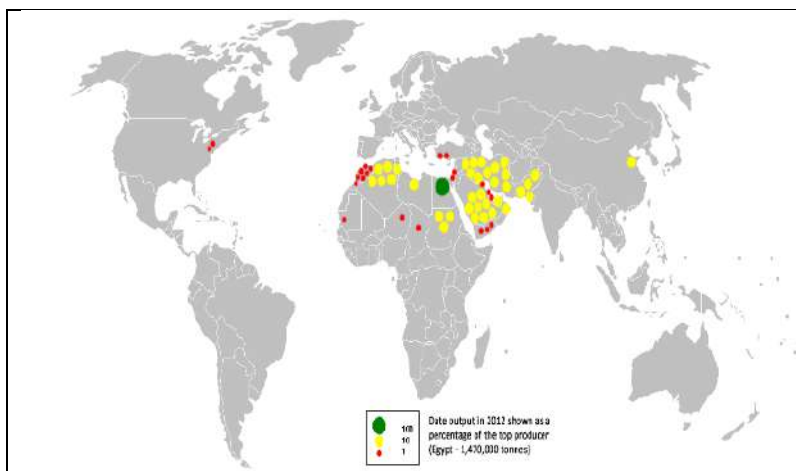
- II -

Iyniwen n tezdayt

Ayniw

Tuẓuḍt¹⁰⁷ n tezdayin d usefrek¹⁰⁸ n yiniwen yuli aṭṭas deg umaḍal deg iseggasen-u imeggura, mani i yuli ajenna n 8 684 512 n ṭun. Timura timezwar deg usefrek n yiniwen ha-ay d Mišr, Iran d Dzayer. (Tunuḡt 1).

Marikan deg 1936 tisi tizdayin xi Dzayer mani gudu tedwel-d xi tmura i yettezznuzun iyniwen netnin d Ccnawa.



1. Mišr	1 562 171	17,99 %
2. Saʕudiya Taʕrabt	1 302 859	15,00 %
3. Iran	1 204 158	13,87 %
4. Dzayer	1 094 700	12,61 %

¹⁰⁷ Zẓu / yeẓzu / tuẓuḍt.

¹⁰⁸ Asefrek (tamaziɣt) : *lintaj*.

5. Lëiraq	614 584	7,08 %
6. Pakistan	471 670	5,43 %
7. Ssudan	440 871	5,08 %
8. Σumman	368 808	4,25 %
9. Imarat Tierabin	345 119	3,97 %
10. Tunes	241 333	2,78 %
11. Libya	176 229	2,03 %
12. Ccnawa	158 294	1,82 %
13. Merruk	111 701	1,28 %
14. Kuwit	96 656	1,11 %
15. Timura-iden	433 721	3,88 %
Total	8 684 512	100 %

*Tunuyt 1 : Takerda teskunu-d asefrek n yiniwen xi tun, aseggas n 2018.
FAOSTAT (FAO).*

1. Timura i yetteznuzun iyniwen (Tafelwit 1)

Dzayer tnekkeḍ ujar n 1 094 700 tun n yiniwen d al tettas-d west¹⁰⁹ 04 deg umyezwer n tmura n umaḍal. Beṣṣaḥ izzayriyen tettin aṭṭas n yiniwen ha-ay d axwin ul netteznuzu-ca aṭṭas (adday n 91 030 tun).

¹⁰⁹ West : tist.

Tafelwit 1 : Timura i yetteznuzun iyniwen deg useggas n 2018.

Timura i yetteznuzun iyniwen 2018 FAOSTAT (FAO) x lkilu		
Tunes	280 129 000	16,5 %
Saæudiya taærabt	201 393 000	12 %
Imarat tierabin	197 553 000	11,6 %
Israël (tamurt n filistin)	160 726 000	9,5 %
Iran	138 814 000	8 %
Pakistan	112 731 000	6,6 %
Læiraq	95 562 000	5,6 %
Dzayer	91 030 000	5,3 %
Marikan	66 141 000	3,9 %
Miřr	49 729 000	2,9 %
Fransa	42 133 000	2,5 %
Timura iḍen	262 862 000	15,5 %
Akit	1 698 803 000	100 %

2. Timura i yettessaḡen iyniwen (Tafelwit 2)

Timura n Asie tessaqent bezzaf n yiniwen d ḡersen ssuq tameqqrant n yiniwen ladḡa surtout Lhend d Malizya.

Tafelwit 2 : Timura i yettsayen iyniwen deg useggas n 2015.

Timura i yettsayen iyniwen 2015 FAOSTAT (FAO)		
Lhend	183 169 000	19 %
Merruk	78 519 000	8,3 %
Fransa	72 090 000	7,6 %
Malizya	47 208 000	5 %
Legliz/Negliz	38 532 000	4,1 %
Indunisya	37 495 000	4,0 %
Marikan	35 688 000	3,8 %
Lalman	31 872 000	3,4 %
Kanada	30 165 000	3,2 %
Rrus	29 527 000	3,1 %
Timura-iɛen	362 134 000	38 %
Akit	946 399 000	100 %

3. Awwaɗ¹¹⁰ n yiniwen

3.1. Afran (Tawlaft 1)

Ameggaru n unebdu tbeddan yiniwen ttsemɗayen-d¹¹¹ xi iɣuran ɣel iɣuran¹¹², amekraz yettisi tazeɓbalt-nnes¹¹³ zdat n wenqar n tafiyyt. Yettili yesmeɗ lhal, yettafa¹¹⁴ x uziwa ɣel uziwa x unqiden¹¹⁵ (ayniw ul temɗi-ca tinni-nnes).

¹¹⁰ Aweɗ / yewweɗ / awwaɗ. yettwawal ɣel At Igli [Aggaɗ].

¹¹¹ Mɗi / ssemɗi / ttsemɗi / ttsemɗanyen-d. Taq. : ssemɗayen-d. Md : aɣ-ak mɗi-yi askaf-u. Taq. : ax ereɗ-iyi askaf-agi.

Di ssuq At Igli ttinin : « tisseɗɗit a tarbaet ca temda ». Taq. : tisseɗɗit a tarbaet la tfukk (yeqqim-d kan uzenzi i win yebɣan ad yay).

¹¹² Xi iɣuran ɣel iɣuran. Taq. : si sɣbeɣ ɣer sɣbeɣ ≠ tameddit. iɣuran (ahat d tiɣriwin n uyaziɗ n tɣebhit ?). ɣru / yeyru / iɣuran. *Les appels*.

¹¹³ Tazeɓbalt. Taq. : taqecwalt.

¹¹⁴ Fafa / ifafa. Taq. : af / yufa. xɗeb / yexɗeb. Yettafa. Taq. : yettnadi.

¹¹⁵ Uqqiden : Taq. : sqeɗ / yesqeɗ / usqiɗ / usqiden.

Ttinin belli deffer n wenqar n tfiyt, afran yettsiqqar aziwa
 ɛla xater iyniwen ttlin ɣman.



Tawlaft 1 : Afran n yeniwen x uziwa (Taaɣliyet).

3.2. Ankaɖ¹¹⁶ (Tawlaft 2)

Ttawɖen-d yiniwen deg umwan, ayur n tuber, mani ul
 yettqimi la d iggen n ulebbuz deg uziwa. Deg wass n
 wenkaɖ, teggen midden twiza, ttisye¹¹⁷ tuttut (leɛwin) id-
 sen d al ttekkalen deg tnezdayin-nsen.

Zdat n weqraɖ n uziwa yettɣeggeɖ¹¹⁸ amekraz tazdayt.
 Itekkes tiffawin timeqqurin d ifekrawen makin¹¹⁹ ad
 yerɣem abrid jar izegmar¹²⁰. Menbeɛd ad kul yergeb x

¹¹⁶ Ankaɖ: aqraɖ / ukkus n uziwa.

¹¹⁷ Ttisye. Taq. : ttawin / teddmn.

¹¹⁸ ɣsegget / yettɣeggeɖ / aɣsegget : itekkes tiffawin d iqelwanen imeqquren.

¹¹⁹ Makin. Taq. : akken / iwakken.

¹²⁰ Izegmer / izegmar : d asennan d ameqqran n tezdayt. inzegmir (in zegmir) : d taddart
 deg Ttwat (Adrar) ibusennanen.

uzeqqur al ulaw n tezdayt x temjert a kul yemjer-d iziwayen n yiniwen d yessehwa-ten x wesyun al wadday. Ttwagrawen yiniwen deg teyrarin¹²¹ d ttwassilyen yel tiddart i wafas¹²². Afas iswi-nnes d asenqeş n hwa gi txabit maki ul ttxemmren-ca yiniwen.



Tawlaft 2 : Ankađ n uziwa xi tazdayt (Taaşliyet).

3.3. Aseggeţ (Tawlaft 3)

Deffer n wenkađ, amekraz yettbedda yettseggeţ tizdayin-nnes, itekkes-as ifekrawen, tiferxatin d tiffawin timeqqurin amu maki ad tneqqa tezdayt d tetthenna xi iţtanen d yixxa¹²³.

¹²¹ Tayrart / tiyrarin : tacekkart tameqqrant xi ixef (cc̣eer) n tyađt.

¹²² Afas/ ufas. Taq. : aefas / weefas.

¹²³ Axxu / ixxa : *insecte(s)*.



Tawlaft 3 : Tugna tesskunu-d aseggeṭat n tezdayt (Taaşliyet).

3.4. Agraw n iḡemma¹²⁴ (Tawlaft 4)

Iḡemma ha-ay d iyniwen d uqqicen i yettuṭṭan¹²⁵ jar tqernaf. Deffer n uşeggeṭ, amekraz yettgerrew-ten makin ul ttsehliken-c tazdayt. Σla xaṭer ttgerwen-d ẏir ixxa¹²⁶ d ijuḍaḍ. Zzeq n ijuḍaḍ yettawi-d tibaktiriyin d igersalen i iteggen aṭan i tezdayt.



Tawlaft 4 : Iḡemma jar tiqernaf (Taaşliyet).

¹²⁴ Qqim / yeqqim / Aḡemma / iḡemma : *les restes*.

¹²⁵ Yuḍa / yettuṭṭanen. Taq. : yeẏli / yetteẏlayen.

¹²⁶ Ixxa. Taq. : ibeeεεac. Ijuḍaḍ. Taq.: igḍaḍ imeqqranen.

4. Tasnelya n yiyniwen

Iyniwen n teydayt ttilin an uḍaḍ, tteggan jar 3 cm al 10 cm. Ttassen-d deg wezrir. Iggen n wezrir yettisi x 15 al 40 n yiniwen. Ttilin tṭfen xi tyuffet, nettlegga-d deg wammas-nnes iggen n yiyes¹²⁷ yer-s iggen n cceqfet. Iyes yettili yegrew xef-s 3 n yelmawen :

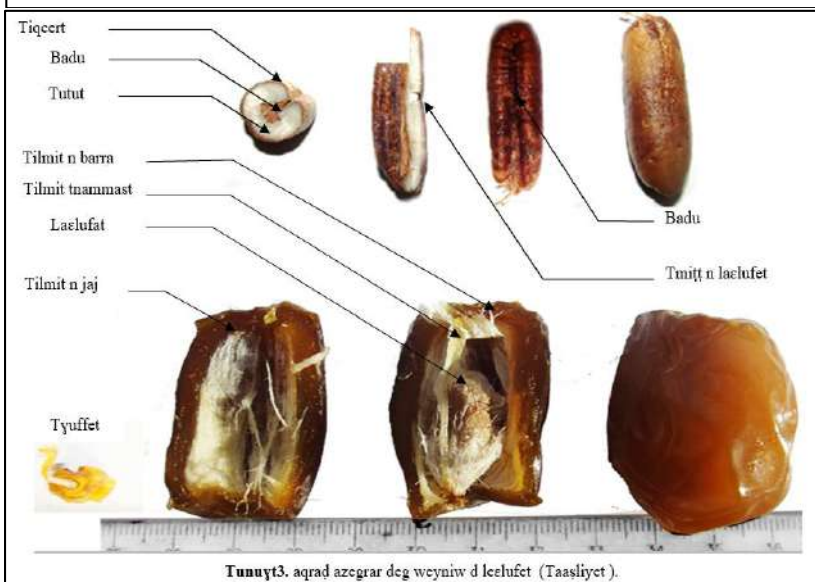
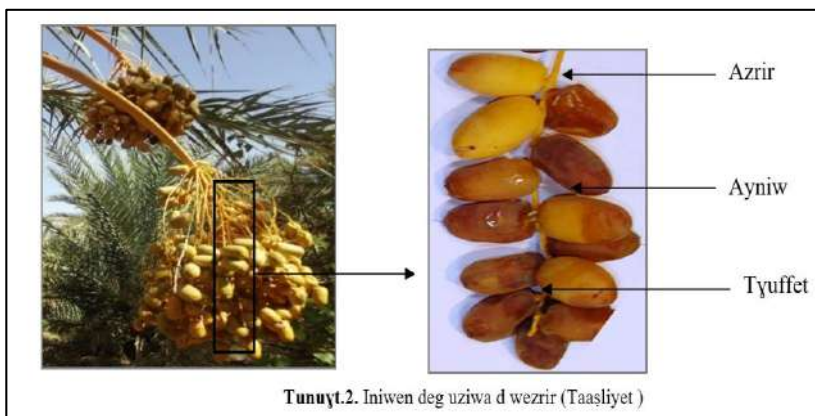
- Ilem n berra (épicarpe) : yettili yeqseḥ, d yeeweṣ¹²⁸ i ulemmez¹²⁹. Sam tameqqrant deg-s hay d ssililuz (cellulose), yettig 08% x lmizan n weyniw.
- Ilem anemmas (mésocarpe) : yettili yel wadday, yehcic, yettwalemmaz fisseε, imum an tamemt. Ittig 55% x lmizan n weyniw.
- Ilem n jaj (endocarpe) : yettarez leelufet, yettas-d d amellal yebheṣ. Yettig 03% x lmizan n weyniw. 44% i d-yeqqimen deg lmizan n weyniw tissit leelufet.

Llun n yiniwen yettneffel xi leḥsab n ddeg : Yetttili d awray deg Buferqqus, d azeggay deg tzeḡgayt (Leḥmira) d Tin-Juher d tikwal yettili d aberkan deg Ccix-Mḥemmed d Bba-Mexluf. La d tmumt n yiniwen temfagam x ddeg yel ddeg, (Tunuyt 2 d Tunuyt 3).

¹²⁷ At Igli qqaren : leelufet. Iyes / iyessan d awal i sen-iruheṇ.

¹²⁸ Yeeweṣ : *difficile à macher*.

¹²⁹ Lmez / alemmez. Taq. : ffez / tuffza.



5. Tawinest¹³⁰ n tinni n yiniwen (Tafelwit 2)

Deffer n umersiq¹³¹ tbedda tnewwart tettuf, makin ad
tuc ayniw deg wammas netlegga-d iysan (leelufat). Ttilin
an iyes, ayniw makin ad d-yawed yexs-as jar 126 d 147 n
wassen x leahsab n ddeg n iyniwen.

¹³⁰ Tawinest : Ɖarret, cycle. At Igli ttinin i txatemt (*bague*) : twinest.

¹³¹ Amersiq. Taq. : ddekkar.

Ger-neɣ yella 05 n twinas i xef-sen i tfat ayniw maki-n a d y aweɣ smiyat n ɗarat yu temfagamen-t x laḥsab n timura (Tafelwit 2).

Tafelwit 3 : Tawinest n tinni n yiniwen.

Tamurt	Tawinest I	Tawinest II	Tawinest III	Tawinest IV	Tawinest V
Tamaziɣt	Ḥebb-Lellu	Alebbuz	Amaɗun	Unqid	Ayniw
Taɛrabt	Loulou	Khelal	Bser	Rotab ou Martouba	Tmar
Lweqt i tissi	1 al	1,6 al	0,8 al	0,5 al	0,5 al
tinni	1,1 yur	2 yur	1,1 yur	1 yur	0,7 yur
yyaren					

5.1.Ḥebb-Lellu

Deffer n umersid tanewwart tuf d al tuc-d ḥebb-lellu yettqama ɣir lkas-nnes yendeɣ¹³² g wezrir n uziwa ha-ay d tyuffet, Tawinest n Ḥebb-Lellu tettaweɣ jar 28 al 35 n iwassen. Ilun-nnes d azeyzaw an tuffa d lmizan-nes ul yettfit-c 06 yegramen. Tettili deg-s tmumt d waman ɗrus.

5.2. Alebbuz¹³³ azeyzaw¹³⁴ (Tawlaft 5)

Ayur¹³⁵ deffer n umersid idewwel-d Ḥebb-Lellu d alebuz d azeyzaw mani i yettzeyyed lmizan-nnes al d ifat 30 yegramen x sebbet n wegraw n waman. Yettili d alebbuz yeqseah aṭṭas d tmumt deg-s druṣ. Tettaweɣ twinest-u jar 45 al 60 n iwassen.

¹³² Yendeɣ. Taq. : yendeɣ.

¹³³ Alebbuz. Taq. : agurbez.

¹³⁴ Azeyzaw. Taq. : azegzaw (y>g).

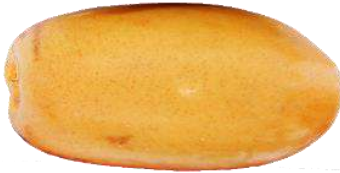
¹³⁵ Ayur : yettwawal ɣel At Igli deg waddad ilelli [yur]. Deg waddad amaruz yettwanna uyur.



Tawlaft 5 : Alebbuz azezwaw (Taaşliyet).

5.3. Alebbuz awray (Tawlaft 6)

Deg umeggaru n twinst n ulebbuz azezwaw tazdayt tbedda tettsukka n Karutin (Karoten) deg ilebbuzen mani i yettneffellun-nseñ d al i iḍeḥḥa-d d awray. Tettzeyyed deg-sen tmunt aṭṭas ḍeḥḥan-d ttwaččan.



Tawlaft 6 : Alebbuz awray (Taaşliyet).

5.4. Urcim (Tawlaft 7)

Al axtin(ḍaynetta) i ixemmer ulebbuz azizaw niy awray x sebbet n wallay n tezyelt (lḥemman) d tifawt d waslay n glukuz i g silla-n, yettneffellun-nseñ d aqehwi d al i tuṭṭa yel tmurt ul d-idewwel-c d ayniw.



Tawlaft 7 : Teskunu-d urcim (Taaşliyet).

5.5. Maḍun¹³⁶ (Tawlaft 8)

Deg wammas n unebdu al axtin i tzeyyed tezyelt ujar n 45 n tfesniwin C° d tafiyyt terša deg uyur n Yulyu, yettbedda weyniw yettawed(yettisi?) x wadday mani i yettili 20% d ayniw d 80% d alebbuz awray. Tawid twinest-u ujar n 20 d 40 wass ha-ay d lweqt mani i yettbedda umekraz afran n yiniwen.



Tawlaft 8 : Teskunu-d maḍun (Taaşliyet).

5.6. Unqıd (Tawlaft 9)

Ammegar u n ubebdu deg uyur n yuct tazıyelt tettıli jar 45 n tfesniwin d 47 tfesniwin. Tmaḍrin daawat mani i d iḍeḥḥa 80% d ayniw d yettqama-d yıir 20% n ulebbuz. Tawinest-u tettig jar 15 d 30 n iwassen.



Tawlaft 9 : Teskunu-d unqıd (Taaşliyet).

5.7. Ayniw (Tawlaft 10)

G u amezwar n xrif yur n ṭuber, unqıd itsmedda tinnines, akul i ḍḥa-d dayniw tnaqeş-n ges aman d al tzayad tmumt mani id ḍharan dguga-i-ḍen an fructose d saccarose. Dadi amekraz i t bedda i t swjad iman-s a d inkaḍ iziwayen.

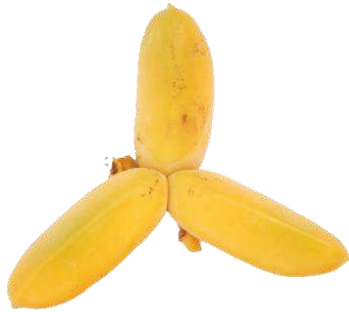
¹³⁶ Awed / yewwed / amaḍun.



Tawlaft 10 : Teskunu-d ayniw i yewden (Taaşliyet).

5.8. Adaw¹³⁷ (parthénocarpique) (Tawlaft 10)

Al axtin ul tgi-ca tezdayt amersiq̄ tgaεad-n-d yiniwen hiḍren d ul deg-sen yelli-ca iḡsan (laεlufet) teqqaren-as dadi i weyniw *adaw*. Xi tarumit ttinin-as (parthénocarpique).



Tawlaft 11 : Teskunu-d adaw (Taaşliyet).

5.9. Uqqic¹³⁸ (Tawlaft 12)

Al axtin i teffud tezdayt, niḡ yeşrem¹³⁹ uziwa x wul n tezdayt, niḡ tella tezdayt thellek, niḡ llan deg-s yixxa yettiqar weyniw zdat mi ad yaweḍ. Dadi ttinin-as uqqic. Amekraz yettnetṭer-ten i walli niḡ yileyman.

¹³⁷ Adaw : alaxetti ul tgi-c tazdayt amersiq̄.

¹³⁸ Uqqic. Taq. : usq̄iḍ.

¹³⁹ Yeşrem : *Fr.* = *glisser*.



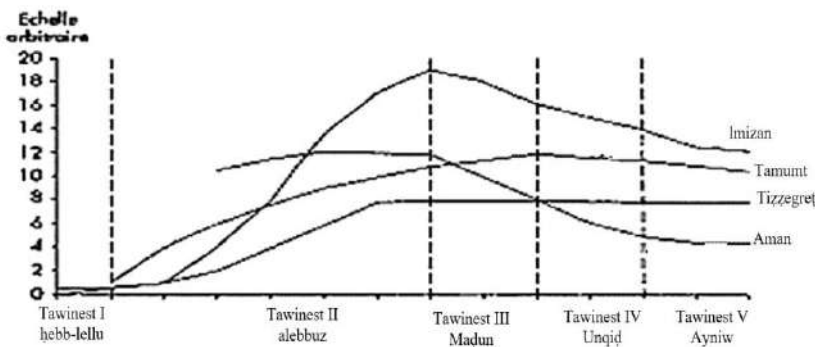
Tawlaft 12 : Teskunu-d uqqic (Taaşliyet).

Tafelwit 4 : tinni n yiniwen (Taaşliyet).

Isem-is xi Tmaziyt	Isem-is xi Taerabt	Anerni n tinni n yiniwen	Tugna
Ḥebb-Lellu	Bounoune Loulou	Tettas-d deffer n umerşiq yettili yuden qae xi tyuffet	
Alebbuz		yetteggam fisseɛ d iyettili d azezwaw deffer n tuttut-nnes ittaggej iggen n tiqqeḥt ela xater yettili deg-s akeb n debay	
Alebbuz Awray		yettili d awray d tuli deg-s tmumt, d lmizan-nnes yettawed lḥedd-nnes	
A-Amadun B- unqid		yedḥa-d ulebbuz ilmeɛ tneqqšen deg-s waman, dadi yettbedda amekraz afran	
Ayniw		Ha-ay d twinest tameggarut, ayniw dadi tbedda tamemt- nnes tgeṭter d al ttneqqšen xef-s waman	
A-Alebbuz awray B- Amadun C- unqid D- Ayniw			

6. Ayniw deg qae twinest n tinni

Deg biochimie n yiniwen tneflen-t tlata n msayel : Imizan, tamumt, tizzegret d waman xi leḥsab n tawniest deg-dwi I yella ayniw (Tunuɣt 4).



Tunuɣt 4 : Anfal n weyniw deg qae twinest n tinni (PEYRON, 2000).

7. Mi ay yellan deg yiniwen

Iyniwen yella deg-sen aṭṭas n ubayur, yettwadker-d deg Weqran belli yettneffeɛ tanzurt¹⁴⁰.

Tafelwit 5 : mokawinat n weyniw deg *Deglet-Nour* (DEVSHONY, 1992).

Mokawinat	Tigmiḍi (%) x wasak n weyniw
Aman	23
Protin	1.5

¹⁴⁰ Tanzurt. Taq. : nnafsa, timnifrit.

Tmumt	72
total	36.1
Saccharose	10.4
Glucose	9.6
Fructose	
Cellulose	7.20
Udi	0.05
Iyađ	1.9

Ayniw ul deg-s yelli-c ƴir tmumt d prutin. Yella deg-s ƴyawsiwin-iđen an viřamin (B6 d C) d yella deg-s ařas n iferdisen n tissent ha-ay d ax win i t-ttuccen i tenzurt.

Tafelwit 6 : Tisent g yiniwen (DEVSHONY, 1992).

Tisent	I 100g n yiniwen i yeqquren
Sudyum	0.017
Putasyum	0.3
Magnisyum	0.036
Fusfur	0.07
Kalesyum	0.1
Uzzal	0.016

8. Aserwes jar weyniw d dđguga-iđen n fakyet

Iyniwen ha-ay d lexyar uřbiř n fakyet, yella deg-s iggen n tezmert tameqqrant mani i yettqewwet bnađem ƴir wařd-s. At zzmen llan teggen-t d tuttut d lařwin i useggas, ttęemmaren xef-s tazeqqa.

Tafelwit 7 : Aserwes jar weyniw d ddguga-iden n fakyet (Randouin, 1961).

Fruit	Azal n tezmert (Kcal)
Tameɣtict	22
Fraise	40
Lim	43
Ččina	48
Imecmac	52
Ixux	52
Ifires	61
Cerise	77
Iqerbac	80
/tibexsisin	
Ađil	81
Lbanan	97
Ayniw	275

9. Tuffra n yiniwen deffer n wenkađ

9.1. Afas n yiniwen

Deffer n wenkađ yettgerraw amekraz iniwen-s deg tyurar d yessili-ten x (s) weyyul-nnes yel tiddart, mani i ten-tessaraden mlih menbeed ttatsen-ten tsednan xi tinsa-nsent¹⁴¹ d al teggent-ten gi txabit deffer n uzeyyed n wazir makin ul ttexşren-c fisseε ul ad naelim-c n txabit (Tawlaft 13. a) ittafes deg tabeţţant (Tawlaft 13. b).

¹⁴¹ Tinsa. Taq. : iđarren.



Tawlaft 13 : a) afas n yeniwen deg txabit, b) afas n yeniwen deg tabeṭṭant.

- III -

Ittānen n tezdāyt

1. Timellalt n tizdayen (lbayud)

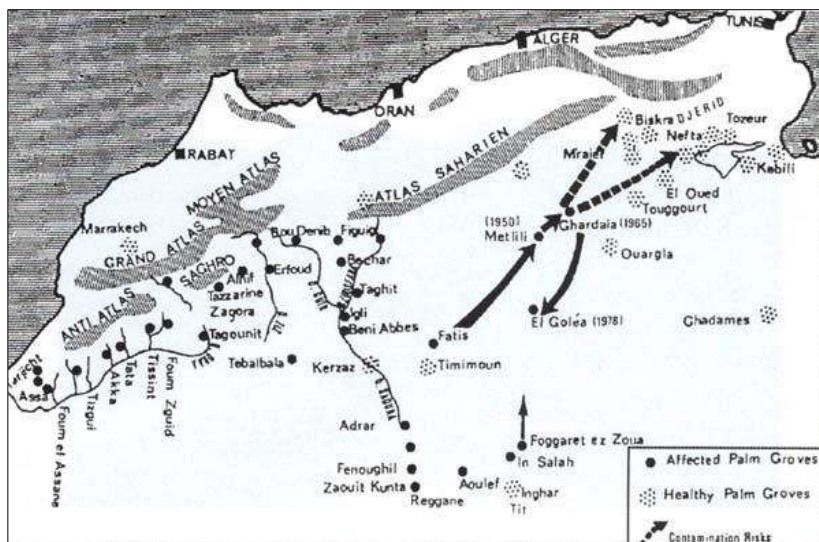
1.1. Amezruy n waṭṭan amellal (Tunuyt 1)

Ttwassnent tmura n tnezruft xi tuzuḍt n tezdayin, mani i xef-s tteklen makin ad ččen, ad irḍen, ad ssken, mani aqimi-nsen gi tmurt inḍed netta d ukur n tezdayt.

Aṭtan n timllelt n tezdayt d iggen xi iṭṭanen i yettneṭṭaren ifadden n ifellahen. Deg Sṣawra mani i sen yenṣu qae tizdayen-nsen d yessgağğ-ten deg tmura-nsen.

Lbayud lgan-t-id tikkelt tamezwart deg Zagura deg Lmerruk deg useggas n 1898. Ihwa-d ṣel Figig, xi figig yutef-d ṣel Beni wennif g dzayer. Jar 1920 d 1950 aṭtan yella yutef ṣel qae tnezdayin n Ssawra.

Jar 1960 d 1978 yewweḍ ṣel Mzab d Lgelya. Aseggas n 1995 ineqqez ṣel tnezdayin n Adrar d Cengiṭ (Muritanya).



Tunuyt 1 : Takerḍa teskunu-d abrid n timllelt n tezdayt (Pereau Leroy, 1958).

1.2. Isyal¹⁴² n waṭṭan (Tawlaft 1)

Aṭṭan yettazeḍ tikkelt tamezwart yel čacit, tanemmast n tezdayt mani i tiqqar iggen n tuffa d al idewwel-d llunnes yecheb an waldun¹⁴³. Tiqqart tettli xi wadday yel tmegna mani tetwabḍa tuffa yel sen n nfuša (izegnan).

Deg umezwar tiqqarent tireṭṭwin n nnefš ayeffus xi wadday yel ujenna. Deffer-s yetmedrin isyel yettas-d ddala n uzgen azelmaḍ xi tmegna n tuffa yel wadday. Ammu dewwel-d tuffa qaε teqqur d llunnes d amellal.

Deg wammas n tuffa i tettdehhar-d iggen n wezrig d aqehwi yetṭef tayda xi wadday yel ujenna. Azrig-u ha-ay d abrid n Fuḍaryum (Fusarium) i iteggen Timllelt n tezdayt. Aṭṭan n lbayud yettbedda yettneqqaz xi tuffa yel tuffa al axtin i yewweḍ yel wul n tezdayt. Dadi yeweḍ wass-nnes ameggaru.

Tazdayt i yetwaṭfen ttilin iḍewran-nnes d izeggayen d hiḍren¹⁴⁴ d ul ttzegrīten-ca aṭṭas.



Tawlaft 1 : a) Isyel n timllelt n tezdayt deg tuffa tazizawt.
b) tazdayt temmu xi lbayud.

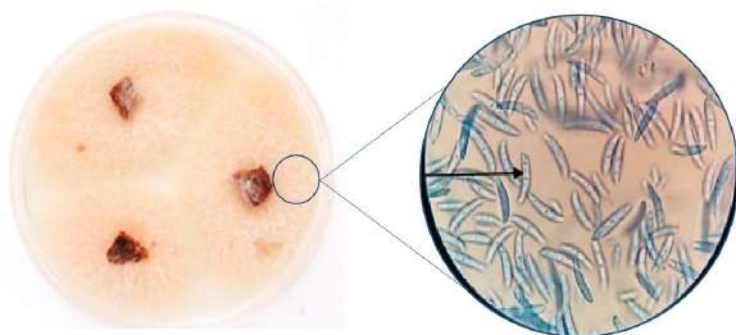
¹⁴² Isyel / isyal, isyalen : *Al aeraḍ, syntômes* (Ref: Tala umaziɣ. Taywalt s tmaziɣt, 2022).

¹⁴³ Aldun : rršaš.

¹⁴⁴ Hiḍren : *faible*.

1.3. Tamentilt n timllelt n tezdayt (Tunuyt 2)

Lbayud tettig-tt tiggent n tfuri¹⁴⁵ ttinin-as *Fusarium oxysporum f.sp.albedinis*. Yettwaæzel-d tikkelt tamezwart aseggas n 1921. Yettwassen aseggas n 1934. Tafuri-u yu ul tettwazra-ca xi tiṭṭ ɛla xaṭer d takeɛbuct aṭṭas, tettwazra xi umicruscup.

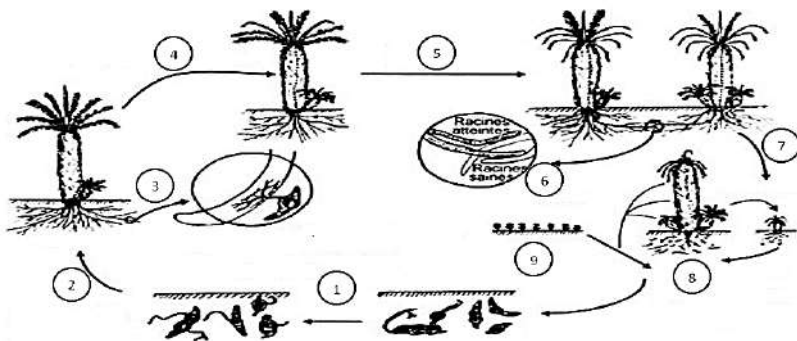


Tunuyt 2 : Tafuri n timllelt n tezdayt adday n umicruscup (Taaşliyet).

1.4. Anḍaḍ n waṭṭan (Tunuyt 3)

1. Acal deg-s les spores n *Fusarium oxysporum f.sp.albedinis* ;
2. Anḍaḍ n waṭṭan ;
3. *Fusarium* iteṭṭef tezdayt xi iḍewran n tissi deffer-s yettebae azeqqur maki ad yaweḍ ɣel tiffawin ;
4. Ttḍehren-d yesɣalen gi tiffawin n čacit tnammast ;
5. Iɣerɣer aṭṭan gi tezdayt d yetɣellebt ;
6. Tazdayt tessendaḍ aṭṭan deg utma-s xi iḍewran ;
7. Temmut tezdayt ;
8. Tiffawin, iḍewran d azeqqur i yeqquren dewwlen-d d leybar deffer-s akul xelḍen netnin d ucal ;
9. Acal ul deg-s yelli-c les spores n waṭṭan.

¹⁴⁵ Tafuri / tifiuriwin : *maladie des champignons*.



Tunuyt 3 : Tawinest n tudert n tezdelt (Sedra, 2003).

Timllelt n tezdayt yettneqq kul aseggas jar 4,5% al 12% n tezdayen i yettwaṭṭfen mani aṭṭan-u yettneqqez xi tezdayt yel tenn-iden deg ucal. Affellaḥ yexṣ-as ad yerzem tiṭ-nnes deg wakud n tuẓudt n tezadyin. ixex-as ul yetteẓẓi-ca tiferxatin ddat-sent (yemmat-sent) deg-s aṭṭan. Akud¹⁴⁶ n wenkaḍ ixexas ad ieqqm amger-nnes.

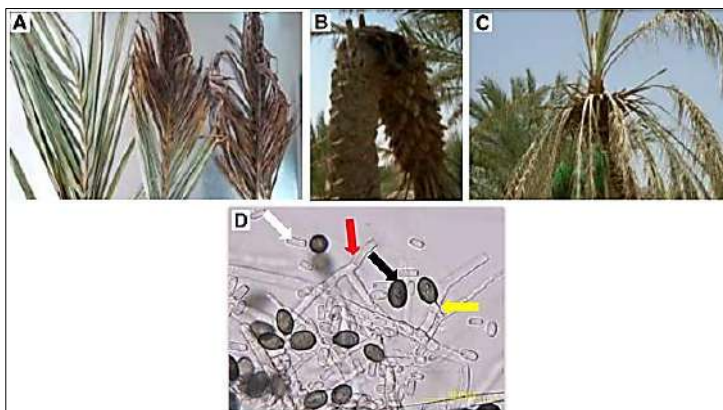
2. Akemmiḍ aberkan (Tunuyt 4)

Ttinin-as deg tamiwin-iden¹⁴⁷ aṭṭan n *hbal*. Yettig-t tiggent n tfuri teqqaren-as *Ceratocystis paradoxal*. Isyal-nnes ttdehren-d lbeε n tiqedqaḍ tiberkanin an wekmaḍ gi treṭwin. Yessegray¹⁴⁸ aymay n tqelwanin d yettxemmer alaw d uzeqqur n tezdayt. Deg wewtem al axti i tedderdeq tqelwant, amersid yettig iggen n rriḥet d tareṣṣaḍt :

¹⁴⁶ Akud : lweqt.

¹⁴⁷ Tama / tamawin : lḡiht / leḡwayeh.

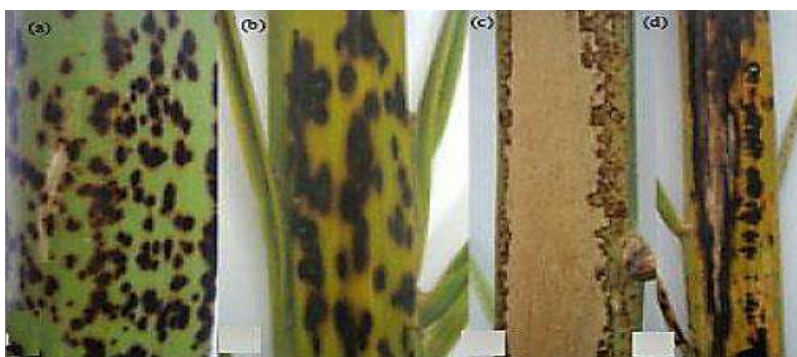
¹⁴⁸ Gri, yegra, agray, ameggaru.



Tunuyt 04. a, b, c) Isyel n akemmiḍ aberkan, d) tafuri n ukemmiḍ aberkan (Fayyadh, 2021)

3. Tiqeḍqad tiqehwiyyin n tuffa (Tunuyt 5)

Aṭṭan-u ssnen-t qae tmura n tuzuḍt n tezdayen deg Tefriqt niḡ tighzirt taerabt. Yettig- t yiggen n tfuri ttinin-as (*Mycosphaerella tassiana*) mani yettiqqaren ḡir yixfawe n trettā tazeyzawt. Maki ad yethenna ufellah xi¹⁴⁹ waṭṭan-u, yexš-as ad iṣeggeṭ deg lexrif qae tiffawin timeqqurin.



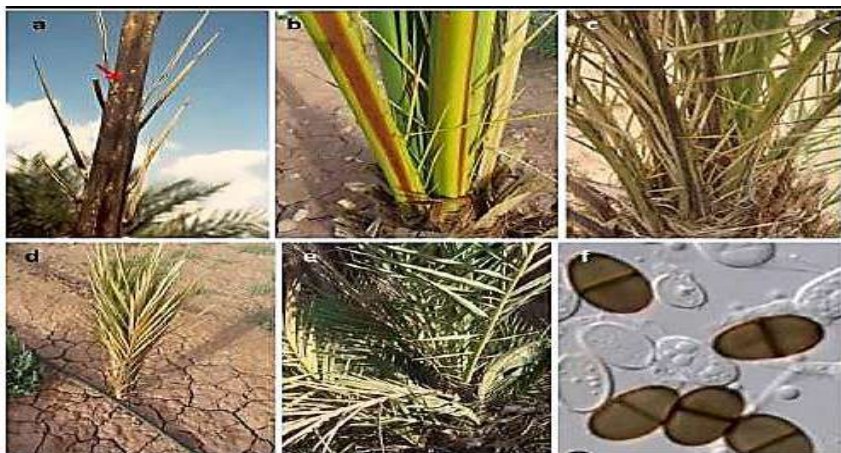
Tunuyt 5 : Isyel n Tiqeḍqad tiqehwiyyin n tuffa (Fayyadh, 2021).

¹⁴⁹ Xi. Taq. : seg.

4. Aṭṭan n *Diplodia* (Tunuyt 6)

Aṭṭan-u tettig-t tiggen n tfuri teqqaren-as *Diplodia phoenicum* (Sacc). Aṭṭan-u ufin-t-id deg 20 ddeg n tezdayin. Deg Ddzayer yettettef laɣa Deglet-Nur. isɣal-nnes ɛehhren-d deg umezwar gi tferxatin tikeɛbac, deffer-s yettafeg ɣel yemma-s kul ɗhan-d gi trettwin-nnes yezrigen d iwrayen teggen xi 15 cm al lmitrat. Deffer-s kul tɛa tiwreyt d yenfel llun-nes ɣel uqehwi. Tazdayt temmut al axtin i yewweɗ waṭṭan ɣel walaw. Anfafad-u¹⁵⁰ yettneṭṭad deg ukud n ukus n tferxatin mani yettageɣ aggas d ameqran deg tazdayt kul yutef-as aṭṭan.

Maki ad tejjɪ tazdayt yexš-as i ufellaḥ ad yezzuzer Benomyl, Methylthiophanate niɣ ddwa deg dwin i yella nhas.



Tunuyt 6 : a, b, c, d) Isɣel n waṭṭan n *Diplodia*, d) tafuri n *Diplodia phoenicum* (Fayyadh, 2021).

¹⁵⁰ Anfafad : virus.

5. Tiqedqad n trettwin n *Graphiola* (Tunuyt 7)

Ittig-t tfuri n *Graphiola phoenicis* (Moug) mani i d-ḍehren yel beɛ n tqedqad xi 1mm al 3mm gi teyda n tuffa i ifaten 3 n isseggasen, dadi yetthewwa leɛmer n tuffa xi 6 n isseggasen al 3 n isseggasen.

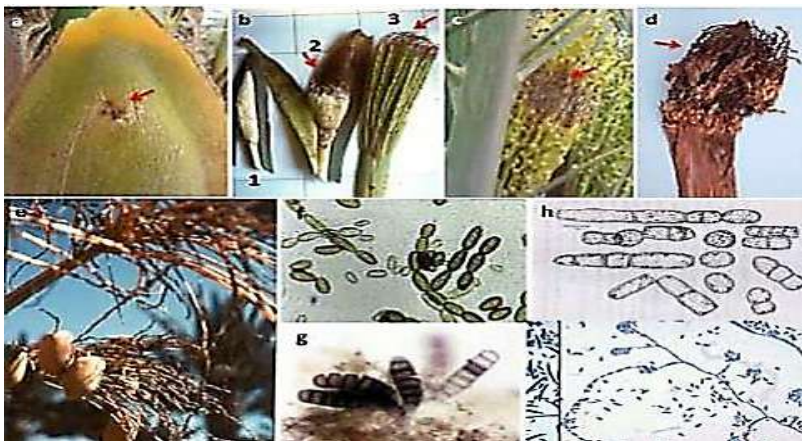
Aṭṭan-u yella aṭṭas deg Misr (Delta d lFayum) mani i tella tuft tettallay aṭṭas, Deg Sɛudiya yettili deg Lqatif d Jedda yeḍher-d isegggasen-u imeggura deg Dzayer. Dawan-t xi Mancozeb d Cupric hydroxide.



Tunuyt 7 : a,b,c) Isyel n *Graphiola* (Fayyadh, 2021).

6. Aṭṭan areṣṣaḍ niḃ azefran (Tunuḃt 8)

Yettigt iggen n tfuri teqqaren-as *Mauginiella scaetiae*, yettxemmer amerṣid dewwel-d ḃres iggen n zefrat terṣeḍ, **ul** yettig-ca n weren¹⁵¹, ammu awtem teṭṭef-t tiggert¹⁵².



Tunuḃt 8 : a, b, c, d, e) Isyel n areṣṣaḍ niḃ azefran, f, g, h) tafuri n waṭṭan areṣṣaḍ (Fayyadh, 2021).

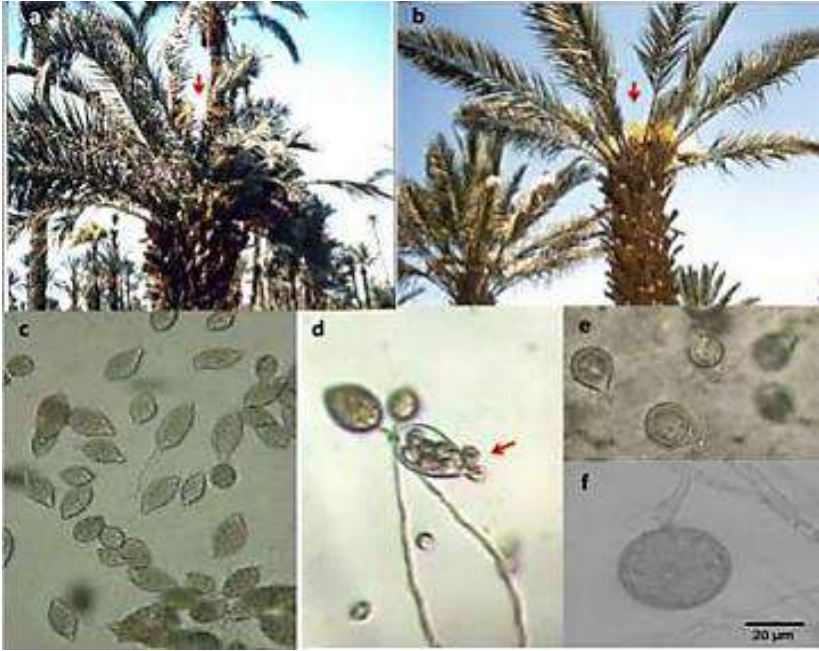
7. Aṭṭan Belæet (Tunuḃt 9)

Aṭṭan-u teggen-t waman al axetti i yutfen ḃel walaw¹⁵³ n tezdayt mani i yettxemmer d zefrat-nnes taṣṣaḍt tcat xi lebeid, sbabb-nnes d iggen n tfuri ttinin-as *Thielaviopsis paradoxa* d *Botryodiplodia theobromae* mani yettsexmaren wul n tezdayt deffer-s xi wassen yettuṭa-d ḃel wadday d kul temmut tezdayt. Aṭṭan-u iæerrem deg Dzayer, Tunes d Lmeruk.

¹⁵¹ Aren / weren : poudre.

¹⁵² Tiggert : stérilité.

¹⁵³ Alaw : ul.



Tunuýt 9 : a, b) Isyel n n waṭṭan Belæet, d) tafuri n *Thielaviopsis paradoxa* (Fayyadh, 2021).

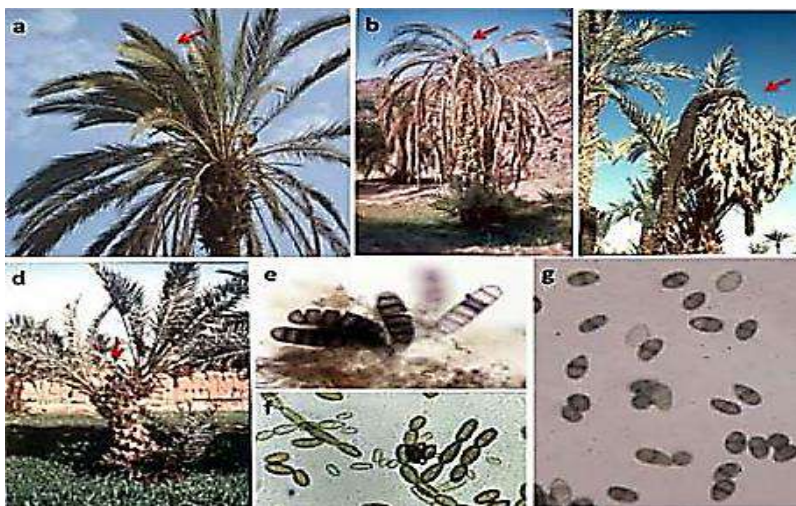
8. Aṭṭan n tmegna¹⁵⁴ i yeknan¹⁵⁵ (Tunuýt 10)

Aṭṭas n uşeggeṭ yettjerreh tazdayt mani i s-ttatfen tfuriwin yel walaw, mani i yettili jemmar deg-s aṭṭas n tmumt kul yufi iman-nnes ibda yettarew mi ay y ittsexmaren jemmar.

Deg waṭṭan-u ttemḥamant tfuriwin n *Thielaviopsis paradoxa*, *Thielaviopsis punctulata*, *Chalaropsis radicolad* *Fusarium moniliforme*.

¹⁵⁴ Tmegna : tête de foliole.

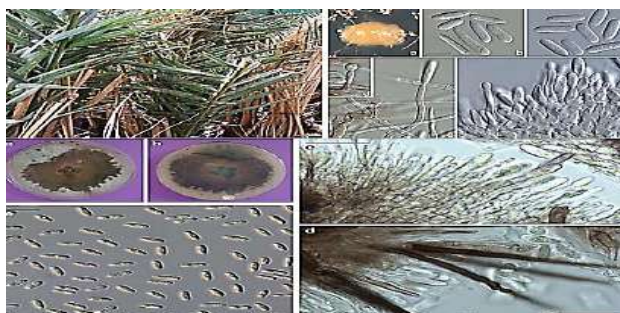
¹⁵⁵ Knu / yekna / kennu : yanḥani.



Tunuyt 10 : a, b, c, d) *Isyēl n waṭṭan n tmegna i yeknan* e, g, h) *tafuri n waṭṭan* (Fayyadh, 2021).

9. Aṭṭan n *Anthracnose* (Tunuyt 11)

Ul yer-s n ḍruret tameqqrant xi tezdayt. ḍehrent-d tretwin ad tinid ttwaqerḍent-d deg-sent tqeḍqaḍ tiqehwiyin. Yeddur xefs-en iggen n ḥacyet d tawrayt d al ttqamant tiberzaz¹⁵⁶. Aṭṭan-u yettig-t iggen n wenfafad (parazit) tinin-as *Colletotrichum gloeosporioides*



Tunuyt 11 : a) *Isyēl n n Anthracnose*, d) *tafuri n waṭṭan* (Fayyadh, 2021).

¹⁵⁶ Tiberzaz = tikeɛbac = timezyanin : Fr. : petite.

10. Tislawt n uziwa (Tunuyt 12)

Aṭṭan-u yeḍher-d tikelt tamezswart tama n Karman n Iran aseggas n 1997. Deffer-s xi 5 n iseggasen yettazeḍ¹⁵⁷ qae ltmura n tuḥut n tezdayin. Deg waṭṭan-u yettwawat ugelwan n uziwa xi wul-nnes mani yetteslawen ilebbuzen zdat n wawwad¹⁵⁸. Aṭṭan n tislawt n uziwa yettig lexsaret tettaweḍ al azgen (50%) n yellet n useggas. Tasleḍt¹⁵⁹ n tezdayin i deg-sent yella waṭṭan sḍehren-d belli deg-sent *Fusarium*, *Alternaria*, *Trichoderma*, *Aspergillus flavus* *Aspergillus niger* d *Thielaviopsis paradoxa*



Tawlaft 2 : Aaziwa yeslaw.

¹⁵⁷ Yettazeḍ : se prolonger.

¹⁵⁸ Aweḍ / yewweḍ / awwad [agaḍ].

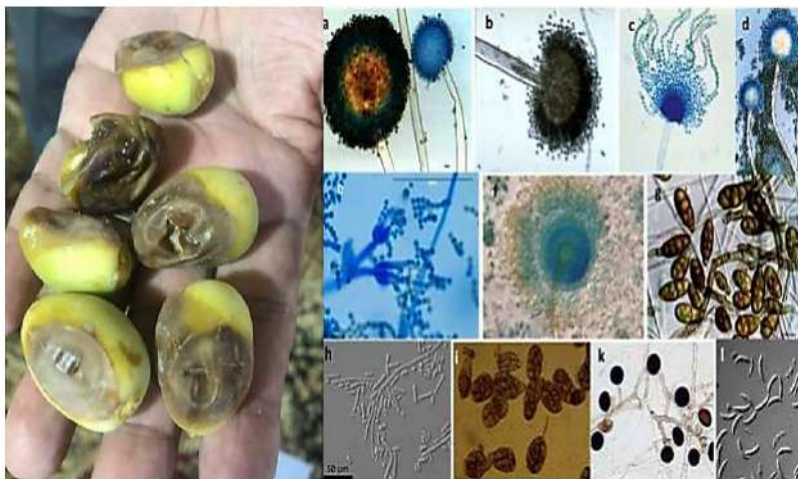
¹⁵⁹ Tasleḍt = analyse.

11. Axmar n ilebbuzen izeyzawen (Tunuyt 13)

Aṭṭan-u yettig lexšaret tameqqrant i ufellaḥ deg tmura n Tefriqt Ugafa d qaε imukan n tuẓuḍt n teẓdayin. Lexšaret tettili jar 5 d 40%. Aṭṭas n tmental¹⁶⁰ yettzayaden aṭṭan-u an :

1. Tičert n uziwa d umḥazam n ilebbuzen
2. Ijerraḥ n yixxa d ijuḍaḍ d tẓuki
3. Allay n tuft n ujenna
4. Amsegraw n waṭṭas n imuḍinen deg teẓdayt

Tifuriwin ha-ay d timental eg teggen aṭṭan : *Alternariaspp* ; *Aspergillus niger* ; *A.phoenicis* ; *Cladosporiumspp* ; *Macrosporiumspp* ; *Phomopsisdiopsyri*.



Tunuyt 12 : a) Isyel n waxmar n ilebbuzen izeyzawen, d) tafuri n waṭṭan (Fayyadh, 2021).

¹⁶⁰ Timental = *facteurs*.

- IV -

Ixxa n tezdayt

1. Qaε ixxa n tezdayt deg Ssawra

Niwwi-d ujar n 12 n iseggasen d necni nettafa¹⁶¹ deg ixxa n Şşawra. Xi Yennayer 2011 al wass n wassu qaε tinezdayin timeqqranin n Şşawra nessers deg-sen tiraččiwin-nney. tetţeft-d xi Tayit al Gşabi, 10 n tnezdayin llant d ayrur n tiyri-u.

Neffey ujar n 108 n tikkal yel tnezdayin n Ssawra. ufuyen-u bđan xi sen ddguga : ayen wass 72 wufuyen x 09h al 12h n iğuran, d ayen deg yiđ 36 n wufuyen, deffer n 20h n yiđ. Nessers aţtas n ddguga n traččiwin :

Tafelwit 1 : Ğermawen n Ssawra.

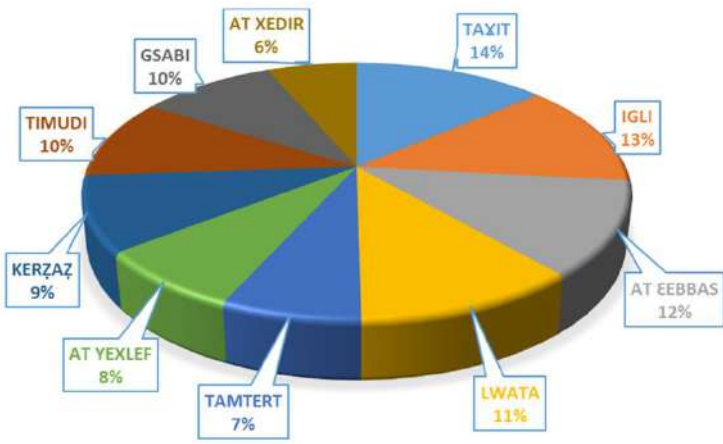
N°	Tanezdayt	Coordinnées	Allay xi ilel	N°	Tanezdayt	Coordonnées	Allay xi ilel
01	Tayit	30°51'43.72" N 02°00'20.66"O	592m	06	At xlef	29°34'26.03" N 01°36'48.00"O	399m
02	Igli	30°26'44.22" N 02°16'36.76"O	498m	07	Kerzağ	29°26'59.36" N 01°24'50.19"O	381m
03	At Ebbas	30°07'29.14" N 02°10'17.85"O	476m	08	Timudi	29°19'10.28" N 01°08'10.32"O	359m
04	Tamtert	29°54'53.10" N 01°54'04.95"O	434m	09	Lwacun n Xedir	29°14'45.47" N 01°02'12.22"O	351m
05	Lwata	29°51'43.39" N 01°49'08.30"O	430m	10	Gşabi	29°05'52.26" N 00°58'17.62"O	335m

Deg wuffuyen-u netţef-d ujar n 6128 n yixxa nfafa-asen xi tafesna, adday N tafesna, Tawacult, Tawsit, Talmest mani i d-nufu 163 n tlemsin ttemfagament.

¹⁶¹ Nettafa / ad naf.

1.1. Asettef xi lehsab ifmiḍiyen n tutṭfa

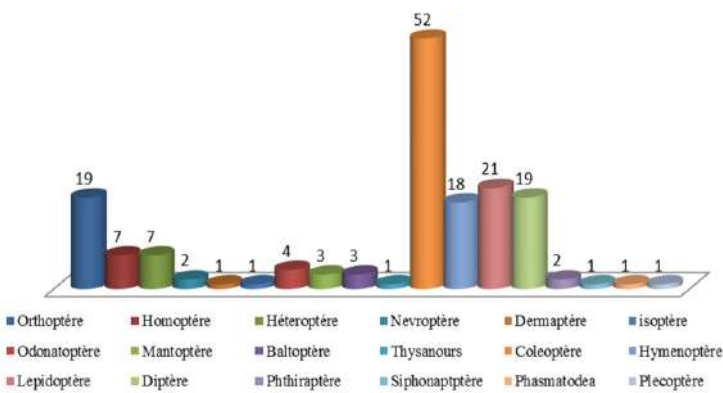
Deffer n weḥsab n ifmiḍiyen n tutṭfa deg qae tɣemmar (tireknin, tixeddiyin.) leqnuta n Ssawra nufu-d nelga-d



Tunuyt 01. Tagrumma² n yixxa i yettwatṭfen di yal tanezdayt (Taaṣliyet)

1.2. Asettef xi lehsab n Tfesna

Deffer n wasun n 163 tlemsin nufu-ten-d ttatfen deg 18 n tfesniwin d 87 twacunin.



Tunuyt 02. isebdad teskumun-d tifesnawin n ixxa deg tinzdayen n Ssawra (Taaṣliyet)

Orthoptera : Tafesna n ixxa yu tilin ifriwen-nsen n ujenna serhen an teyda : Burxes, Tmuryet, Tibliṭṭi.

Ism n Tamaziyt : Tmuryet Twacult : Acredidae Talmest : <i>Anacridium aegyptium</i> Akud n tuṭfa : 11/06/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifrayer tizizawin	
Ism n Tamaziyt : Tibliṭṭi Twacult : Acredidae Talmest : <i>Oedipoda caerulescens</i> Akud n tuṭfa : 12/03/2009 Amkan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifrayer tizizawin	
Ism n Tamaziyt : Burxes La Twacult : Acredidae Talmest : <i>Truxalis nasuta</i> Akud n tuṭfa : 12/04/2009 Amkan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifrayer tizizawin	

Ism n Tamaziɣt : Burxes La Twacult : Acredidae Talmest : Ul yetwassen-c Akud n tuɣfa : 12/02/2009 Amkan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	
Ism n Tamaziɣt : Burxes La Twacult : Acredidae Talmest : non identifier Akud n tuɣfa : 12/04/2009 Amkan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	

Ism n Tamaziɣt : Burxes Twacult : Acredidae Talmest : <i>Acrida</i> <i>ungarica</i> Akud n tuɣfa : 22/04/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifray tizizawin	
Ism n Tamaziɣt : Burxes Twacult : Acredidae Talmest : <i>Hippopedon</i> <i>capito</i> Akud n tuɣfa : 11/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifray tizizawin	

Ism n Tamaziyt : Tibliṭti Twacult : Acredidae Talmest : <i>Arphia pseudonietana</i> Akud n tuṭfa : 12/04/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifrāy tizizawin	
Ism n Tamaziyt : Burxes Twacult : Acredidae Talmest : <i>Calliptamus barbarus</i> Akud n tuṭfa : 11/03/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifrāy tizizawin	

Ism n Tamaziɣt : Burxes Twacult : Acredidae Talmest : Ul yetwassen-c Akud n tuɣfa : 11/03/2009 jusqu à 08/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	
Ism n Tamaziɣt : Tmurɣet Twacult : Acredidae Talmest : <i>Schistocerca gregaria</i> Akud n tuɣfa : 13/04/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	

Ism n Tamaziyt : Burxes Twacult : Rhaphidophoridae Talmest : <i>Ul</i> <i>yetwassen-c</i> Akud n tuṭfa : 01/03/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifrāy tizizawin	
Ism n Tamaziyt : Burxes Twacult : Romaleidae Talmest : <i>Tropidacris</i> <i>collaris</i> Akud n tuṭfa : 15/06/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifrāy tizizawin	

Ism n Tamaziɣt : Burxes Twacult : Phaneropteridae Talmest : <i>Scuddscudderia</i> <i>furcata</i> Akud n tuɣfa : 14/05/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	
Ism n Tamaziɣt : zerza n waman Twacult : Gryllotalpidae Talmest : <i>Neocurtilla</i> <i>hexadactyla</i> Akud n tuɣfa : 02/07/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	

Ism n Tamaziɣt : abzzir aberkan Twacult : Gryllidae Talmest : <i>Gryllus spp.</i> <i>Linnaeus, 1758</i> Akud n tuɣfa : 04/07/2009 ar 08/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	
Ism n Tamaziɣt : abzzir Twacult : Gryllidae Talmest : <i>Acheta</i> <i>domesticus (Linnaeus)</i> Akud n tuɣfa : 04/07/2009 jusqu à 08/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	

Hemiptera : Tafesna n ixxa yu tilin ifriwen-nsen n ujenna ha-ay d azgen n in wadday.

Ism n Tamaziɣt : Iwetɛtten n tezdayt Twacult : Phoenicococcidae Talmest : <i>Phoenicococcus marlatti</i> Cockerell, 1899 Akud n tuɣfa : tqama qae aseggas Ameɛkan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tzemmem aman n tsekla	
Ism n Tamaziɣt : Tilcet n tezdayt Twacult : Diaspididae Talmest : <i>Parlatoria blanchardi</i> Tag Akud n tuɣfa : tqama qae aseggas Ameɛkan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tzemmem aman n tsekla	

Ism n Tamaziyt : Tilcet n ixxa Twacult : Cimicidae Talmest : <i>Cimex lectularis</i> Akud n tuṭfa : 11/03/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tzemmem aman n tsekla	
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Homoptera : Tafesna n ixxa yu tilin ifriwen-nsen n ujenna d in wadday kif-kif.

Ism n Tamaziyt : Abzzir Twacult : Trapiduchidae Talmest : <i>Ommatissus binotatus</i> Akud n tuṭfa : 05/06/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tzemmem aman n tsekla	
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Ism n Tamaziyt : Iwetṭen n tizzizewt Twacult : Aphididae Talmest : <i>Cinara</i> <i>tujafilina</i> Akud n tuṭfa : 27/03/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tzemmem aman n tsekla	
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Heteropta : Tafesna n ixxa yu tilin ifriwen-nsen n ujenna d in wadday mfangamen in wadday zegerṭen x in ujenna.

Ism n Tamaziyt : izi n fakyet Twacult : Rhopalidae Talmest : <i>Brachycarenum</i> <i>tigrinus</i> Akud n tuṭfa : 04/04/2009 Adeg n tuṭfa : 30.446498, -2.277135 Tuttut-nes : lfakyet	
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Ism n Tamaziɣt : Twacult : Lygaeidae Talmest : <i>Lygaeus</i> <i>creticus</i> Akud n tuɣfa : 19/07/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	
Ism n Tamaziɣt : Twacult : Cydnidae Talmest : <i>Cydnus</i> <i>aterrimus</i> Akud n tuɣfa : 04/05/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	

Ism n Tamaziɣt : Tanattart Twacult : Pentatomidae Talmest : <i>Acrosternum</i> <i>heegeri</i> Akud n tuɣfa : 04/05/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tzemmem aman n tsekla	
Ism n Tamaziɣt : Twacult : Alydidae Talmest : <i>Alydus</i> <i>calcaratus</i> Akud n tuɣfa : 19/07/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	

Neuroptera : Tafesna n ixxa yu tilin ifriwen-nsen deg-sen izewran an tratca.

Ism n Tamaziɣt : Kurtata Twacult : Myrmeleontidae Talmest : <i>Curvatus</i> <i>navas</i> Akud n tuɣfa : 04/04/2009 jusqu'à 08/2009 Ameɛkan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : ixxa iɛen (aketɣuf)	
Ism n Tamaziɣt : Twacult : Chrysopidae Talmest : <i>Crysopa</i> <i>vulgaris</i> Akud n tuɣfa : 11/05/2009 Ameɛkan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : ixxa iɛen	

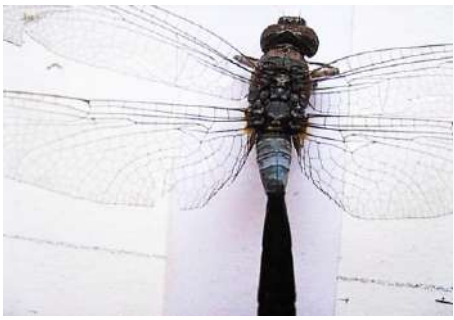
Dermaptera : Tafesna n ixxa yer-sen ifriwen an ilem.

Ism n Tamaziyt : wen-tmezɣin Twacult : Forficulidae Talmest : <i>Forficula auricularia</i> Akud n tuɣfa : 13/04/2009 Ameɛkan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	
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Isoptera : Tafesna n ixxa ifriwen-nsen usen-d gaddan d in uɣenna tisɣen-asen in wadday.

Ism n Tamaziyt : akeɣɣuf amellal Twacult : Termitidae Talmest : <i>Microcerotermes diversus</i> Akud n tuɣfa : 09/04/2009 Ameɛkan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : aqeccuɣ n tsekla	
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Odonatopta : Tafesna n ixxa ifriwen-nsen yer-sen lḥeskat an tymas.

Ism n Tamaziɣt : Taberbillut Twacult : Libellules Talmest : <i>Sympetrum fonscolombii</i> Akud n tuṭfa : 11/03/2009 jusqu à 08/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : ixxa iḍen	
Ism n Tamaziɣt : Taberbillut Twacult : Libellules Talmest : <i>Orthetrum chrysostigma</i> Akud n tuṭfa : 11/03/2009 jusqu à 08/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : ixxa iḍen	

Ism n Tamaziyt : Taberbillut Twacult : Coenagrionidae Talmest : <i>Ischnura elegans</i> Akud n tuṭfa : 15/07/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : ixxa iden	
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Mantodea : Tafesna n ixxa tinsa n zzat yer-sen timeqranin teskunun-tid an malek yella yetzalla.

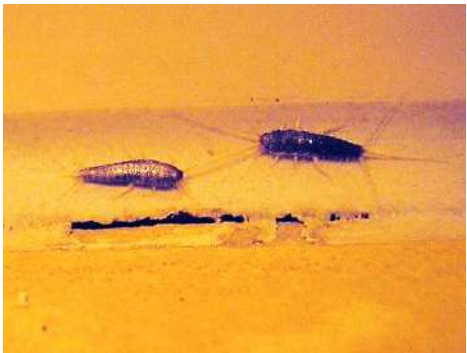
Ism n Tamaziyt : εeggarnsa Twacult : Mantidae Talmest : <i>Blepharopsis mendica</i> Akud n tuṭfa : 04/05/2009 07/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : ixxa iden	
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Ism n Tamaziɣt : εeggarnsa Twacult : Mantidae Talmest : <i>Mantis religiosa</i> Linnaeus Akud n tuɣfa : 11/05/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : ixxa iɣen	
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Blattoptera : Tafesna n ixxa yu tilin ifriwen-nsen tassen-d qewsen.

Ism n Tamaziɣt : Zebluk Twacult : Blattellidae Talmest : <i>Parcoblatta sp</i> Akud n tuɣfa : 21/04/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay	
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Zygentoma : Ixxa n tafesna ul yer-sen yelli-c n ifriwen.

Ism n Tamaziyt : Aslem n tfray Twacult : Lepismatidae Talrest : <i>Divers</i> <i>divers</i> Akud n tuṭfa : 29/05/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifrāy	
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Coléoptera : Tafesna n ixxa yu tilin ifriwen-nsen n ujenna qeṣṣan d al tasen-d an isfer tefren in wadday.

Ism n Tamaziyt : Tamyigrant sebɛa n tqeḍqad Twacult : Coccinellidae Talrest : <i>Coccinella</i> <i>septempunctata</i> Akud n tuṭfa : du 13/03/2009 jusqu'à 05/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : ixxa iden	
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Ism n Tamaziɣt : Tamyigrant ḥdeɛec n tqedqaḍ Twacult : Coccinellidae Talmest : <i>Coccinella</i> <i>undecimpunctata</i> Akud n tuṭfa : 22/06/ 2009 jusqu à 07/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifray n tsekla	
Ism n Tamaziɣt : Tamyigrant n tezdayt Twacult : Coccinellidae Talmest : <i>Pharoscyrnus</i> <i>ancharago</i> Akud n tuṭfa : 10/07/2009 jusqu à 08/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : ixxa iḍen	
Ism n Tamaziɣt : Tajlujet Twacult : Tenebrionidae Talmest : <i>Pimelia consobrina</i> Akud n tuṭfa : 11/02/2009 jusqu à 07/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifray tizizawin	

Ism n Tamaziɣt : Tajlujet Twacult : Tenebrionidae Talmest : <i>Pterolasia squalida</i> Akud n tuɣfa : 20/02/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	
Ism n Tamaziɣt : Tajlujet Twacult : Tenebrionidae Talmest : <i>Adesmia montana</i> Akud n tuɣfa : 25/02/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	
Ism n Tamaziɣt : Tajlujet Twacult : Tenebrionidae Talmest : <i>Pimelia canescens</i> Akud n tuɣfa : 2/04/2009 jusqu à 08/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	

Ism n Tamaziɣt : Tajlujet n rebbi Twacult : Tenebrionidae Talmest : <i>Erodis gibbus</i> Akud n tuɣfa : 29/01/2009 jusqu à 06/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɛy tizizawin	
Ism n Tamaziɣt : Buɣrandɛ Twacult : Tenebrionidae Talmest : <i>Tribolium castaneum</i> Akud n tuɣfa : 10/03/2009 jusqu à 05/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : yeniwen	
Ism n Tamaziɣt : Tajlujet n rebbi Twacult : Tenebrionidae Talmest : <i>Zophosis bicarinata cognata</i> Akud n tuɣfa : 28/02/2009 jusqu à 05/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɛy tizizawin	

Ism n Tamaziɣt : Tajlujet Twacult : Tenebrionidae Talmest : <i>Lyptonychus curvicornis</i> Akud n tuɣfa : 28/02/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	
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Ism n Tamaziɣt : Tajlujet Twacult : Hybosoridae Talmest : <i>Hybosorus illigeri</i> Akud n tuɣfa : 10/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : aqeccuɣ n tsekla	
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Ism n Tamaziɣt : Buɣɣanda n yiniwen Twacult : Cucujidae Talmest : <i>Oryzaephilus surinamensis</i> Akud n tuɣfa : 10/03/2009 jusqu à 05/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : iyniwen	
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Ism n Tamaziɣt : Tajlujet n iyes n yiniwen Twacult : Scolytidae Talmest : <i>Coccotrypes dactyliperda</i> Akud n tuɣfa : 10/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : iysan n yiniwen	
Ism n Tamaziɣt : scarapel Twacult : Buprestidae Talmest : <i>Julodis aequinoctialis</i> Akud n tuɣfa : 18/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrayer tizizawin	
Ism n Tamaziɣt : Tajlujet n tmezɣin Twacult : Carabidae Talmest : <i>Abax ovalis</i> Akud n tuɣfa : 04/04/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrayer tizizawin	

Ism n Tamaziɣt : Tajlujet n tmezɣin Twacult : Carabidae Talmest : <i>Broscus Laevigatus Illustris</i> Akud n tuɣfa : 15/04/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : ixxa iɣen	
Ism n Tamaziɣt : Tajlujet n tmezɣin Twacult : Carabidae Talmest : <i>Broscus nobilis</i> Akud n tuɣfa : 28/04/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : ixxa iɣen	
Ism n Tamaziɣt : Tajlujet n tmezɣin Twacult : Carabidae Talmest : <i>Calathus libanensi</i> Akud n tuɣfa : 10/04/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	

Ism n Tamaziɣt : Tajlujet n tmezɣin Twacult : Carabidae Talmest : <i>Harpalus affinis</i> Akud n tuɣfa : 21/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : ixxa iɣen	
Ism n Tamaziɣt : Tajlujet n tmezɣin Twacult : Carabidae Talmest : <i>Carabus hemrioli</i> Akud n tuɣfa : 21/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifray tizizawin	
Ism n Tamaziɣt : Tajlujet taqedqaɗt Twacult : Carabidae Talmest : <i>Cyminidis tabida</i> Akud n tuɣfa : 10/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifray tizizawin	

Ism n Tamaziyt : Bujluj aqeqdaq Twacult : Carabidae Talmest : <i>Anthia sexmaculata</i> Akud n tuṭfa : 18/06/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : ixxa iḍen	
Ism n Tamaziyt : Tajlujet Twacult : Carabidae Talmest : <i>Calosoma olivier</i> Akud n tuṭfa : 10/03/2009 jusqu à 05/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : ixxa iḍen	
Ism n Tamaziyt : Tajlujet n ikeṭṭufen Twacult : Carabidae Talmest : <i>Cicindela hybrida</i> Akud n tuṭfa : 10/02/2009 jusqu à 08/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : ixxa iḍen (keṭṭufen)	

Ism n Tamaziyt : Tajlujet Twacult : Carabidae Talmest : <i>Diacromus germanus</i> Akud n tuṭfa : 18/02/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifray tizizawin	
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Ism n Tamaziyt : Tajlujet n tmezɣin Twacult : Carabidae Talmest : <i>Odocnemis caudatus</i> (Allard, 1876) Akud n tuṭfa : 15/04/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifray tizizawin	
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Ism n Tamaziyt : Buker Twacult : Curculionidae Talmest : <i>Coniocleonus planidorsis</i> Akud n tuṭfa : 10/07/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifray tizizawin	
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Ism n Tamaziɣt : Buker Twacult : Curculionidae Talmest : <i>Rhytideres plicatus</i> Akud n tuɣfa : 10/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	
Ism n Tamaziɣt : Bab n iccawen Twacult : Cerambycidae Talmest : <i>Pseudophilus testaceus</i> Akud n tuɣfa : 01/04/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : aqeccuɣ n tsekla	
Ism n Tamaziɣt : Tajlujet taqehwit Twacult : Cerambycidae Talmest : <i>Polyarthron pectinicornis</i> Akud n tuɣfa : 29/07/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : aqeccuɣ n tsekla	

Ism n Tamaziɣt : Tajlujet Twacult : Chrysmolidae Talmest : Europtron gracile Akud n tuɣfa : 10/05/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	
Ism n Tamaziɣt : Tajlujet Twacult : Chrysomelidae Talmest : Titubea tredicimpunctata Akud n tuɣfa : 08/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	
Ism n Tamaziɣt : Tajlujet n ibawen Twacult : Cetoniidae Talmest : Leucocelis crinita Akud n tuɣfa : 25/02/2009 jusqu à 04/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	

Twacult : Cetoniidae Talmest : <i>leucocelis crinita</i> Akud n tuṭfa : 25/02/2009 jusqu à 04/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifrāy tizizawin	
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Ism n Tamaziyt : Bujluj n uccaw Twacult : Scarabaeidae Talmest : <i>Oryctes rhinoceros</i> Akud n tuṭfa : 26/07/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : aqeccuḍ n tsekla	
Ism n Tamaziyt : Tajlujet Twacult : Scarabaeidae Talmest : <i>Ul yetwassen-c</i> Akud n tuṭfa : 20/02/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifrāy tizizawin	

Ism n Tamaziɣt : Tajlujet Twacult : Scarabaeidae Talmest : <i>Cyphonoxia kircheri</i> Akud n tuɣfa : 10/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣy tizizawin	
Ism n Tamaziɣt : Tajlujet Twacult : Scarabaeidae Talmest : <i>Aplidia chaifensis</i> Akud n tuɣfa : 01/07/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : aqeccuɣ n tsekla	
Ism n Tamaziɣt : Arduz Twacult : Scarabaeidae Talmest : <i>Scarabaeus cristatus Fabricius, 1775</i> Akud n tuɣfa : 28/06/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : leyɣbar	

Ism n Tamaziɣt : Arduz wacult : Scarabaeidae Talmest : <i>Pentoden bispinosus</i> Akud n tuɣfa : 28/06/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : leybar	
Ism n Tamaziɣt : Tajlujet Twacult : Nitidulidae Talmest : <i>Cybocephalus palmarum</i> Akud n tuɣfa : 10/07/2009 jusqu à 08/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : ixxa iɣen (ssam)	
Ism n Tamaziɣt : Tajlujet Twacult : Nitidulidae Talmest : <i>Carpophilus hemipterus</i> (Linnaeus) Akud n tuɣfa : 03/05/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifrɣay tizizawin	

Ism n Tamaziɣt : Tajlujet Twacult : Dascillidae Talmest : <i>Dascillus cervinus</i> Akud n tuɣfa : 18/06/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : leybar	
Ism n Tamaziɣt : Tajlujet n waman Twacult : Dytiscidae Talmest : <i>Acilius sulcatus</i> Akud n tuɣfa : 03/05/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifray tizizawin	
Ism n Tamaziɣt : Tajlujet Twacult : Histeridae Talmest : <i>Saprinus aegyptiacus</i> Akud n tuɣfa : 07/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : tifray tizizawin	

Ism n Tamaziyt : Tajlujet Twacult : Melyridae Talmest : <i>Zygia sp</i> Ameikan n tuṭfa : 07/03/2009 Ameikan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifray tizizawin	
Ism n Tamaziyt : Tajlujet Twacult : Meloidae Talmest : <i>Diaphorocera hemprichi</i> Akud n tuṭfa : 07/03/2009 Ameikan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tifray tizizawin	
Ism n Tamaziyt : Tajlujet n tiffawin Twacult : Bostrichidae Talmest : <i>Phonapate frontalis</i> Akud n tuṭfa : 07 /03/2009 Ameikan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : aqeccuḍ n tsekla	

Hymenoptera : Tafesna n ixxa yu tilin ifriwen-nsen deg-sen izewran.

Ism n Tamaziyt : Bužanzan (waṛṛẓann) izeyren Twacult : Apidae Talmest : <i>Xylocopa pubescens</i> Akud n tuṭfa : 19/03/2009 jusqu à 07/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : Ixxa iḍen	
Ism n Tamaziyt : waṛṛẓan amessal Twacult : Vespidae Talmest : <i>Delta emarginatum</i> Akud n tuṭfa : 11/05/2009 jusqu à 08/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : Ixxa iḍen	
Ism n Tamaziyt : Bužanzan (waṛṛẓann) Twacult : Vespidae Talmest : <i>Eumenes sp</i> Akud n tuṭfa : 29/05/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : Ixxa iḍen	

Ism n Tamaziɣt : Buɣanzan (waɣɣann) Twacult : Sphecidae Talmest : <i>Ammophila</i> <i>zanthoptera</i> Akud n tuɣfa : 12/04/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : Ixxa iɣen	
Ism n Tamaziɣt : Buɣanzan (waɣɣann) n rtila Twacult : Pompilidae Talmest : <i>Agenioideus humilis</i> Akud n tuɣfa : 18/04/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : Ixxa iɣen	
Ism n Tamaziɣt : Tizit Twacult : Scoliidae Talmest : <i>Megascolia maculata</i> Akud n tuɣfa : 07/05/2009 jusqu à 06/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : Tifray n tizzizewt	

Ism n Tamaziɣt : Tizit Twacult : Scoliidae Talmest : <i>Elis sp</i> Akud n tuɣfa : 22/04/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : Tifray tizizawin	
Ism n Tamaziɣt : Tizit Twacult : Andrena Talmest : <i>Andrena agilissima</i> Akud n tuɣfa : 13/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : Tifray n tizzizewt	
Ism n Tamaziɣt : Akettuf faraɣi Twacult : Formicidae Talmest : <i>Cataglyphis bicolor</i> Akud n tuɣfa : 2/03/2009 jusqu à 08/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : Tifray n tizzizewt	

Ism n Tamaziɣt : Akettuf arumi Twacult : Formicidae Talmest : <i>Cataglyphis bombycina</i> Akud n tuɣfa : 29/02/2009 jusqu à 08/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : Tifray n tizzizewt	
Ism n Tamaziɣt : Akettuf akeɛbuc Twacult : Formicidae Talmest : <i>Messor barbarus</i> Akud n tuɣfa : 2/03/2009 jusqu à 08/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : nécrophage	
Ism n Tamaziɣt : Akettuf akeɛbuc Twacult : Formicidae Talmest : <i>Monomorium salomonis</i> Akud n tuɣfa : 17/03/2009 jusqu à 08/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : nécrophage	

Ism n Tamaziyt : Akettuf n ɗur Twacult : Formicidae Talmest : <i>Pheidole_pallidula</i> Akud n tuɣfa : 2/05/2009 jusqu à 08/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : nécrophage	
Ism n Tamaziyt : Tamɣa Twacult : Ichneumonodae Talmest : <i>Enicospilus spp</i> Akud n tuɣfa : 19/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : Ixxa iɗen	
Ism n Tamaziyt : Tamɣa Twacult : Ichneumonodae Talmest : Ul yetwassen-c Akud n tuɣfa : 19/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : Ixxa iɗen	

Lepidoptera : Tafesna n ixxa yu tilin ifriwen-nsen deg-sen tikbert.

Ism n Tamaziɣt : Afreṭṭu Twacult : Danaidae Talmest : <i>Danaus chrysippus</i> Akud n tuṭfa : 28/04/2009 jusqu'à 08/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : ameršid	
Ism n Tamaziɣt : Afreṭṭu Twacult : Nymphalidae Talmest : <i>Vanessa cardui</i> Akud n tuṭfa : 18/03/2009 jusqu'à 07/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : ameršid	
Ism n Tamaziɣt : Afreṭṭu Twacult : Pieridae Talmest : <i>Pontia daplidice</i> Akud n tuṭfa : 10/03/2009 jusqu'à 05/2009 Amekan n tuṭfa : 30.446498, -2.277135s Tuttut-nes : ameršid	

Ism n Tamaziyt : Afreṭṭu Twacult : Pieridae Talmest : <i>Pieris brassicae</i> Akud n tuṭfa : 17/02/2009 06/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : amersid	
Ism n Tamaziyt : Afreṭṭu Twacult : Pieridae Talmest : <i>Colias philodice</i> Akud n tuṭfa : 18/05/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : Amerşid	
Ism n Tamaziyt : Afreṭṭu Twacult : Lycanidaes Talmest : <i>Lampides boeticus</i> Akud n tuṭfa : 10/05/2009 jusqu à 06/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : amersid	

Ism n Tamaziɣt : Ablullu Twacult : Geometridae Talmest : <i>Crocalis elinguaris</i> Akud n tuɣfa : 28/06/2009 Amekan n tuɣfa : parcelle 04 Tuttut-nes : amersiqd	
Ism n Tamaziɣt : Ablullu Twacult : Sphingidae Talmest : <i>Hippotion osiris</i> Akud n tuɣfa : 10/04/2009 Amekan n tuɣfa : parcelle1 palmier 2 Tuttut-nes : Tifray tizizawin	
Ism n Tamaziɣt : Ablullu Twacult : Sphingidae Talmest : <i>Hyles livornica</i> Akud n tuɣfa : 18/04/2009 Amekan n tuɣfa : la parcelle 04 Tuttut-nes : Tifray tizizawin	
Ism n Tamaziɣt : Afrettu n tketca n yiniwen Twacult : Pyralidae Talmest : <i>Ectomyelois ceratoniae</i> Zeller Akud n tuɣfa : 02/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : iyniwen	

Ism n Tamaziɣt : Afrettu n tzeqqa Twacult : Pyralidae Talmest : <i>Ephestia elutella</i> Akud n tuɣfa : 02/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : iyniwen	
Ism n Tamaziɣt : Afrettu Twacult : Arctiidae Talmest : <i>Utethesia pulchella</i> Akud n tuɣfa : 03/04/2009 jusqu'à 05/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : amersidj	
Ism n Tamaziɣt : Ablullu Twacult : Noctuidae Talmest : <i>Dysgonia algira</i> Akud n tuɣfa : 10/06/2009 jusqu'à 08/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : amersidj	

Ism n Tamaziyt : Ablullu Twacult : Noctuidae Talmest : <i>Euclidia glyphica</i> Akud n tuřfa : 06/06/2009 Amekan n tuřfa : 30.446498, -2.277135 Tuttut-nes : amersiqđ	
Ism n Tamaziyt : Ablullu Twacult : Noctuidae Talmest : <i>Amphipyra tragopoginis</i> Akud n tuřfa : 10/07/2009 Amekan n tuřfa : 30.446498, -2.277135 Tuttut-nes : amersiqđ	
Ism n Tamaziyt : Ablullu Twacult : Noctuidae Talmest : <i>Ul yetwassen-c</i> Akud n tuřfa : 10/07/2009 Amekan n tuřfa : 30.446498, -2.277135 Tuttut-nes : amersiqđ	

Ism n Tamaziɣt : Ablullu Twacult : Geometridae Talmest : non identifier Akud n tuɣfa : 06/06/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes Tifray tizizawin	
Ism n Tamaziɣt : Ablullu Twacult : Hepialidae Talmest : <i>Hepialus humoli</i> Akud n tuɣfa : 06/06/2009 Amekan n tuɣfa : parcelle 2 palmier 03 Tuttut-nes : Tifray tizizawin	

Diptera : Tafesna n ixxa yu tili ɣel-en ɣir sen n ifriwen.

Ism n Tamaziɣt : Izi n tedmayt Twacult : Asilidae Talmest : <i>Eutolmus rufibarbis</i> Meigen Akud n tuɣfa : 20/03/2009 jusqu à 07/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : Ixxa iɣen	
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Ism n Tamaziɣt : Izi bab n iles Twacult : Bombyliidae Talmest : <i>Villa sp</i> Akud n tuɣfa : 11/03/2009 Amekan n tuɣfa : parcelle 04 palmier 10 Tuttut-nes : Tifray tizizawin	
Ism n Tamaziɣt : Izi azizaw Twacult : Calliphoridae Talmest : <i>Lucilia sericata</i> Akud n tuɣfa : 11/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : Izɣan	
Ism n Tamaziɣt : Izi Twacult : Calliphoridae Talmest : <i>Calliphora vicina</i> Akud n tuɣfa : 06/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : Izɣan	
Ism n Tamaziɣt : Izi n tiddart Twacult : Muscidae Talmest : <i>Musca domestica</i> Akud n tuɣfa : 11/03/2009 Amekan n tuɣfa : 30.446498, -2.277135 Tuttut-nes : Izɣan	

Ism n Tamaziyt : Tadisit Twacult : Chaoboridae Talmest : Ul yetwassen-c Akud n tuṭfa : 11/05/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tzemmem idammen	
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Ism n Tamaziyt : Tamza n izan Twacult : Syrphidae Talmest : <i>Ischiodon aegyptius</i> <i>wiedemann 1830</i> Akud n tuṭfa : 17/06/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : Ixxa iḍen	
Ism n Tamaziyt : Tamza n izan Twacult : Syrphidae Talmest : <i>Ceriana conopsoides</i> Akud n tuṭfa : 30/03/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : Ixxa iḍen	

Ism n Tamaziyt : Izi awray Twacult : Agromyzidae Talmest : <i>Opomyza florum</i> Akud n tuṭfa : 11/03/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : fakyet	
Ism n Tamaziyt : Tadisit Twacult : Culicidae Talmest : <i>Culex pipiens</i> Akud n tuṭfa : 31/05/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : tzemmem idammen	
Ism n Tamaziyt : Izi azerṛig Twacult : Calliphoridae Talmest : non identifier Akud n tuṭfa : 21/03/2009 Amekan n tuṭfa : 30.446498, -2.277135 Tuttut-nes : Leybar	

2. Ixxa i yessaḍanen tazdayt

2.1. Tilcin n tezdayt niy tillit tamellalt n

Ssam *Parlatoria blanchardi* Targ

Ires aṭṭas n yismawen xi tmurt al tayed :

Tafelwit 2 : yismawen n *Parlatoria blanchardi* deg timura tierabin.

Tamurt	Isem n waxxu	Tamawt
Dzayer	Ssam, jrib d lmen	
Tunes	Jmil	
Cengit/Muriṭanya	Wrifis d wklifis	
Merruk	Neqqub, Jemla, Tlixt, Tabexut, Taslact	

Tillit tamellalt d iggen xi yixxa i yettēddmen tazdayt n yiniwen. Xi iggen n tama ittaden qae tazdayt, d xi tama-iden ul yuggeḡḡ-c la d iggen n tmurt. Qae tinzdayen yella deg-sen. Tṭef-tt x lqaf al qaf n Ssawra ul yemennek deg la d iggen uyrem. D al iteṭṭef qae ddguga n tezdayin xi tumliḥa al wewtem.

Tettwassen Tillit tamellalt tikkell tamezwart deg useggas n 1868. Amezwar i xefs yeyren ha-ay d M.-E. Blanchard deg yiggen n tnezdayt n yiṣzer n Fir, saddu n Lemṣeyyer deg tnezruft n Dzayer.

2.1.1. Asettef n *taxonomie*

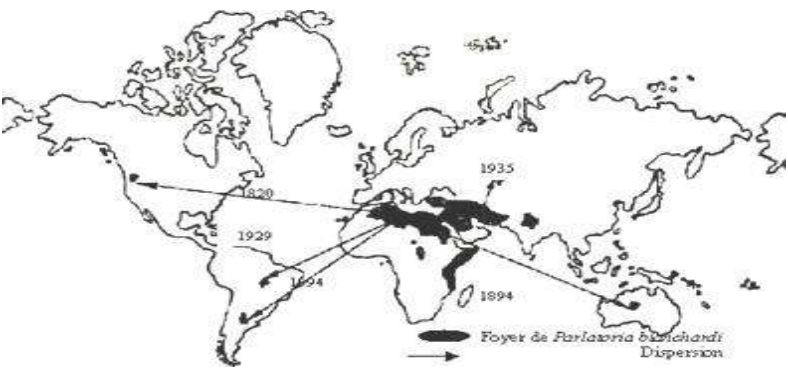
Tillit tamellalt ttinin-as xi Tlatinit *Parlatoria blanchardi*. Tutef deg Tgelda n hwayec (animalia), adur n hwayec i iēlmen lemfaṣel deg tinsa-nsen (Arthropoda), Asmil n

yixxa (Insecta), Tafesna n yixxa i iɛelmen nnefɛɛ n wefri (Hemiptera) d Tawacult n Diaspididae.

Tafelwit 3 : Asettef n taxonomie n tillit tamellat.

Tagelda:	Animalia
Adur:	Arthropoda
Asmil:	Insecta
Tafesna:	Hemiptera
Adday N Tafesna:	Sternorrhyncha
Tawacult:	Diaspididae
Tawsit:	Parlatoria
Talmest:	P. blanchardi
<i>Parlatoria blanchardi</i> (Targioni Tozzetti, 1892)	

2.1.2. Asrebbu n tillit tamellalt n ssam



Tunuyt 3 : Takerɛa sskunu-d asrebbu n tillit tamellalt n ssam deg tmurt umaɗal (DOUMANDJI, 1981).

2.1.2.1. Deg tmurt umaɗal

Parlatoria blanchardi Targ., yuden qaɛ imukan i deg-sen tekker tezdayt. Yeɗher-d tikkelt tamezswart deg Tmurt-N-Sen-Iyezran deg Lɛiraq. X siden ineqqaz yel Iran d tmurt n

Lhend, εawed tasebbabt terru-t-d yel Sɛudiya d ujenna n Tefriqt. Tillit tamellalt ul telli-c yir deg Marikan mani i xef-s thennan x useggas n 1936. (BROYDEN, 1941)

2.1.2.2. Deg Dzayer

Deg Dzayer, qaε tinezdayin ttwaṭṭfent x Tillit tamellalt n ssam mani yettligi-d cra n imukan deg-sen ḍrus d cra n imukan-iden deg-sen aṭṭas, laɣya al axti i tella tuft n ujenna tuli niy tizdayin qar bent jar-atsent aṭṭas d tiffawin-nsent igget tneqquer i tenniḍen ma ay yettsefsusin affag n yixxa. Tinezdayin n ujenna n Wargla d Biskra deg-sent Ssem ujar n Tnezdayin n wadday n Adrar d Tinduf.

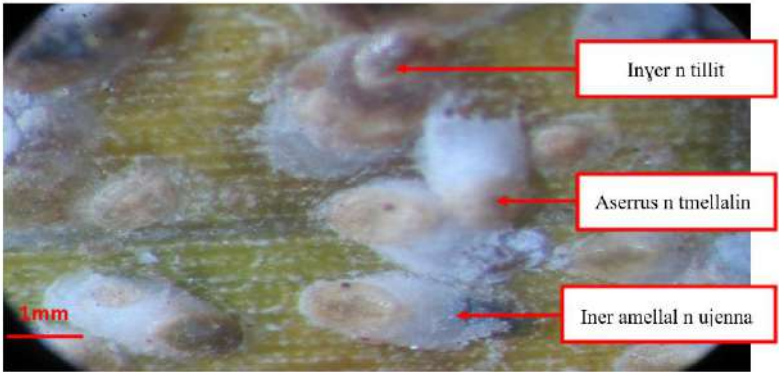
Tutef-d Tillit tamellalt yel Beni Wennif deg Dzayer xi tamurt n Figig d Tfilalt deg Lmerruk jar 1937 d 1938. Deg Cengiṭ d tama n Adrar, Tunyad d Terjit deg useggas n 1949. Ammas n tenzruft : Fezzan, Ṭasili d Timimun deg useggas n 1912. Deg Beccar Tayit, Igli At Ṣabbas deg useggas n 1920.

2.1.2.3. Deg Ssawra

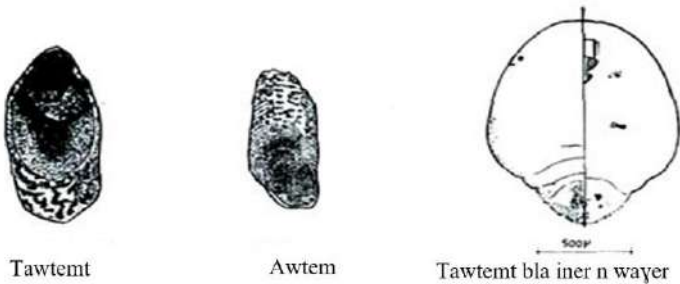
Tinezdayen n ujenna n Ssawra: Igli, Tayit d At Ṣebbas netnint i deg-s illan aṭṭas n Tillit tamellalt mani i yettfat 220 n tyiggent¹⁶² deg 1cm² n treṭṭwin. Adday n Lewwata, tinezdayen n wammas, Idiy, Tamtert d Ugerzim ntlegga-d udun jar 60 d 160 n tyeggnin deg 1cm² n treṭṭwin. Takeεbuct qaε nlegga-tt-d deg tnezdayin n Kerzaz d Legṣabi d Tbelbalt.

¹⁶² Tayiggent / tiyeggnin : *unité(s)*.

timeqqurin yel tiffawwin tizeyzawin n wul n tezdayt. Awtem n Tillit tamellalt yettili d akeεbuc, xi tewtemt deffer n uqarreq yettig ifriwen maki ad yefferfer.



Tunuyt 5 : Tasnelya n Ssam (Taaşliyet).



Tawlaft 1 : Tillit adday n umikruscup X 40(Taaşliyet).

2.1.4. Twinest n tudert n Ssam

2.1.4.1. Tizdal n tillit tamellalt n ssam

Xi leḥsab n SMIRNOFF (1954), tizdelt n waxxu-u tusi-d teẓẓegreṭ, d llun-nnes d axuxi, ugum¹⁶⁶-nnes yettig 0,04 mm d wadan-nnes yusi-d ihcic, tegrew-ten tewtemt adday n wayer n yiner amellal. Tettaṛew tewtemt jar 11 d 20 n tzedlin, tqarreq xef-sen x 03 al 05 n wassen.

2.1.4.2. Takečča i yetthinciwen

Deffer n wefqas tteffey-d tkečča xi tezdelt, tiẓẓegreṭ-nnes tettawed al 0,3 mm d llun-nes yettili d azeggaṭ. Deg umezwar tettili tewtemt yer-s 06 n tinsa, d al thinciw tafa xi tireṭṭwin tizeyzawin. Al axtin yettxeṣṣ amkan treṣṣa d al ttseḍrus tikli-nnes. Tikwal tettawi-d x cra n sswayeε al 03 d nettat tettafa¹⁶⁷ x¹⁶⁸ imukan maki ad treṣṣa.

2.1.4.3. Takečča i ineḍden¹⁶⁹

X¹⁷⁰ 02 al 03 n iwassen, takečča i yetthinciwen treṣṣa d al tneṭṭed deg treṭṭa. deffer-s kull tuden xi yiggen n usufey d amellal yeknu x yiner, teqqaren-as inyer amezwar. Deffer-s takečča tessenfal ilem-nnes ha-ay d mi ay yesttinin aderdeq (mue). Deffer-s kull tessufey-d iner-iden maki ad tesseknu inyer ameggaru deffer-s tuṭṭant tinsa maki ad ttentṭed deg treṭṭa. Dadi ad nezmer ad nmeyyez jar wewtem d tewtemt n Ssam.

¹⁶⁶ Ugum / iguma. *Diamètre (d'un cercle) = (dict. Chafik).*

¹⁶⁷ Tattafa. Taq. : tettnadi.

¹⁶⁸ X imukan : Taq. : yef imukan.

¹⁶⁹ Nṭed / neṭṭed / ineṭṭden [ineḍden]. D tamsertit jar (ṭ + ḍ) = (ḍ + d).

¹⁷⁰ X-sen : Taq. : yer-sen.

2.1.4.4. Tawtemt

Tawtemt n tillit tamellalt n ssam, takerεbuct, tettili d tazeggayt tebhēt, tzeḡga al tneffel nettat d ugma-nnes. Tizzeḡreṭ-nnes al axti i tiweḍ tettili jar 1,2 d 1,4 mm d inyer-nnes yettig 1,6 mm n tizzeḡreṭ d 0,3 mm n tehrew (yer) berra tettili an tizdelt d al teffertāḥ, llun-nnes yedwel-d aqehwi. (BALACHOWSKY & MESNIL, 1935)

2.1.4.5. Awtem

Awtem yettili d akeεbuc xi tewtemt yettig la d netta inyer d amellal. Tizzeḡreṭ-nnes tili jar 0,8 al 0,9mm. Al axti i yiweḍ awtem, yedwel-d llun-nnes d azeggay iza yel tiwrey. Inyer n wewtem yettas-d d azegrar, yer-s jar 0,8 d 0,9 mm d yettili yer-s sen (02) n yifriwen llun-nsen d afrawan¹⁷¹, 06 n tensa¹⁷² d tyuga n iccawen n wakay d sent n tiṭṭawin an iεeqqayen. Awtem aqmu-nnes yettili yeldem ul yetetti walu mani i itedder yir maki ad yemlek, ad yaṛew deffer-s kull yemmut. (SMIRNOFF, 1954).

2.1.5. Tawinest n tudert n tillit tamellalt n ssam

Parlatoria blanchardi

Tawinest n tudert n wewtem temfagam aṭṭas d tun n tewtemt (TOURNEUR & HUGUES, 1975).

SMIRNOFF (1954) yessaweḍ ad d-yini, belli, iwetman i yer-sen ifriwen, ttreclen tiwetmin i inettḍden deg trettḍwin n tuffa ul n yerzem-ca qaddi tiwetmin-u ussen-d xi tuffa-iddḍen i deg-sent llan aṭṭas n Ssam. Deffer n wercal, tawtemt n tillit tamellalt n ssam taṛew tizdal-nnes (jar 7 d 12 n tezdal) adday n yinyer-nnes i yeknun x yiner mani i xef-sen

¹⁷¹ Afrawan / afrawanen : *transparent* (Ref : awawal n Tmaziyt tatrart).

¹⁷² Tansit / tinsa. Taq. : aqejjir / iqejjiren. aḍar / iḍarren.

tqerreq LAUDEHO & BENASSY (1969). Tizdal i d-yettwarṛwen ttwagerwent xi iggen n tamemt trezzem-tt-d tewtemt maki ad ten-tesseantəḍ adday-nnes d ul tuṭṭint-ca aqerraq yettawi-d jar 3 al 5 n iwassen. (SMIRNOFF, 1957).

Deffer n wefrurex teffyent-d tkeččawin i yetḥinciwen (L1). Ttilin deg umezwar d tixuxiyin yetzegga l uzeggay al axti i ufin tareṭṭa tajdidt d tazeyzawt texwa tneṭṭeḍ deg-s. deffer-s kull terzem-d iggen n yiner d amellal maki ad tesseknew inyer-nnes menbaed yettneffel llun-nsen yedwel-d d aqehwi yettzegga al tiberkent. Deg weswir-u yettili muḥal ad tettassen tewsit n tkečča (SMIRNOFF, 1954 ; SMIRNOFF, 1957 ; BALACHOWSKY & KAUSSARI, 1956 ; BALACHOWSKY, 1951 et BALACHOWSKY, 1953).

07 n iwassen deffer n wefrurex n tezdal, tikeččawin i yetḥinciwen (L1) ttsenfalent ilem¹⁷³ d al dewwlent-d tikeččawin timneṭṭdin (L2). Kull uḍant tinsa-nsent. Dadi ad nezmer ad nessen awtem xi tawtemt. Takečča tawtemt L2 n tillit tamellalt n ssam tettissi aṭṭas i tmeqqrant, yir tili takeebuct xes akeb d ul yettili-ca yer-s la valve. Awtem n tkečča yettili d azeḡrar mani i s-d-teffyent sent (02) n tqeḍqaḍ deg tameḡna-nnes ha-ay d tiṭṭawin.

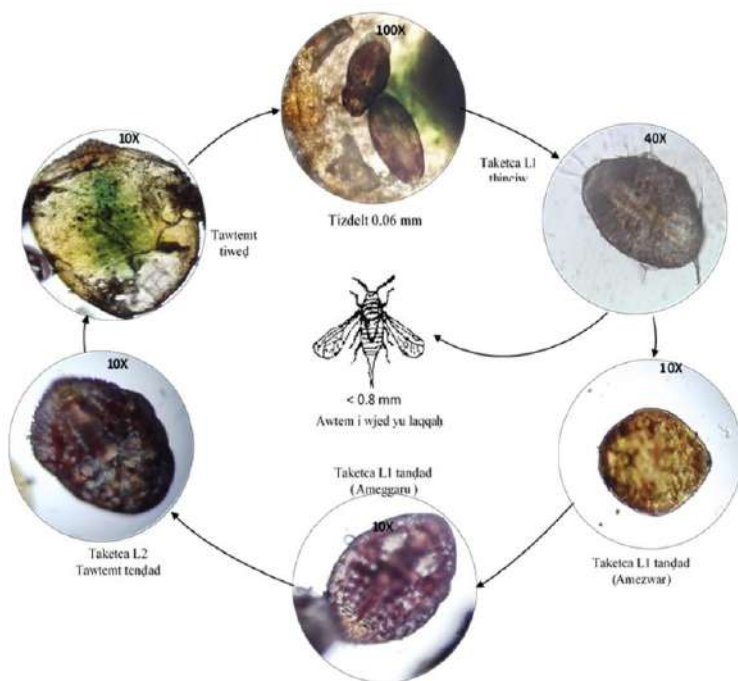
Qae iwetman n tikeččawin d tawtemt yella yer-sen iggent n tadist, xef-s i d-trezzem iner-nnes maki ad tesseknu inyer ittaden-t d al itteteḥ-xes isenga, lḥemman d wejric.

Amilas deffer, takečča n L2 teawed tdewwel tsenfal ilem-nnes, maki ad tedwweel tewtemt tameqqrant tiwweḍ, mani i yetteggam inyer-nnes d al tessewjed iman-nnes i tarwa n tizdal.

¹⁷³ Ilem. Taq. : aglim.

Awtem lweqt-u yettfat x bezzaf n wenfal. Teffyen-d tiṭṭawin-nnes, d al ttzegriṭṭen yifriwen-nnes menbaed teffyen-d 03 n tyugiwin n tinsa d yicawen n wakay¹⁷⁴, deg umeggaru yettuf wadis-nnes kull temfagam xi tamegna, awtem 05 n tikkal d netta yettsenfal ilem-nnes maki ad isi talɣa-nnes tameggarut.

Taɣawsa tameggarut i yettinin belli awtem iweḍ ha-ay d ayanim n wercal, qaε anfal-u yettšerra adday n yinɣer. Ufuy n tillit tamellalt n ssam x inɣer yettili xi iggen n texbut tusi-d deg nnefš azgen n uzagur-nnes (MADKOURI, 1975).



Tunuyt 6 : Tnast n tudert n tilcet n tezdajt (Taašliyet).

¹⁷⁴ Akiy / yattakiy / akay.

2.1.6. Tɛɛbayeε n tillit tamellalt n ssam

Tilcin n tezdayt (Ssam) tteqqasent anezwu i yeqquren d tuft i ihewwan (takeεbuct). Deg tmura n utaram mani i yella wejric d tuft tuli aɛɛas, tillit tamellalt n ssam ul itedderca niy yettkenna yettsugum ad yali lhemman.

Tilcin n tezdayt ul tugjen la d iggen n wemkan deg tezdayt, kullci tettefent an tiffawin n wadday an tiffawin n wul an teyda an izewran n tetteɛa an uziwa an iyniwen xi iman-nsen.

2.1.7. Anerni n wenɛaɛ n waɛɛan deg tezdayt

Tillit tamellalt yettig aɛɛas n tsutwin¹⁷⁵ deg useggas, tillit tamellalt ssam tettafa lebda xi tiffawin i d-iɣemyent tijdidin n wul. Deg umezzwar tettafa tillit xi imukan i yexwan, adday n tetteɛa. Deffer-s kull tuzed ɣel ujenna x siden tettafa ad tawed ɣel qae tetteɛwin n tezdayt ladɣa tin walaw. Tayda ha-ay d imkan imeggaru i itetteffen tilcin n ssam deg tuffa. Deffer n tiffawin tilcin n ssam tneqqzent ɣel iziwayen deg wul n tezdayt, tebeent aqelwan kull ufint-d iman-sen deg uziwa.

Anɛaɛ n tilcin n ssam ttemfagam x lehsab n wezwu d wakud d imkan :

- Tarwa tameqqrant deg yur n Tuber d Furar al wataf n tefsut deg Meyres.
- deg wataf n unebdu yettreggeb lhemman teshewwa tillit n ssam tarwa-nnes d al tkenna.
- Deg tejrect mani i tettali tismeɛt jar yur n Nnayer d Furar, tillit treɣɣel xi iman-nnes deg yinɣer n yiner.
- Deg tesmeɛt tameqqrant adday n 0C° tdewwel tillit, tkenna adday n yinɣer-nnes ujar n iggen n yur.

¹⁷⁵ Tasuta / tisutwin : lɣil / leɣyal.

- Tikeččawin L_2 n tillit tamellalt i yewwđent, tizzarent yel tarwa n tizdal amezwar n tefsut. Tasuta tamezward (T_1) n telcin tđher-d amezwar n unebdu deffer tarwa n tezdal xi 03 n iyyarren.
- Tasuta west sent (T_2) tilcin-nnes ttefggament x tazza, ttawyent-d jar 2 al 2,5 n iyyaren.
- Tasuta west krađ (T_3) tali-d ameggaru n Γ uct ammas n ššmayem.
- Tasuta west kuž (T_4) tilcin-nnes ttalyent-d netnint d wejric n Wember ixef-sent itseyyef akenni kull tťsent ul d-tekkrent-ca al tafsut i d-yusen.

Tiwetmin n telcin ttawđent deg uyur n Meyres, ttzemrent ad gent al kuž (04) n tsutwin deg useggas. Krađ (03) n tisutwin xef-sent ttwaggent yir deg 75 n iwassen x tazza. D Tasuta west 04 tameggarut tawi-d al 150 n iwassen ɛla xater tuța-d nettat d tismeđt n tejrect.

(MADKOURI, 2000)

2.1.8. Izellayen n tillit tamellalt gi tazdayt

(*Parlatoria blanchardi*)

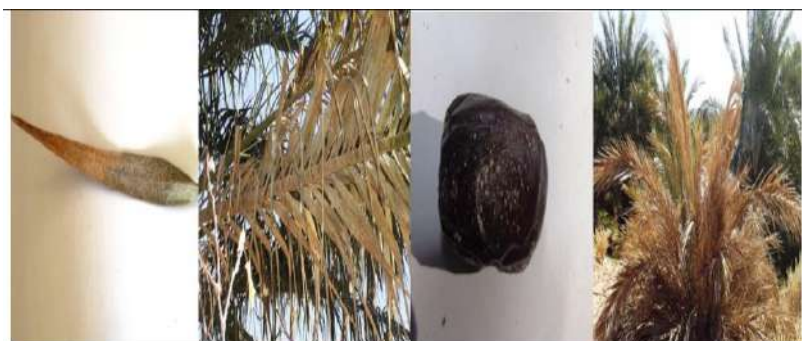
Tilcin d ddeg n yixxa tedrent xi uzemmem n uselyay¹⁷⁶ n tezdaiy, al axti i yujer wemđan n tilcin al 220 n tilcin deg 1cm^2 / trettwin n tezdaiy. Ttwadden qae trettwin ha-ay d ma ul nttuggeğ-ca tifawt ad tawed yel trettwin mi ay yettesreyden la photosynthèse, d taneffut n tezdaiy. Dadi tarețta tbedda tettiwiřy, deffer-s temmut.

Tikeččawin n tillit tamellalt n ssam n tsutwin L_1 d L_2 maki aya čcent ttneqbent tilmit n ujenna n trettta x yiggen n yisten deg waqmu-nsent. Deffer-s ttrezment ammas n trettta lbae n Inzimen i yettrežžan la chlorophyle maki ad

¹⁷⁶ Uselyay : *extrat, séve*.

tzemmen yexfif. Inzimen-u¹⁷⁷ ttædmen tareṭṭa mani i tiwriy d al tzeḡga tizzeyzewt-nnes deffer-s tiqqar qae tuffa al d-taweḍ ddala n tezdajt.

(BALACHOWSKY, 1932). (SMIRNOFF, 1954).



Tiwreyt n trettwin → tiqqart n tuffa → anṭad g winiw → tazdayt temmu

Tunuyt 7 : Izellayen n tillit tamellalt gi tazdayt (Taaşliyet).

Iyniwen i yettwaṭṭfen xi tillit-u yettnefffel llun-nsen d al ḍefsen eḷa xater ttwazemmen qae aman-nsen menbaed ttreyḍen x ugma d al ttiwriyen. Deffer-s kull yeqqaren d al ttuṭṭan zdat n tinni-nsen. Iyniwen-u la d matta iwḍen tinni ttqaman meeyubin ha-ay d ma ul ten-yettuggeḡḡ-ca ad nzen niy yetteshewwa-yasen ssumet-nsen, kul buren¹⁷⁸.

2.1.9. Iberdan n imenyi n tillit tamellalt n ssam

I tmenyiwt n « tillit tamellalt n ssam » *Parlatoria blanchardi*, ixeşş-aneḡ ad neḍfer¹⁷⁹aṭṭas n iberdan n imenyi xsen physique, chimique d biologique.

¹⁷⁷ Anzim / inzimen : *Enzyme(s)* (Ref : auteur).

¹⁷⁸ Kul buren : wa ad buren.

¹⁷⁹ Ḍfer / yeḍfer / ineḍfer.

2.1.10. Imenɣi amḥaq n tillit tamellalt n ssam

Maki ay thenna imekreɣ xi tillit tamellalt n ssam ixexṣ-as lebda ad isenqa tizdayin-nnes, laɣya deffer n wenkaɗ mani i s-ixexṣen ad isiggeɗ tiffawin i i yettawttfent d yessekmeɗ-tent¹⁸⁰. Matta tanezdayt tella tettwattef aṭas ixexṣ-as aseɣgeɗ ad yaweɗ al walaw, ad qeyment ɣir tiffawin ul nettwattef-ca. Ixexṣ-as i ufellaḥ ad igrew iɣemma i yettqaman jar tqernaf əla xater la d netnin tneṭṭden deg-sen tilcin.

Afrag n tiffawin timeqqurin yettawttfen d iggen x sebbet i ttagjent tilcin tamellalt n ssam ad ɛerment (ketrent) əla xater yettesmedda deg-s twinest n tudert-nnes deg-s. Besgeɗ i yettwakksen ixexṣ-as ad yettwassekmeɗ fisseɛ.

Deg Lɛiraq iwɗen al useryi n qaɛ tizdayin i yettawttfent maki yetthannan xi tilcet-u.

(LEPESME, 1947) (DELASSUS & PASQUIER, 1931). (LEPESME, 1947)

2.1.11. Imenɣi *chimique* n tillit tamellalt n ssam

Axxu y n tilcin iqsaḥ gi iman-nes əla xater ɣres iggen inɣer n iner yettagej-t ijhed d al ittefert x dwayat n imkreɣ, dadi imkreɣ ixesas adidawa tizdayen-nes x les produit ig twaxeldən netnin d waman maki adyatfen x izwran, ammu aten tzemmem tillit tamellalt n ssam gi tareṭṭa.

Lḥal west sen hayd axnag n ticet adday n inɣer-nes x urecci n zyutat. Aṭas n dwayat ig tneqqen tilcin n tzdayt DIMETHOATE, BOUILLIES SULFOCALCIQUES xi 7% D ACIDE N LKEBRIT d SULFATE N UZDAL KEHAT en 1968. Bla dawayt yu anezmer annig zyutat n uzdal an zit awray d zit amellal, zyutat yu (DELASSUS & PASQUIER, 1931). (IDDER, 2011).

¹⁸⁰ Kmeɗ / ssekmeɗ / akmaɗ = rɣu / sseryi / aseryu.

Maki ul nettu-ca belli lxedmet lebda x dwayat n ixxa itsexsar cal d waman n wadday n tmurt d itsehlak afellaḥ x lkanser d lahlukat n tneffut uxcinen.

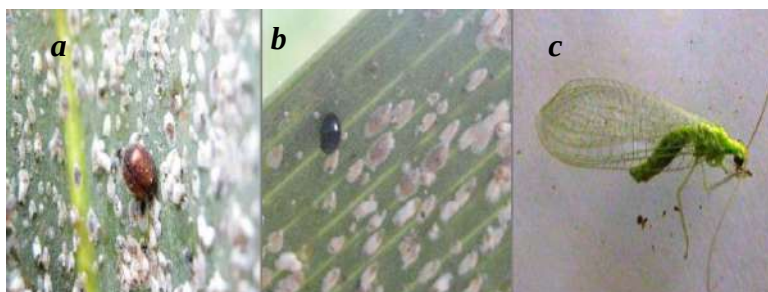
2.1.12. Imenḃi *biologique* n tillit tamellalt n ssam

G ddeg yu n imenḃi n txeddem x u acenḃu (ennemi) n uxxu ig tēedmen tazdayt, ɛla xater gi tanzdayt kul axxu ḃres axxu iḃen d asenḃu-nes yettetit hayd may ig tagejen les effictifs nes rṣan.

2.1.13. Isenga n tillit tamellalt n ssam

(*Parlatoria blanchardi* Targ)

Yella aṭṭas n isenḃa n tillit tamellalt n ssam g tinzdayin n Igli, tamezward xi twacult n Coccinellidae tinin-as *Pharoscyrnus ancharago* Faim., d ten-iḃen xi twacult n Nitudilidae tinin-as *Cybocephalus seminulum* Baudi., west telata xi twacult n Coccinellidae tinin-as *Chilocorus bipustulatus* L., imezwar it d i lgan hayd TOURNEUR et LECOUSTRE g useggas n 1975 g tmurt n MURITANYA. IDDER g useggas n 1992 ilga-d gi tinzdayen n Wargla 04 n isenḃa iḃen *Chrysopa vulgaris*, *Cybocephalus seminulum*, *Pharoscyrnus semiglobosus* d *Aphytis mytilaspidis*.



Tunuyt 8 : Isenḃa n tillit tamellalt n ssam g tinzdayen n IGLI (Taaṣliyet).

2.2. Takečča n yiniwen

Ectomyelois ceratoniae Zeller

Takečča n yiniwen d igget xi ixxa l yettedren ammas n yiniwen, tḍeyyeε afellaḥ bezaf εla xater ul trezzem i lad-ca tizar xi tizdayen l rumman l taxerrubt.
(DOUMANDJI, 1981).

Tafelwit 4 : Taxonomie n takečča n yiniwen.

Scientific classification	
Tagelda	Animalia
Adur	Arthropoda
Asmil	Insecta
Tafesna	Lepidoptera
Tawacult	Pyralidae
Tawsit	Ectomyelois
Talmest	<i>Ectomyelois ceratoniae</i>

2.2.1. Tasnelya n tkečča n yiniwen (Tunuyt 10)

2.2.1.1. Afreṭtu i yewḍen

Ectomyelois ceratoniae Zeller., d afreṭtu d akeεbuc ittig x 6 al 12 mm n tizzegreṭ d x 16 al 22 mm jar ifriwen-nes (DHOUIBI, 1991). X leḥseb n DOUMANDJI-MITICHE aseggas n 1977 & DOUMANDJI aseggas n 1981, yella sen n ddguga n ifriṭṭa n yiniwen x leḥsab n imukan. Tissi lkulur armadi g tell beṣṣaḥ iḍeḥḥa-d lkulur-nes d amellal irrez gi tinzdayen. Ifriwen n ujenna l u afriṭtu yu tilin cebḥen x laxṭuṭa ḍehren ama ifriwen n wadday tili-gsen iggen tesfift n ḥrir. (DHOUIBI, 1991).

2.2.1.2. Tizdelt

Tig jar 0,6 d 0,8 mm. Tili tamellat g umezwar menbaεd duggel-d taxuxit 24 saεa deffer n tarwa-nes.
(DOUMANDJI, 1981).

2.2.1.3. Takečča

Deffer n ufuy-nes xi tizdelt tili tizzegret-nes yir 1mm, tawed al 18 mm g iwassen-nes imeggura zat m ad tedwel d afrittu (DHOUIBI, 1991), (Tafelwit 5). Gi tudert-nes tqama takečča jar 06 n simanat al 06 n iyyeren x lehsab n humman (LE BERRE, 1978). Tfuwet tudert-nes teffer jar ayniw d iyes-nes d me'a iwasse ittitar ayen x iberriden-nes. (DOUMANDJI-MITICHE, 1977).

Tafelwit 5 : Tasnelya n tkečča n yiniwen.

Takečča	L1	L2	L3	L4	L5
Amɗan n tukɗin	8	12	15	32	35
Tizegreɗ (mm)	1,1à 2	2 à 2,3	3,3 à 5,6	6,9 à 12,4	12,3 à 14,6
Cɕacit n tmegna 1/10 mm	2,98	4,46	6,35	10,25	15,43

2.2.1.4. Tenukna n tkečča (Chrysalide)

Tenukna n tketca tig 8mm n tizegreɗ, tused an cornet illa yres 07 n sigat kul sigt illa-ges sen n usennanen d illa yres sen tukɗin g sigat n udis-nes. Tamegan-nes tused taqehwit teqsaḥ ges sen n icawen d tiymas-nes usent-d an laggat. Takečča zat ma tken teddar jwan n ḥrir. (DHOUIBI, 1991).



Tunuyt 9 : Tasnelya n tkečča n yiniwen (Taaşliyet).

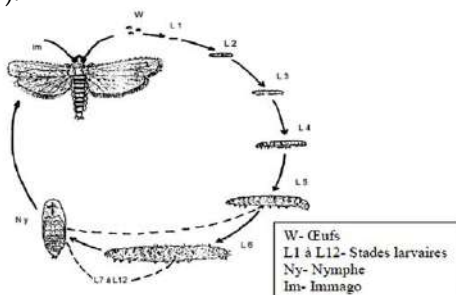
2.2.2. Twinest tasnudert n tkečča n yiniwen

Takečča n yiniwen tella g qae tinzdyen n Sawar mani l teŭtef aŭtas iyniwen d rumman. *Ectomyelois ceratoniae* Zeller., tig x iggen (01) al xemsa (05) n tisutwin x leḥsab n tutut d les conditions climatiques (NAY, 2006).

GOTHILF g useggas n 1969, inna belli ufuy n ifritŭta i g iwḍen itili g nefš amezwar n yeḍ (giḍ) Ƴir kul iffay-d ittizar l tewtemt maki attileqqaḥ d ittawid aŭtas n waqt ajenna-nes maki ay semda. (WERTHEIMER, 1958)

Tawtemt tesrusu jar 60 d 120 tizdelt tqamant Ƴir 04 n i wassen maki a feqsent (LE BERRE, 1978). X leḥsab n WERTHEIMER, (1958), inna belli takečča i di fƳen tajdidt xi tizdelt tafa x uxbu mani a teffer d ixesas tutut attil sadwes. Kul terzu taflut jar ayniw d iƳes-nes maki a teffer. Defres tbedda teddar tratca-nes x leḥrir. Agma n tketca iqqen x lḥemman d tutut. Tbedda takečča tsenfal ilem al teggam. Illa mani i tsenfal ilem-nes 12 tikkelt ittawid ayu jar 42 al 240 n iwassen (VILARDIBO, 1975). G iwassen-nes imeggura takečča taden iman-nes d al tkenna g iggen n uxbu tinin-as tenukna. Tig takečča tameгна-nes ljihet n urzam n tyufet maki ad yas ufuy-nes ihiyen. (DOUMANDJI- MITICHE, 1977).

Ittawid akenni n tketca jaj n tenukna x 03 al 05 n iwassen (LE BERRE, 1978).



Tunuyt 10. Twinest tasnudert n tkečča n yiniwen.
(DOUMANDJI - MITICHE, 1983).

2.2.3. Izellayen

Takečča n yiniwen d igget x ixxa ig tesrun afellaḥ εla xater ul tugejent-ca ayniw ad i twaffer si tazeqqa. Ter-naε g Iqli ujar n 45% n inwen n ukmas tateften takečča d ul tzemred-ca aten tzzenzed gi iyemma d yiniwen ig uḍan l tmurt ujar n 75% gsen qaε tilin twaṭfen.

Gi tmurt n Wargla nisbet n uṭuf tali l ujar n 42,5% gi tinzdayen d al taffeg al 64,7% gi tizqawin n ukmas. BENADDOUN (1987), *le taux* n uṭṭuf yir g ddeg n deglet nur ittli jar 27% d 67,50%. (RAACHE (1990)

2.2.4. Ttawilat n imenyi n tkečča n yiniwen

Tugda tameqqrant n ifellahen gudu hayd ataf n tketca yut l tinzdayen-nsen, amala ḍhan-d tebεen kul iberdan maki a tenyen

2.2.5. Imenyi x ddwa n tekečča n yiniwen

G iseggasen n 80 d 90 dzayer tella tzuzur aṭṭas n dwa n DDT dwa yu illa itqetṭa tṭawṣa gi takečča beṣaḥ illa ititig bezzaf n i waṭṭanen i bnadem hayd axuin it menεen.g useggas n 2006 (WERTHEIMER, 1958 & WARNER, 1988).

X useggas n 2006 l gudu dewwlen-d tzuzuren dwayat iḍen an : MALATHION al 2%, PARATHION 1,25%, PHOSALONE 4%. D gi tizeqwin n ukmas dewwlend teggen amequs x BROMURE N METHYL.

2.2.6. Imenyi asnudar n tekečča n yiniwen

Isenya n tketca n yiniwen εermen, illa indi tettin tizdal, lbaε iḍen tettin takečča, d lbaε tneqen afriṭu qaε inu adnefεen maki adsenqšen les effectifs n tketca bla ma nssehlek afellaḥ.

Tafelwit 6 : Imenḡi *biologique* n tekečča n yiniwen.

Tafesna	Tawacult	Talmest
Hymenoptera	Braconidae	<i>Phanerotoma flavitestacae</i> , <i>Phanerotoma dentata</i> , <i>Phanerotoma</i> sp. <i>Habrobracon brevicornis</i> , <i>Habrobracon hebetor</i> . <i>Bracon mellitor</i> <i>Microbracon pembertonii</i> <i>Apanteles lacteus</i> , <i>A. myeloenta</i> , <i>Apanteles</i> sp. <i>Ascogaster</i> sp. <i>Rhogas testaceus</i>
	Pteromalidae	<i>Anisopteromalus mollis</i>
	Bethylidae	<i>Perisierola gallicola</i> , <i>P. emigrata</i> <i>Goniozus legneri</i>
	Encyrtidae	<i>Pentalitomastix plethoricus</i>
	Ichneumonidae	<i>Pristomerus vulnerator</i> , <i>Nemeritis canescens</i> <i>Horogenes</i> sp. <i>Gelis</i> sp. <i>Herpestomus arridens</i>
	Chalcididae	<i>Brachymeris aegyptiaca</i> , <i>Brachymeris</i> sp. <i>Antrocephalus mitys</i>
	Perilampidae	<i>Perilampus tristis</i> (hyper-parasite d' <i>Apanteles</i> pupae)
	Trichogrammatida	<i>Trichogramma embryophagum</i> , <i>Trichogramma</i> sp
Diptera	Tachinidae	<i>Clausicella suturata</i>
Acari	Pyemotidae	<i>Pyemotes ventricosus</i>

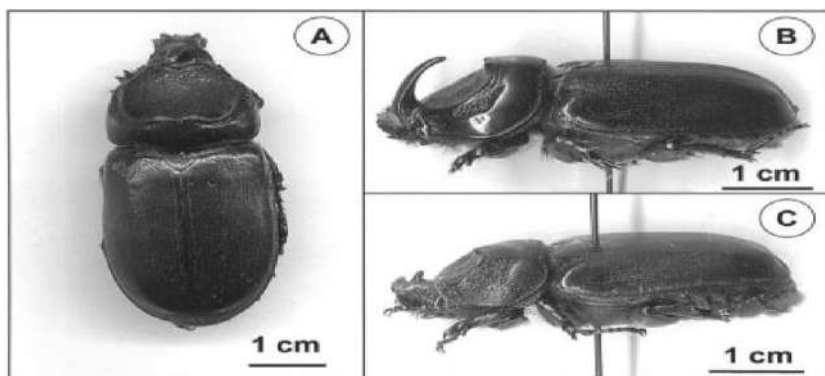
2.2.7. Imenɣi afiziki n tekečča n yiniwen

Dadi afellaḥ itgerw qae lemsayel maki uli tig-ca dwayet, ammu i tḥawal x tanemmirt-nes d l'écosystème gi tinzdayen qae adnini dadi lbae n les techniques :

- Asneqqu n tinzdayen d usekmeḍ n yiniwen d tiffawin ig twaṭfen-t ;
- Adan n iziwayen xi tcukar maki ul ittirew-ca afrettu g uziwa (BOUKA et al., 2001) ;
- Anya n tizdal x les radiations (Gamma) lad matta feqsen-t tefyen-d iwetman egren d'*Ectomyeloides ceratoniae*. (BEN ADDOUN, 1987).
- Ukus n tizarayen, iqelwanen, iyemma ig tqaman g uzeqqur deffer n unkaḍ.
- Asegget n tqernaf d ukus n usan maki ul teṭfen-ca iyemma d l kerfi.
- Agraw n yiniwen i guḍḍan gi tmurt
- Asneqqu n tizeqqa n ukmas zat n waggaḍ n xrif (IDDER, 2008).

2.3. Bujluj (*Oryctes rhinoceros* L.) (Tunuyt 11)

Oryctes rhinoceros L., ittatef netta d ixxa yer-sen ifriwen quren (coléoptère) twacult n Scarabaeidae adday n twacult Dynastina yres iggen n lkulur taqehwit tza l tzweyt i tseqsiq aday n tfiyt, tizegreṭ-nes taweḍ al 40 mm. Tawtemt lebda temfagam netta-t d uwtem. Awtem ittili yres iggen n uccaw gi tameгна-nes idfes l deffer. Bujluj yu itteqas imukan mani ittili ahiluf n tizdayen bezaf.



Tunuyt 11 : *Oryctes rhinoceros* (A) awtem ig iwɗen x ɗher-nes - (B) awtem x nnaɗer (C) tawtemt x nnaɗer. (DAVE MOORE, 2005).

2.3.1. Tarwa n tɗdal (Tunuyt 12)

Saddu n izewran n tezdayt ig tilin xemren x waman bezaf. Tsekka tawtemt n bujluj tiddart-nes x uluɗ defres tesrusu ges tizdal-nes tilint timellalin d timqranin. (DAVE MOORE, 2005)

2.3.2. Takečča n bujluj (Tunuyt 12)

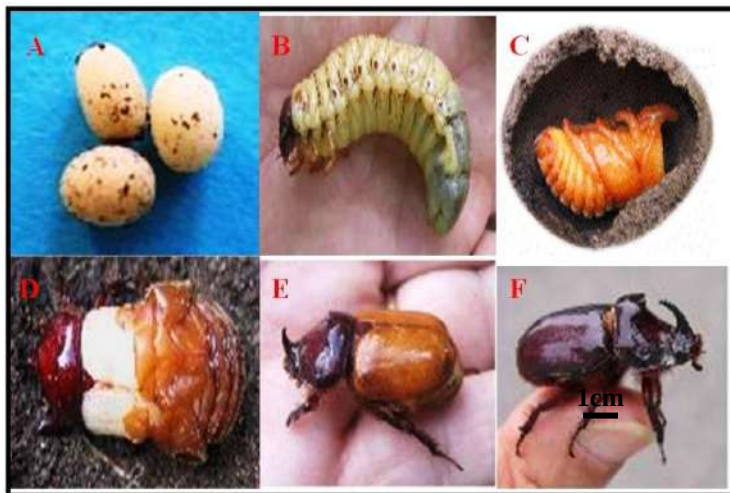
Tasse-d tqawes an uyur tetti yir izeyran d iqecɗan tqama takečča x sen (02) al tlata (03) n yaren. Takečča tawɗ al deggel-d d bujluj deffer n 05 n yeren.

2.3.3. Tenukna (Tunuyt 12)

Alaxtin itiweɗ takečča x 3,5 al 4,5 cm n tizegret teqqaz iggen 20 cm adday n ucal d al tregel xi-manes g iggen n tqellut n uluɗ anecten n tizdelt-nes. Tsenfal ilem-nes 05 n tikkal. Iddegel lkulur-nes d amellal itzegga al itzwiɣ g umeggaru takečča tuc-d-anae bujluj dameqran imda aggaɗ-nes. (SCHOONHOVEN, 2005)

2.3.4. Bujluj i yewden (Tunuyt 12)

Alaxti I d iffay bujluj xi tenukna ittli illuz. Maki ad igem ixes-as ad yetec aţtas. Ul ittugj la azizaw la ameqqur. Dadi itæeddem tazdayt ladya azeqqur d izewran. Tawtemt teqqaz ixuba ammas n izewran n tezdayin tizizawin hayd ma un tugj-ca aman ad alyen l tuffawin defer kul temmu.



Tunuyt 12 : Iweswiren n ukur n bujluj (A) tizdal -(B) Takečča - (C) Tenukna - (D) asenfel n ilem - (E) awtem d miyyay - (F) awtem iweđ. (BRIAN MORRIS, 2004)

2.4. Butranđa bab n tiymas

(*Oryzaephilus surinamensis* L.)

Teqaren-as butranđa bab n tyimas əla xater đher-nes ittli itcur xi tyimas, ittisyas I butranđa n farina yir netta tiqehwit-nes tili tqsaḥ, ittig jar 2,5 d 3,5 mm, ittass-d itarreq x wadday. Tinin laşel-nes amezwar xi Tamarikant n wadday, ul ittagej lad iggen n umkan gi tmurt ittef qae imenzawen. Ittarew aţtas mani igget n tawtemt niε sent tucen-d jehd n twacunin. Tiwinest n tudert-nes atezmer attig g 20 n iwassen. (AL – DOSARY, 2007)

2.4.1. Twinest n tudert (Tunuyt 14)

Tawtemt n butranđa bab n tɣmas tawid x 06 al 10 n iwassen d nettat tarew gi tizdal ammas n winiw. 17 n wassen deffer n tarwa tfeqsen-t tizdal teqqasen-t lhemman adili jar 25 d 30°C, talyen-d xi tizdelt tkitcawin tikaɛbac teggen-t udun n 0,9 mm. Takečča tili tkeckuc aṭṭas tafa xi tutut maki atgem jar 30 al 49 n iwassen tsenfal ilem-nes x sent al xamesa n tikkal. Alaxti i teḍha tizzegret-nes 03mm takečča tkenna. (LEPESME, 1947).

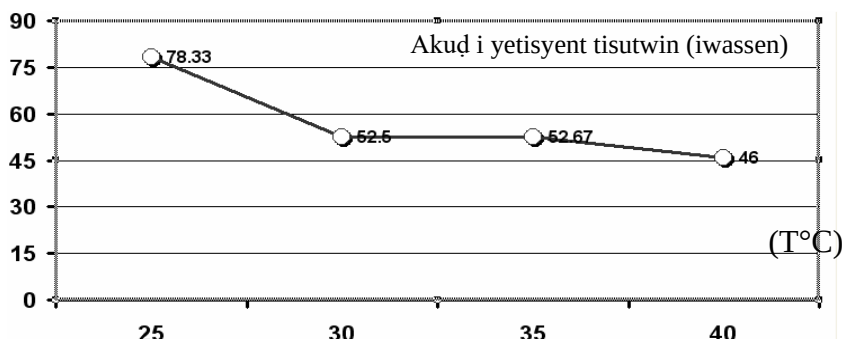
Butranđa bab n tɣmas ul tɛeddem-ca tizdayen g iman-nen beṣṣaḥ idyee yiniwen lwaqt n ukmas-nen g tzaqqa.

Tafelwit 7 : N tɣyelt xi tarwa n butranđa bab n tiɣmas.
(AL – DOSARY, 2007)

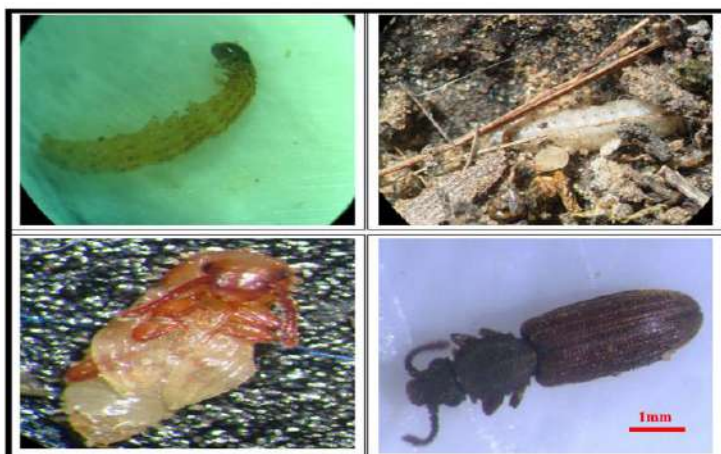
(T°C)	Zzat n tarwa (iwassen)	Anecten n tezdal g iggen n tarwa	Aqerrag xi tizdal (iwassen)	Amfidi n ufqas (%)
25	18.67	15.67	10	23.5
30	9.33	38.67	09	78.67
35	12.33	80.67	07	59.33
40	10.67	53.33	04	38.73

Tafelwit 8 : Asemdu n tɣyelt xi tkiččawin butranđa bab n tiɣmas d ukenni-nsent (AL –DOSARY, 2007)

(T°C)	Anecten id tawi takečča (iwassen)				Tamuttrt	Anecten id tawi akenni
	Stade 1	Stade 2	Stade 3	Stade 4		
25	6	5.67	7.5	12.5	31.67	13.33
30	5.4	5.67	6.5	7.43	25	5.67
35	5.2	4.67	5.7	8	23.5	5.33
40	3.34	5.5	6.33	6.5	21.67	6.67



Tunuyt 12 : Asemdu n tzyelt x ukud i tisyent tisutwin.
(AL -DOSARY, 2007)



Tunuyt 14 : Iweswiren n ukur butraṇḍa bab n tiymas.
(*Oryzaephilus surinamensis* L.) (Taaşliyet).

2.5. *Ommatissus binotatus* Fieb.

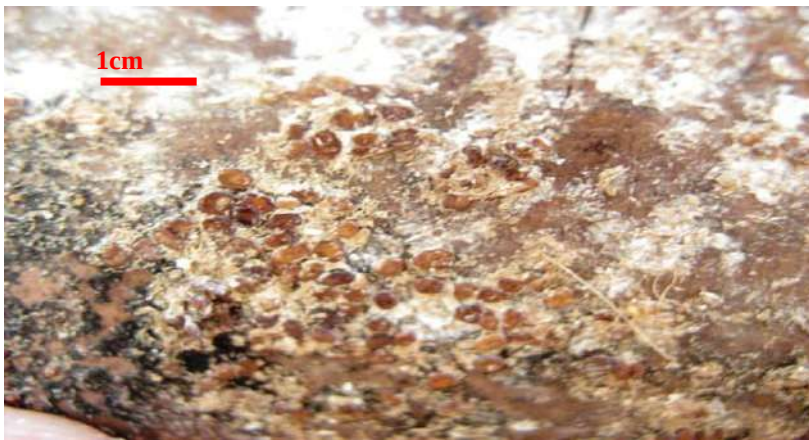
D iggen n ddeg n ifritṭa tatfen netnin d ixxa ifriwen-nsen kif kif (Homoptères) twacunt n Trapiduchidae tḍyeḥ aṭṭas tazdayt an :

- Tkitcawin-nes d tindi ig iwḍent tzemment aselḥay xi tireṭṭwin d tiyedwin n tuffa. taqelwant n uziwa d izriren-nes ul tugej ladca taweḍ al yiniwen ;

- Deffer n uzemmem n uselyay trezzem iggen n tamamt tmum agena n trettwin, my d tjebden abeččaču d la bactérie ig teggen iṭṭanen.
(BUXTON, 1920).

2.6. *Phoenicococcus marlatti* Ckll. (Tawlaft 2)

Tinḍfert n tillit tamellalt n ssam lebda tetbaet ; tneṭaḍ adday n tqarnift lkulur-nes d aqehwi iza l tizwayt trezzem lad nettat iggen n timlelt an iner.



Tawlaft 2 : Teskunu-d *Phoenicococcus marlatti* (Taaṣliyet).

2.7. Bab n Iccawen *Pseudophilus testaceus* Gah., (Tawlaft 3)

Tajlujet tameqrant, tatef nettat d ixxa ifriwen-nsen quren, d twacult n Cerambycidae, iffay-d tikkelt tamezwart g Leiraq ittili bezzaf sadu n iyezran, tintedayen i gsen alli tuft aṭṭas l ujenna :

- Tsihiḍiren tazdayt terrant-d tmiyyayt ;
- Duggel-d tazdayt thiyyen y uṭṭu d terzi ɛla xater aziqqur-nes ittli inqeb ;

- Tidras tarwa n tezdayt d ul tig-ca n iziwayen ;
- Aṭṭas n inyuren g uzeqqur ul tugjen-ca aman a d alyen l ujenna hayd ma ig tsqqaren tiffawin n tzdayt.



Tawlaft 03. Teskunu-d Bab n Iccawen *Pseudophilus testaceus* (Taaşliyet).

2.8. *Phonapate frontalis* Fahraeus

Xi tijlujin tikeɛbac ittafef netta d les coléoptères adday n twacult n Bostrichidae teqqaz la d nettat ixuba gi tuffa, kul tedweld tehci d al trezza fisæ.

2.9. *Apate monachus* Fabricius

Ittlili lkulur-nes d aqehwi iqseḥ tig jar 1.5 d 2 cm tatef l wammas n tayda n tuffa mani iges tarew tizdal-nes deffer n ufqas tefyen-d tikitcawin-nes tettint azizaw d umeqqur may ig tneqqan tuffa.

2.10. *Tmuryi Schistocerca gregaria* F. (Tawlaft 04)

Tatef nettat d ixxa isen teqqaren Orthoptères (ifriwen ig serḥen) ul i tuggej lad iggen umkan gi tazdyt kulci ttetit an tuffa, aziwa, amersid, azeqqur... alaxti itutef tmuryi l tintzdayen itqamayaw ufellah yir tzallit d tutra i mulana.

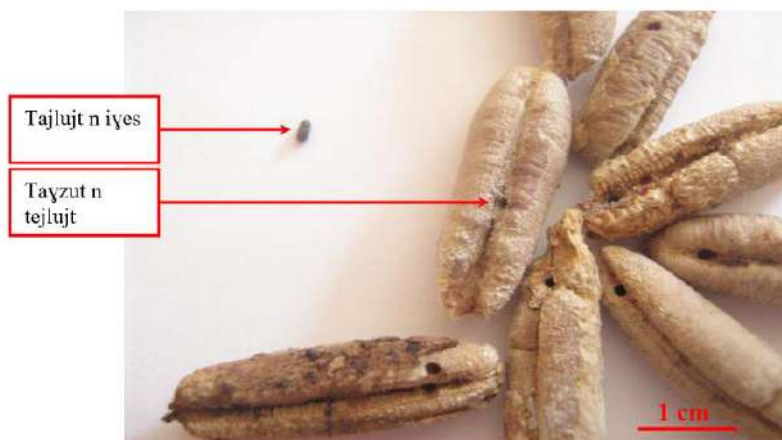


Tawlaft 4 : Teskunu-d izellayen tmurji *Schistocerca gregaria* (Taaşliyet).

2.11. Tjlujt n iyes *Coccotrypes dactyliperda* (Tawlaft 5)

Lkulur-nes d aqehwi iza l tizweyt. Ittatef axxu yu netta d at ifriwen ig qeşen (coléoptère) d twacunt n Scolytidae tizegret-nes ul tfit-ca 1.5 mm, awtem ittili d akeebuc xi tawtemt. Tawtemt lwaqt n tarwa tajlujt yu teqqaz g iyes n winiw, deffer n usersi n tizdal takečča-nes tawid x 05 al 10 n iwassen maki ad teffey.

Tukna n tketca i tawid jar 5 d 7 n iwassen, deffer-s teffey-d tifert tiweđ tafa x uwtem maki at leqqah. Tiwinest n tudert n tejlujt yu ul tfit-ca 70 n iwassen.
(BUXTON, 1920)

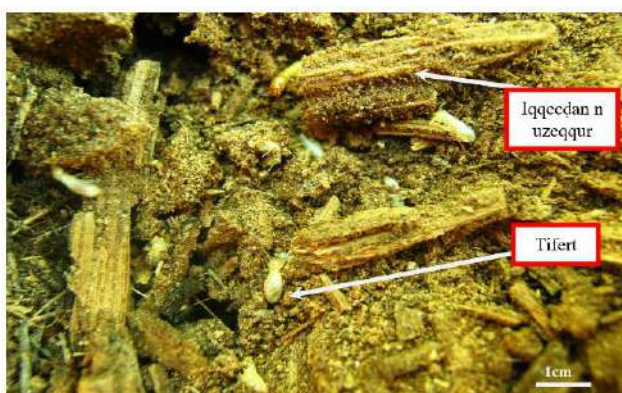


Tawlaft 5 : Teskunu-d tajlujet n iyes (Taaşliyet).

2.12. Tifert *Microcerotermes diversus* Sil. (Tawlaft 6)

Tifert tatef gi tafesna n isopteres (ifriwen kif-kif) teder tifert gi tirbaein yer-sent tagelllidt tsigur-ten. Tifert tili aţtas g tinzdayen timyarın d tiwessarın. Tifert tetti luħ n uzeqqur d izewran hayd ma iddi terran tazdayt tehciċ :

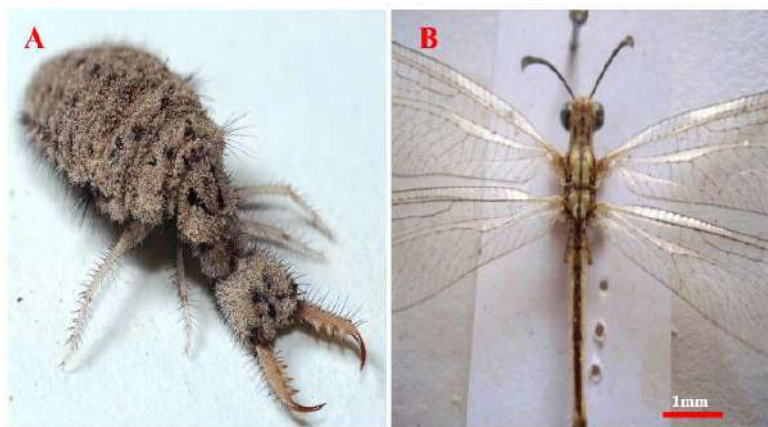
- Tifert tsehlak tazdayt d al tsenqas tarwa n iziwayen
- Tazdayt deggel-d tehciċ d al tuţta mata illa ađu ijhed



Tawlaft 6 : Teskunu-d Tifert *Microcerotermes diversus* (Taaşliyet).

2.13. Kurtata imenÿi n tifert d iket  fen (Tawlaft 7)

Tatef gi tafesna n les n  vropt  res. Kurtata hayd imenÿi n tifert, tedder adday n tezdayen g ucal, mani itteqqaz iggen te  zut tmeddi ges i yeket  tufen d tiffert. Alaxti i tuki kurtata belli tu   a ca n   ajet gi ta  zut-nes tbedda tne   ar-xes cal maki at da te   ef.



Tawlaft 7 : Teskunu-d kurtata *Curvatus navas* a) take   a (b) ameqran (Taa  liyet).

2.14. Takula *Oligonychus afrasiaticus* (Tawlaft 8)

Takula d ddeg n rrtila tame  zyant a   as ul yettwazra-ca xi     . Yetttig jar 0,2 d 0,4 mm. Ttinin-as xi ta  rabt "Buferwa". Tatef deg twacunt n Tetranychidae. Al axtin i tutef y  l tinezdayin ul tugge   la d igget n tezdayt tezzar-d q    iziwayen te   teftent ta  yent (l  ybret).

Tawtemt-nnes tettarew-d a   as n tezdal-nnes. Deg uziwa ttawyen-d xi 3 al 5 n iwassen maki ad feqsent. Takula tettarew tizdal-nnes deg yiggen n weglef n le     r, tleggi-t xi uziwa. Ammu tbedda tgerrew xef-s l  ybret ha-ay d ma i   ey   en iyniwen. (SEDRA, 2003).



*Tawlaft 8 : Izellayen (lexşarat) i tettig Takula n iyiwen, afxas d uzella n tamemt.
(Taaşliyet).*

- V -

Anafuy (Ifuras) n tezdayt

1. Tamamt n yiniwen (Reb) (Tawlaft 1)

Sskenun-t xi yiniwen iqdimen niε uxcinen¹⁸¹ ul n tnez-ca, tseterturen-ten deg waman ca 4 n swayee alxtin i reẓmen tamamt-nsen deffer-s kul setyen-ten¹⁸².

Tamamt n yiniwen tucen-t i tenẓurt¹⁸³ tendi i yerwen tajdidt maki ad tter idamen-nes deffer n tarwa εla xater yella-deg-s aṭṭas n uzdal¹⁸⁴.



Tawlaft 1 : Teskunu-d tamamt yetwaxdem xi yiniwen.

2. Lagmi (Tawlaft 2)

Llun-nes yettli d amellal iḡebber, tsufuyent-id xi walaw¹⁸⁵ n tezdayt tawessart¹⁸⁶. yetreggeb afellaḥ l wul n tezdayt kul yekkes tiffawin n walaw, alxti i yewweḍ l

¹⁸¹ Uxcinen : Fr. : *mauvais*.

¹⁸² Setyen / astay : ṣaffan : Fr. : *filtration*.

¹⁸³ Tenẓurt : tameṭuṭ i yerwen.

¹⁸⁴ Azdal : Fr. : *le fer*.

¹⁸⁵ Walaw / alaw = wul : Fr. : *coeur*.

¹⁸⁶ Tawessart = tamẓart niε tameqqrant.

jmmar ad yeyez deg wammas-nes iggen ubadu maki a xef-s di serseb lagmi an terga deffer-s tgerwen-t deg tanwart maki a yeqyem d asemmaḍ əla xaṭer matta yehma yetxemmer d iḍḍaḥḥa-d d ccrab¹⁸⁷.

Lagmi tgerwen-t aṭṭas deg uyur n Remḍan əla xaṭer yettekes fad, d yetsarad¹⁸⁸ adan¹⁸⁹ d dsa¹⁹⁰



Tawlaft 02. Lagmi n walaw n tezdajt.

3. Mahya (Tunuḡt 1)

Tinin-as aman n tudert, la sskenun-t udayen n Seferdym ig lan dren deg Sawra lan teggen-t xi uḍil, iqerbac¹⁹¹ d yiniwen. Udayen lan tsexmaren iyniwen deg iggen n uqlil n ufexar, deffer-s kul stertren-t deg ilemsi, irugga i d tefyen tfatent deg tiyu d al tsmuḍun-t xi waman maki ad yuc tigeṭṭar n alcool txelḍḍen-t netta d jus n iqerbac.

¹⁸⁷ Ccrab : *Fr. vin.*

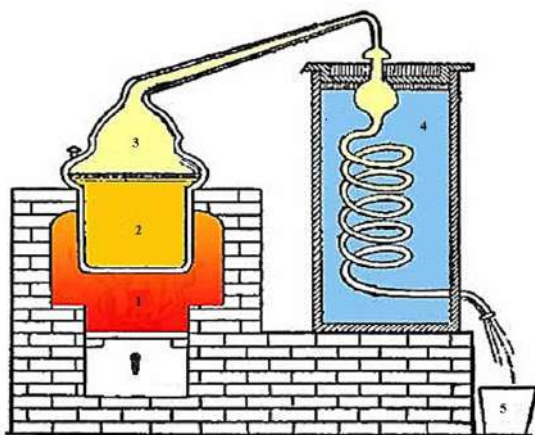
¹⁸⁸ Sired / yetsarad : *Fr. nettoyer.*

¹⁸⁹ Adan : *Fr. intestin.*

¹⁹⁰ Dsa : *Fr. le foie.*

¹⁹¹ Iqerbac / tazart : *Fr. figue.*

- 1- Timsi
- 2- Iyniwen ikdimen d uqqicen
- 3- Taseksut tgerew leffar n uterter.
- 4- Aman isemmaçen
- 5- Mahya



Tunuyt 1 : Asufay n mahya x yiniwen.

4. lqehwa n iyes n yiniwen (Tawlaft 3)

Ddeg yu n lqehwa teggent xi iyes n yiniwen themšen-ten, deffer-s kul zden-ten d xelden-ten netnin d ukeb n lqehwa. Tawed sumet lqahwa-yu deg Fransa al 39,96 € i iggen n kilo, Lqehwa tqetṭa tyawṣa deg ashewwu n ufmiḍi n tmunt d lkulestirul deg idammen.



Tawlaft 3 : Lqehwa xi iyesan n yiniwen.

5. Tutut n welli (Tawlaft 5)

Deffer n unkaḍ amekraz yettseggeṭ tizdayen-nes mani ig tekkes qaε tiffawin timeqqurin d al yetgrew iɣemma d uqqicen. Terran-ten d tutut n welli d ileyman mani iten tezzaḍen deg tasirt d al zayden-xsen irden niε taceqlubit¹⁹².



Tawlaft 4 : Tasirt n tiffawin.

¹⁹² Taceqlubit = akbal.

6. Tisksa (Tawlaft 5)

Gsen-t ay tsalayen uċċu¹⁹³, sskenun-ten xi tiretṭwin tilemmaḍin n wul n tezdayt.

Deg umezwar tṛezmen aqelwan l izriren an tnelwin deffer-s tlegyen-xsen tirtṭwin tneqben-as xi isten¹⁹⁴. Tagjen-as adday-nes irzem maki as dren traċa¹⁹⁵.



Tawlaft 5 : Taseksut n usili n iċċu.

7. Tagnint (Tawlaft 6)

Tagnint tissi-as i tseksut yir nettat yres tmadunt-nes, tesrusun deg-s šwaleḥ n cbaḥ n tsednan an tazult, lebxur, lḥanni.

¹⁹³ Uċċu : sekksu.

¹⁹⁴ Isten : tsarut tameqqrant.

¹⁹⁵ Traċa : cebket.



Tawlaft 6 : Tigninin n tzult d lebxur.

8. Zilem (Tawlaft 7)

D ddeg n isywan sskenun-t xi usan n tezdayt, deg umezwar teddaren iggen n tejdilt deffer-s tgerwen-tent qae maki ad seqshen zilem ela xater tagmen xef-s deg uyrur ixes-as adili iqsaḥ maki ul itqereḍ-ca.



Tawlaft 7 : Aseknu n zilem.

9. Asyun

Yettisy-as¹⁹⁶ an uzilem walayenni d azewwar d al txedmen-t xi tiggi, asan, taɣduft n walli niε ixeff n iɣayɗen.

10. Tirkasin (Tawlaft 8)

Tirkasin hayd tisila twaxdamen-t xi tiretɣwin



Tawlaft 8 : Tirkasin n traɣwin.

11. Taklut (Tawlaft 9)

Teddarent tisednan¹⁹⁷ xi traɣwin n wul n tezdayt tizizawin yettisi deg-s imekrez tutut-nes l tertut alxtin yeg xs ad ikel¹⁹⁸ itxeddem.

¹⁹⁶ Yettisy : tisit /Fr. : *ressemble*.

¹⁹⁷ Tisednan : timeɣtuɣin.

¹⁹⁸ Kel/ akla / yettekkal : deffer n umekli, ansa : *passé la nuit*, akla : *passé l'après midi*.



Tawlaft 9 : Taklut deg dwi yettefren iyniwen d tuttut n umekraz.

12. Zinbil (Tawlaft 10)

Yettwadra xi tiraṭwin, tsaraḍent yu ayyul tisyen deg-s tiṣayt. Tesrusun-t ajenna n teḥlast maki ul yettig-ca n tmejeuṭ ¹⁹⁹yu ayyul.



Tawlaft 10 : Zenbil n uyyul.

¹⁹⁹ Tamejeuṭ : ḍebret / Fr. : *blessure*.

13. Ajertil (Tawlaft 11)

Ha-ay d taḥsirt n at zzmen tawyen-d deg-s jar iggen d sen iwassen teddaren deg-s, tsun-t gi tanwalt.



Tawlaft 11 : Ajertil n traṭwin.

14. Azefli (Tawlaft 12)

Tidrin n zzmen twaskan-t xi luḍ, alx̣tin yewḍen l uzeḥli tesrusun izeṭran n tezḍayt an ifeggaḡen, deffer-s tesrusun ajenna-n̄sen iggen n srir teddaren-t xi tiỵedwin n tffawin n tezḍayt deffer-s tleḍmen-t xi luḍ.



Tawlaft 12 : Izeṭran n uzeḥli ajenna-n̄sen srir n tiỵedwin.

15. Tayda n uzeṭṭa

Tayda n wammas n uzeṭṭa tawyent-d xi tiffawin tizeggar n tazdayt, teksen xef-s tiraṭwin.

16. Tazwet d tṛewwaḥt (Tawlaft 13)

Tazwet ččaten xef-s izan txedment iwessaren xi tiraṭwin.



Tawlaft 13 : Trewwaḥt d tazwet n izan.

17. Lemḍel (Tawlaft 14)

Iteddart imekrez xi tiraṭwin d yetteṛeḍ-t deg uneḍu yettissi xef-s tfuyt, tirḍent aṭṭas iwassen n ukraz d umjar.



Tawlaft 14 : Lemḍel n traṭwin.

18. Izeṃran n uska (Tawlaft 15)

Teggen xef-sen ifeggaḡen n uzefli, tawret, d al teggen-t-n y uqimi



Tawlaft 15 : Izeṃran n uska.

19. Afrag (Tawlaft 16)

Ssefregen xef-s tanzdayt yessenqes aḍu. D ittefer turtut.



Tawlaft 16 : Afraq n turtut.

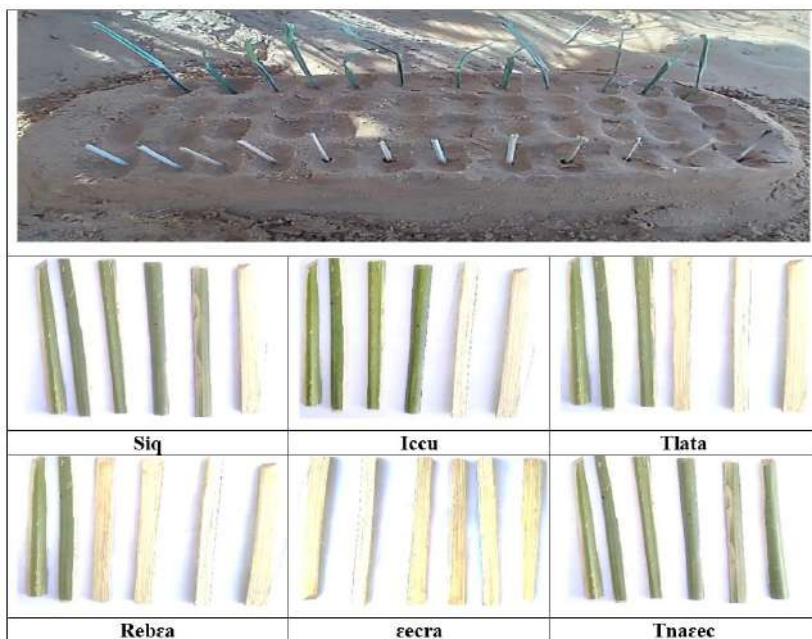
- VI -

Uṛaṛen deg tnzdayt

1. Siq

Turarent tisdnan gi tanzdayt, g umezwar txedmen-t iqecdan n siq xi tagda n tuffa n tezdayt, tbeřant l 06. Kel igget tilli ges ljihet tamellat d teniřen tazizawt. Defres kul xedmen-t řaret. Ittili ges rebęa n řfufa kel řef yella-ges 12 n tayzut teqqaren-asen iřelsan.

Tqamant tisdnan i ga-d yurar sent tinin-asen tıřfigin d sent tineřen tinin-asen timenřar. Tıřfigin teggent 12 n uzegmer hayd lwacun-nsent, d timnřar teggent 12 n uqeccuř hayd lwacun-nsent... Itbedda n urřar x unřar n iqecdan n siq gi tmurt d tsugmed anecten ak d yuc ; iccu, siq, sent n tidrin, tlata tidrin rebęa n tidrin ęacra n tidrin d tnaęec n tidrin.

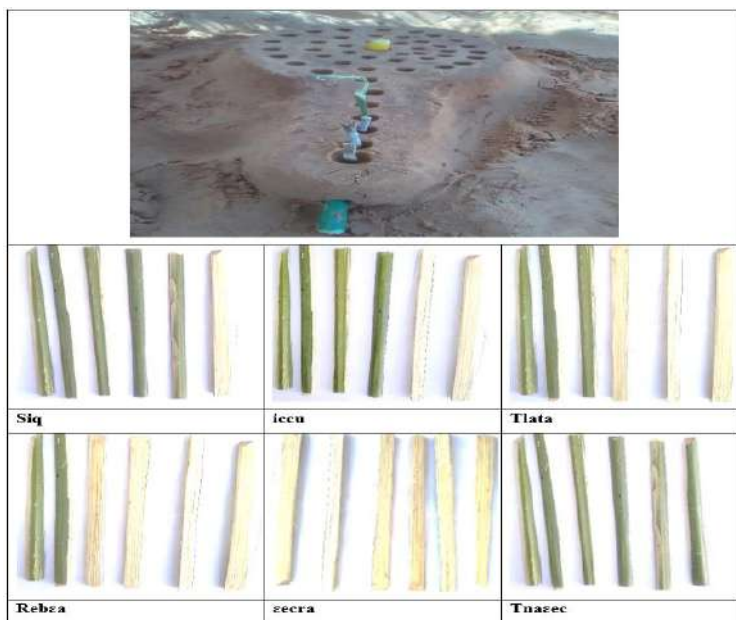


Tunuyt 1 : řaret n siq x lwacun-nes d uslgn n urar (Taařliyet).

G umezwar maki at sekeckec tameţuţ lwacun-nes ixes-
as ad tawdi siq , εacra niε tnaεec, dadi tini-as ca tsmeklew
lwacun-nes amala ad zemren ad iguren, x leħsen n unċar n
siq, atzmer tameţuţ attegrew qaε lawcun-nes d iguren g
iggen n tikkelt dadi teqaren-as tajrut εla xater atezmer
attafeg. Tisdnan id yewċant l tidrin n timenċar-nsent
tbeddant tneqqan-asent lwacun-nsent i tinniċent wi ig
semdant timezwar hayd tinidi ig yelbent.

2. Amza

Turart n umza tisyas i siq mfaġament yir g ċaret.
Turarent 05 n midden, 04 tillit d lwacun d umeggaru a d
yuṛar i umza, lwacun ixessa-sen aduṛaren maki aya rewl
wi igiwċen l uyellus n ulebbuz ad islek beşşah indi iten di
tṭef amzza aten iteċ.



Tunuyt 2 : Ċaret n umza x lwacun-nes d uslgn n uṛar (Taaşliyet).

3. Buzegmer

Turařen-t sen n imenđar, tafañ x wi iyes-illan tamella. Ittili ges 10 n ilebbuzen 09 x ddeg d iggen x ddeg idđen, tmeṭlen qaε ilebbuzen adday n uca. Defres tbeddan trecgen azegmer x ugenna, wi id isufren alebbuz ig mfagamen netta ig εelmen tamella.

	
09 nilebbuzen n boufequs d iggen imfagam d wen uherđan	Amđal n ilebbuzen adday n ucal
	
Ixser uyers illi-ca n tamella	Irbah yres tamella

Tunuyt 3 : Turařt n buzegmer d uslgn n urař (Taařliyet).

4. Zerragraq

Ittwaxdam x uqelawan azizaw n uziwa itwaqraq maki itella tseḍhar-d teṣwired, tuṛaren-xes lwacun laxrif deffer n unkaḍ ittig lḥas an waqqas n ufus.



Tunuyt 4 : Tuṛart n zerragraq (Taasliyet).

5. Lemccet (Takuba)

Turaren xes lwacun imezyanen timenyi jarat-sen, txedmen-t xi taqelwant n uziwa azizaw d al teknen-t xi tiretṭwin g ufus-nes.



Tunuyt 5 : Tuṛart n takuba n wacun (Taasliyet).

6. Takuret

Tkuret tuṛaṛen-t tarbaet n irgazen, tisseri i l hockey n guddu, deg-s iggen tkuret teskennun-t x usan n tazdayt tlegyen-t mliḥ x iggen usyūn, tiyriyen n tiyta tskenun-ten xi tayda n tuffa n tezdayt ddeg n TUMLIHA εla xater tayda-nes yettili teqsaḥ aṭṭas, tqqaṛen-as lmekcaḥ. Inuṛaṛen tilin xemsa d xemsa iden d imenḍar we ig yeswen aṭṭas netta ig rebḥen.



Tunuyt 6 : a) takuret n wasan b) lmekḥac n tiyta (Taaṣliyet).

7. Taylula

Tuṛaṛent lwacun jar tizdayen yeg qerbent aṭṭas, teqnen sent n tiffawin, iggen a yeni d wenniḍen at yedmer. Iyzer n Ssawra yetbedda deg Igli, ha-ay d amlaga n sen iyezran Gir d Zuzfana, deg lejnubat n Ssawra d xi tizzegreṭ n 200 Km netlegga-d aṭṭas n tinzdayen.

- VII -

Tinzdayin n Ssawra

1. Ayrem n Igli

Yused l ujenna n at εebbas jar-a-tsen 75 Km,

Tafelwit 1 : Tinzdayen n uyrem n Igli.

Tanzdayt	ṭṭaşilt n smiyyet
01 Akebli	Tanezdayt yu tused l uṭṭu n tfyt ljihet n l qeblet
02 Turtut	Amkan n ukraz
03 Timmi	Tused smiyet yu xi timmi (timmiwin) n tiṭ,
04 Tamezrart	Ul twasen-c
05 Tagwirtit	Ul twasen-c
06 Acenqur ameqran	Ul twasen-c
07 Buεemda	Smiyyet taεrabt tinni isebdad
08 Aṭṭerṭas	Ha-ay-d lwesset n tiddart yettili sadu n turtut
09 Tellemsi	Ul twasen-c
10 Tbuxirant	Ul twasen-c
11 Tawerda	Awal yebḍa ṭel sen n iwawalen amezwar ha-ay-d tawert= bab da= dadi
12 Tamizit	Tused x umezzi n waman, dyella wi igtinin ha-ay-d iṭes n tmezmact
13 Amxisir	Ul twasen-c
14 Lexbaṭi	Ul twasen-c
15 Bukeddara	Ul twasen-c
16 Iyzer n at Brahim	Tanezdayt tused saddu niyzer
17 Taḍrawit	Tamda tedra
18 Taxerbint	Ul twasen-c
19 Belcheb	Yella deg-s aṭṭas n waḍu

20	Tawessart	Tizdayen timɣarin
21	Aɣrem n uyil	Ul twasen-c
22	Tisuyar	Ul twasen-c
23	Tinuxecci	Tizdayen-nes usent-d deg umsitet
24	Tabuzit	Ul twasen-c
25	Tisɣar	Ul twasen-c
26	Abuhu	Ul twasen-c
27	Tanut tajdidt	Tannut niε annu
28	Lxebtet	Ul twasen-c
29	Balyazi	Ul twasen-c
30	Buqeddara	Ul twasen-c
31	Dayur	Ul twasen-c
32	Aɣzu n eli u lɛarbi	Aɣzu yused xi taɣzut tinit yu asehwa
33	Taɣzut n at baɣyir	Aɣzu yused xi taɣzut tinit yu asehwa
34	Taberkat n at deħan	Taberkat d dden n tsekla
35	Tanut n tɣanimt	Yella deg-s aṭṭas n tɣanimt
36	Tin tlat	Ul twasen-c
37	Tin maεtalla	Ul twasen-c
38	Idmamen	Ul twasen-c
39	Σedlun	Ul twasen-c
40	Xebbuza	Ul twasen-c
41	Ifli	Yella deg-s aṭṭas n waman
42	Targa	Tiṭ n waman tegguren
43	Tamɣart	Tameqrant tella deg-s iggen n tezdayt tawessart
44	Tamma n Hella	Ul twasen-c
45	Iflan	Deg-s aṭṭas n waman
46	Lebbared	Deg-s aman ajenna
47	Tasa	Ul twasen-c
48	Agerzim	Ul twasen-c

49	Azurez ameqran	Tiṭ n waman tameqrant
50	Tfasut	Ul twasen-c
51	Afud	yez leḥsab n ufud atelgid aman,
52	Bdar	Ul twasen-c
53	Tuzdit	Tamurt tserraḥ amkan n tuzuṭ n tizdayen
54	Σebbu	Ul twasen-c
55	Σmmi mbarek	Ul twasen-c
56	Iyeryer	Aman deg-s jlan, ixεaḥ atteyzed aṭṭas
57	Imekrez	Amkan n ukraz d tuzut n irden

2. Ayrem n Mazzer

Yused l wadday n uyrem n Igli jar-a-tsen 15 Km.

Tafelwit 2 : Tinzdayen n uyrem n Mazzer.

	Tanzdayt	ṭṭaṣilt n smiyyet
01	Iwetman	Tinzdayt deg-s aṭṭas n iwetman
02	Tiṭṭawin	Deg-s tiṭṭawin n waman
03	Ifli	Amkan n waman
04	Baṛan	Ul twasen-c
05	Tbaṛant	Ul twasen-c
06	Tijent n wadday	Tijent ha-ay d amsegraw n waman
07	Tameḥzunt	Ul twasen-c
08	Urten	Ul twasen-c
09	Aεebbu	Ul twasen-c
10	Teεebbut	Ul twasen-c
11	Azru	Ul twasen-c
12	Ifli n εisa	Amkan n waman

13	Bunedraṛ	Ul twasen-c
14	Azerquq	Ul twasen-c
15	Lweset	Amkan yerzem
16	Tqabbayt	Ul twasen-c
17	Asergeb	Tused x urgab
18	Bubrik	Ul twasen-c
19	Merkan	Ul twasen-c

3. Ayrem n At Sebas

Yused l wadday n uyrem n Igli jar-a-tsen 75 Km.

Tafelwit 3 : Tinzdayen n uyrem n At Sebas.

	Tanzdayt	ṭṭaṣilt n smiyyet
01	Turgin	Ul twasen-c
02	Tifezzawin	Ul twasen-c
03	Turmas	Tused x urmas (triplex)
04	Fedriyen	Tused xi tidrayt n waman
05	Admam	Ddeg n tizdayen
06	Warurut	Ul twasen-c
07	Mealla	Ul twasen-c
08	Frawen	Tused xi tafrawt
09	Bent omran	Ul twasen-c
10	Bḥira	Ul twasen-c
11	Lebtaḥ	Ul twasen-c
12	Jerf	Ul twasen-c
13	Tamtert	Ddeg n tsakelt ephydra alata
14	Taşennaret	Ul twasen-c
15	Tameezuzet	Ul twasen-c

16	Hemmuca	Ul twasen-c
17	Tifezzawin	Ul twasen-c
18	Taneqqalet	Ul twasen-c
19	Dehlu	Ul twasen-c
20	Qetla	Ul twasen-c
21	Buqazi	Ul twasen-c
22	Badriyan	Ul twasen-c
23	Fteṭis	Ul twasen-c
24	Feṣṣa	Ul twasen-c
25	Kaḍi ben allal	Ul twasen-c
26	Jnanzawiya	Ul twasen-c
27	Kaid ben setman	Ul twasen-c
28	Fetṭus	Ul twasen-c
29	Ferwan	Ul twasen-c
30	Qerqasi	Ul twasen-c
31	Wad lylul	Ul twasen-c
32	Wad james	Ul twasen-c
33	ḥewisi	Ul twasen-c
34	Tuylilin	Ul twasen-c
35	Tureggin	Ul twasen-c
36	Kermet ceḥbun	Ul twasen-c
37	Jnan Saru	Ul twasen-c
38	Gulead	Ddeg n uluḍ
39	ṭasra	Ul twasen-c
40	Dennu	Ul twasen-c
41	Bujemea	Ul twasen-c
42	Reg lexḍer	Ul twasen-c

4. Ayrem n Lwata

Yused l wadday n uyrem n Igli jar-a-tsen 103 Km.

Tafelwit 4 : Tinzdayen n uyrem n Lwata.

	Tanzdayt	ṭṭaşilt n smiyyet
01	Ammas	Tused deg ammas n uyrem
02	Idiy	Ul twasen-c
03	Anfid	Ul twasen-c
04	Majja	Ul twasen-c

5. Ayrem n Kerzaz

Yused l wadday n uyrem n Igli jar-a-tsen 167 Km.

Tafelwit 5 : Tinzdayen n uyrem n Kerzaz.

	Tanzdayt	ṭṭaşilt n smiyyet
01	Tiyerday	Tused x uyerda
02	Azeggar	Tused xi tazeggaret

6. Ayrem n Wlad Xeḍir

Yused l wadday n uyrem n Wlad Xeḍir jar-a-tsen 218 Km.

Tafelwit 6 : Tinzdayen n uyrem n Wlad Xeḍir.

	Tanzdayt	ṭṭaşilt n smiyyet
01	Telemsu	Ul twasen-c
02	Tabelbal	Ul twasen-c
03	ṡarmseid	Ul twasen-c
04	Tuḥrir	Ul twasen-c
05	Qerruc	Ul twasen-c
06	Asraf	Ul twasen-c
07	Taqlaqal	Ul twasen-c
08	Lenbak	Ul twasen-c

Taggrayt tamatut

Deg Ssawra teqqyem-d yir Igli d Mazzer i yettssawalen Tamaziyt-Tabeldit. Kull iɣermawen yenniɗen yeẓwa yiles-nsen. Ujar n wezgen n smiyat n imukan, d tnezdayin deg iɣermawen ul nettsiwil-ca tamaziyt. Qqeyment al gudu, walayenni anamek-nsen yeyba.

Tazdayt d tnezdayt ha-ay d isebdad n iɣermawen deg taneẓruft. xef-s tuttut-nsen, netnin d welli-nsen, d xef-s irraɗ-nsen d x izeyran-nnes d tiffawin tsekkan tidrin-nsen. D al tetteshewwa tazɣelt n unebdu, ha-ay d axwin At Tnezruft učen-as amkan d ameqqran gi tudert-nsen.

Izewran n tezdayt yreyren deg ucal. Tetṭef iman-nnes maki ul tt-yettneṭṭer-ca waɗu. Maki i tetṭef iman-nnes, tetṭef id-s Tabeldit, awal n At Igli.

Al idennat Tabeldit tella ul tettwari-c. Ha-ay d ma i yuggjen aṭṭas n iwawalen ad ẓwan. Ass-u aɗlis-u i yeryey xi tezdayt d leqɗem (asurif) d ameqqran ɣel zdat. Ad yenfeε Tabeldit deg Igli deg umezwar d yenfaε Tamaziyt qaε.

Aḍlis-u yeččat sen n ijuḍaḍ xi iggen n tɣunt.
Amezwar ad yefreg Tabeldit x uɣbar. D west sen
ad yesselmed i At tnezruft maki ad zɣyen tazdayt
d thellan deg-s.

Yella iggen n waggaj n at zzman an : weyrur,
tsaylalt, tafeggat, ixess-aneɣ ad yeqqyem ul
yettɣebba-ca. D maki ad yeqqyem ixess-aneɣ ad
nari aṭṭas x yiles n Weyrem n At Igli.

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